Stochastics

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Chapter 1

Introduction

Stochastics is the branch of Mathematics that deals with randomness. As we all know, there are many phenomena in our today's world which involve some randomness. Thus, Stochastics is an important and interesting subject within Mathematics, but it is also highly relevant for many other disciplines, ranging from the natural sciences and engineering to the social sciences.

Stochastics consists of Probability Theory and Mathematical Statistics. In Probability Theory, some underlying probabilistic model is given and one aims to predict and understand its behaviour. In Mathematical Statistics, the underlying probabilistic model is partially unknown, but some observations are given. The aim is to draw conclusions about the probabilistic model from the data. Let us illustrate this difference with an example. A coin is called fair if both sides, 'head' and 'tail', have the same probability when we flip the coin. In Probability Theory, one considers the situation that a fair coin is tossed 100 times and asks for the probability that 'head' occurs 60 times. In Mathematical Statistics, the related problem is that some coin was tossed 100 times and that 'head' occurred 60 times. From this information, one tries to learn whether the coin is fair or whether it favours some side.

One can think of Probability Theory as a more theoretical subject than Mathematical Statistics. On the other hand, from the example above one can see that Mathematical Statistics requires a sound understanding of Probability Theory. For this reason most of this lecture is devoted to Probability Theory, before we study some questions from Mathematical Statistics at the end of the semester.

Historically, much research in Probability Theory was motivated by gambling. One was interested in understanding the chances of a player in some game or lottery. Throughout this class we will often study examples such as coin tosses and dice rolls since they are rather instructive. In the following we briefly describe some more involved applications of Stochastics to real-world problems.

Let us model the queue in a post office with a single counter. One can think of the arrival times of customers as a random process and the service time of a customer can also be assumed to be random. This is the easiest example of a queuing system. More involved queuing systems arise in the context of computer systems or production systems.

Stochastic models are essential to compute the premium of an insurance contract and to control the risk exposure of an insurance company. In finance it is often assumed that asset prices follow random processes such as the so-called Brownian motion, which was first used to describe the random movements of particles in Physics.

A highly relevant statistical question, which arises in particular in medicine, is to decide if one treatment is better than another. Here, the treatment outcomes of two groups of patients are given, one group received a new drug, while the other group received an existing drug or a placebo. The goal is to decide if the new drug leads to better results. Usually the outcome of a treatment with a drug is at least to some part random. Thus, the challenge is to understand if a new drug is really advantageous or if it only showed slightly better results due to random fluctuations.

A further important problem, which is a key topic in Machine Learning, is to predict some dependent variable y from independent variables $x_1, \ldots, x_m$. Here, $x_1, \ldots, x_m$ could be some medical data and y could be the indicator whether a patient has some disease or not. Another example would be to predict the income y from some other data $x_1, \ldots, x_m$ like age or education. The goal is to find some model how y is determined by $x_1, \ldots, x_m$. For the described examples it might be obvious that there exists no deterministic formula that allows to compute y from $x_1, \ldots, x_m$. This problem can be resolved by modelling y as a random quantity, whose distribution depends on $x_1, \ldots, x_m$.

Chapter 2

Probability spaces

2.1 Discrete probability spaces

A random experiment is an experiment whose outcome is not predictable and which we can repeat as often as we want. Examples of random experiments are

1. throwing a die
2. tossing a coin n times.

The set of all possible outcomes of a random experiment is called sample space and usually denoted by Ω. For the examples above we have $\Omega = \{1, 2, 3, 4, 5, 6\}$ and $\Omega = \{0, 1\}^n$, respectively, where we encode heads and tails by ones and zeros. The elements of Ω are called elementary events. We denote the subsets of Ω as events. We say that an event $A \subseteq \Omega$ occurs in a random experiment if the observed elementary event ω belongs to A. For throwing a die, $A = \{2, 4, 6\}$ is the event that the die shows an even number, while $B = \{2, 3, 5\}$ is the event that the outcome is a prime number. Then, $A \cup B = \{2, 3, 4, 5, 6\}$ is the event that the die shows an even number or a prime number, whereas $A \cap B = \{2\}$ is the event that we throw an even number and a prime number at the same time.

The set of all possible events is the power set $\mathcal{P}(\Omega)$ of Ω, which is the set of all subsets of Ω and which is given by

$$\mathcal{P}(\Omega) := \{A : A \subseteq \Omega\}.$$

For example, we have $\mathcal{P}(\{a, b\}) = \{\emptyset, \{a\}, \{b\}, \{a, b\}\}$.

For the moment we will only consider the discrete case where the sample space is at most countable. We say that a set is at most countable if it has only a finite number of elements or if its elements can be indexed by the natural numbers $\mathbb{N}$. In the next definition we introduce probabilities.

Definition 2.1 (Discrete probability space)**.** A discrete probability space $(\Omega, \mathbf{P})$ consists of an at most countable non-empty set Ω and a function $\mathbf{P} : \mathcal{P}(\Omega) \to [0, 1]$ such that

(i) $\mathbf{P}(\Omega) = 1$

(ii) for all disjoint $A_1, A_2, \ldots \in \mathcal{P}(\Omega)$ (i.e., $A_i \cap A_j = \emptyset$ for $i \neq j$),

$$\mathbf{P}(\cup_{k=1}^{\infty} A_k) = \sum_{k=1}^{\infty} \mathbf{P}(A_k). \tag{2.1}$$

The set Ω is denoted as sample space and the function $\mathbf{P}$ is called probability or probability measure.

Remark 2.2. (i) The condition (2.1) is called σ-additivity.

(ii) It follows from the σ-additivity that $\mathbf{P}(\emptyset) = 0$.

(iii) If Ω is a finite set (i.e., it has only a finite number of elements), σ-additivity is equivalent to

$$\mathbf{P}(A \cup B) = \mathbf{P}(A) + \mathbf{P}(B)$$

for all disjoint $A, B \subseteq \Omega$ (i.e., $A \cap B = \emptyset$).

Throughout this class, we usually denote the cardinality (i.e. the number of elements) of a finite set M by $|M|$.

Example 2.3 (Rolling a die)**.** The random experiment of throwing a die can be described by the probability space $(\Omega, \mathbf{P})$ with $\Omega = \{1, \ldots, 6\}$ and

$$\mathbf{P}(A) = \frac{|A|}{6} \quad \text{for} \quad A \subseteq \{1, \ldots, 6\}.$$

Then, we have, for example,

$$\mathbf{P}(\text{‘even number’}) = \mathbf{P}(\{2, 4, 6\}) = \frac{|\{2, 4, 6\}|}{6} = \frac{3}{6} = \frac{1}{2}.$$

Note that Definition 2.1 is a formal definition, which does not tell us if the discrete probability space given in Example 2.3 is the correct one for the random experiment of throwing a die. For example $\Omega = \{1, \ldots, 6\}$ with

$$\mathbf{P}(A) = \begin{cases} 1, & 6 \in A, \\ 0, & 6 \notin A, \end{cases} \quad \text{for} \quad A \subseteq \{1, \ldots, 6\}$$

would be a discrete probability space as well.

So in order to find a suitable probability space for a given random experiment, we have to understand what the probability of an event should be. Assume that a random experiment is given and that we are interested in an event A. Now we repeat our random experiment many times and count how often the event A occurs. Let $k_n(A)$ be the number of outcomes in the first n trials that belong to A. The ratio $k_n(A)/n$ is called relative frequency. Now one can observe in experiments that $k_n(A)/n$ converges to some limit as $n \to \infty$. This limit is the desired probability of A.

Example 2.4 (Tossing a fair coin n times)**.** The random experiment of tossing a fair coin n times for $n \in \mathbb{N}$ is described by the probability space $(\Omega, \mathbf{P})$ with $\Omega = \{0,1\}^n$ and

$$\mathbf{P}(A) = \frac{|A|}{2^n} \quad \text{for} \quad A \subseteq \{0,1\}^n.$$

Here, we denote 'head' by 1 and 'tail' by 0. Then, we have, for example,

$$\mathbf{P}(\text{'exactly one head'}) = \mathbf{P}(\{(1,0,\ldots,0),\ldots,(0,\ldots,0,1)\}) = \frac{n}{2^n}.$$

The discrete probability spaces in Example 2.3 and in Example 2.4 are special cases of a more general class of discrete probability spaces.

Example 2.5 (Laplace probability space)**.** Let Ω be a finite non-empty set. Then $(\Omega, \mathbf{P})$ with

$$\mathbf{P}(A) = \frac{|A|}{|\Omega|} \quad \text{for} \quad A \subseteq \Omega$$

is a probability space.

Next we provide an example of a discrete probability space that is not a Laplace probability space.

Example 2.6 (Tossing an unfair coin)**.** Let $p \in [0,1]$, let $\Omega = \{0,1\}$ and let $\mathbf{P}$ be such that

$$\mathbf{P}(\emptyset) = 0, \quad \mathbf{P}(\{0\}) = 1-p, \quad \mathbf{P}(\{1\}) = p, \quad \text{and} \quad \mathbf{P}(\{0,1\}) = 1.$$

This discrete probability space describes the random experiment of tossing a coin whose probability for 'head', which is encoded as 1, is p. If $p \neq \frac{1}{2}$, the coin is unfair.

Remark 2.7. Let $(\Omega, \mathbf{P})$ be a discrete probability space and let $A, B \subseteq \Omega$, i.e., A and B are events.

- The intersection $A \cap B$ is the event that A and B both occur, whereas the union $A \cup B$ is the event that at least one of them occurs.
- The complement $A^c := \Omega \setminus A$ of A in Ω is the event that A does not occur.
- If $A \subseteq B$, then B occurs whenever A occurs.

The rules in the following lemma will be very helpful to compute probabilities.

Lemma 2.8. Let $(\Omega, \mathbf{P})$ be a discrete probability space.

(i) For disjoint $A_1, \ldots, A_n \subseteq \Omega$,

$$\mathbf{P}(\cup_{k=1}^n A_k) = \sum_{k=1}^n \mathbf{P}(A_k).$$

(ii) For $A \subseteq \Omega$,
$$\mathbf{P}(A^c) = 1 - \mathbf{P}(A).$$

(iii) For $A, B \subseteq \Omega$,
$$\mathbf{P}(A \cup B) = \mathbf{P}(A) + \mathbf{P}(B) - \mathbf{P}(A \cap B).$$

(iv) For $A \subseteq B \subseteq \Omega$,
$$\mathbf{P}(A) \leq \mathbf{P}(B).$$

(v) For $A_1, \ldots, A_n \subseteq \Omega$,
$$\mathbf{P}(\cup_{k=1}^n A_k) \leq \sum_{k=1}^n \mathbf{P}(A_k).$$

Proof. (i) Define $A_k := \emptyset$ for $k > n$ so that by the σ-additivity (see (2.1)) and $\mathbf{P}(A_k) = \mathbf{P}(\emptyset) = 0$ for $n > k$,
$$\mathbf{P}(\cup_{k=1}^n A_k) = \mathbf{P}(\cup_{k=1}^\infty A_k) = \sum_{k=1}^\infty \mathbf{P}(A_k) = \sum_{k=1}^n \mathbf{P}(A_k).$$

(ii) Since $A \cap A^c = \emptyset$ and $A \cup A^c = \Omega$, it follows from (i) that
$$\mathbf{P}(A) + \mathbf{P}(A^c) = \mathbf{P}(A \cup A^c) = \mathbf{P}(\Omega) = 1.$$

(iii) By rewriting the events and applying (i), we obtain
$$\begin{aligned}
\mathbf{P}(A \cup B) &= \mathbf{P}((A \cap B^c) \cup (A^c \cap B) \cup (A \cap B)) \\
&\overset{(i)}{=} \mathbf{P}(A \cap B^c) + \mathbf{P}(A^c \cap B) + \mathbf{P}(A \cap B) \\
&= \mathbf{P}(A \cap B^c) + \mathbf{P}(A \cap B) + \mathbf{P}(A^c \cap B) + \mathbf{P}(A \cap B) - \mathbf{P}(A \cap B) \\
&\overset{(i)}{=} \mathbf{P}((A \cap B^c) \cup (A \cap B)) + \mathbf{P}((A^c \cap B) \cup (A \cap B)) - \mathbf{P}(A \cap B) \\
&= \mathbf{P}(A) + \mathbf{P}(B) - \mathbf{P}(A \cap B).
\end{aligned}$$

(iv) We have
$$\mathbf{P}(A) \leq \mathbf{P}(A) + \underbrace{\mathbf{P}(B \setminus A)}_{\geq 0} \overset{(i)}{=} \mathbf{P}(A \cup (B \setminus A)) = \mathbf{P}(B).$$

(v) Let $B_k := A_k \setminus \cup_{j=1}^{k-1} B_j$ for $k \in \{1, \ldots, n\}$ so that $B_k \subseteq A_k$, $\cup_{k=1}^n A_k = \cup_{k=1}^n B_k$, and $B_1, \ldots, B_n$ are disjoint. Then, we derive
$$\mathbf{P}(\cup_{k=1}^n A_k) = \mathbf{P}(\cup_{k=1}^n B_k) \overset{(i)}{=} \sum_{k=1}^n \underbrace{\mathbf{P}(B_k)}_{\overset{(iv)}{\leq} \mathbf{P}(A_k)} \leq \sum_{k=1}^n \mathbf{P}(A_k),$$
which is the desired inequality.

□

Part (iii) of the previous lemma is generalised by the following result.

Theorem 2.9 (Inclusion-exclusion principle)**.** For $A_1, \ldots, A_n \subseteq \Omega$ with $n \geq 2$,

$$\mathbf{P}(\cup_{k=1}^n A_k) = \sum_{k=1}^n (-1)^{k-1} \sum_{1 \leq i_1 < \ldots < i_k \leq n} \mathbf{P}(\cap_{j=1}^k A_{i_j}).$$

Proof. We prove the assertion by induction over n. The base case $n = 2$ is Lemma 2.8 (iii). So it remains to show that the statement for $n+1$ follows from that for n. From Lemma 2.8 (iii) we deduce

$$\begin{aligned}\mathbf{P}(\cup_{k=1}^{n+1} A_k) &= \mathbf{P}((\cup_{k=1}^n A_k) \cup A_{n+1}) = \mathbf{P}(\cup_{k=1}^n A_k) + \mathbf{P}(A_{n+1}) - \mathbf{P}((\cup_{k=1}^n A_k) \cap A_{n+1}) \\ &= \mathbf{P}(\cup_{k=1}^n A_k) + \mathbf{P}(A_{n+1}) - \mathbf{P}(\cup_{k=1}^n (A_k \cap A_{n+1})).\end{aligned}$$

Now it follows from the induction hypothesis that

$$\begin{aligned}\mathbf{P}(\cup_{k=1}^{n+1} A_k) = &\sum_{k=1}^n (-1)^{k-1} \sum_{1 \leq i_1 < \ldots < i_k \leq n} \mathbf{P}(\cap_{j=1}^k A_{i_j}) + \mathbf{P}(A_{n+1}) \\ &- \sum_{k=1}^n (-1)^{k-1} \sum_{1 \leq i_1 < \ldots < i_k \leq n} \mathbf{P}(\cap_{j=1}^k (A_{i_j} \cap A_{n+1}))\end{aligned}$$

Since

$$\begin{aligned}&\mathbf{P}(A_{n+1}) - \sum_{k=1}^n (-1)^{k-1} \sum_{1 \leq i_1 < \ldots < i_k \leq n} \mathbf{P}(\cap_{j=1}^k (A_{i_j} \cap A_{n+1})) \\ &= \mathbf{P}(A_{n+1}) - \sum_{k=1}^n (-1)^{k-1} \sum_{1 \leq i_1 < \ldots < i_k < i_{k+1} = n+1} \mathbf{P}(\cap_{j=1}^{k+1} A_{i_j}) \\ &= \mathbf{P}(A_{n+1}) + \sum_{k=2}^{n+1} (-1)^{k-1} \sum_{1 \leq i_1 < \ldots < i_k = n+1} \mathbf{P}(\cap_{j=1}^k A_{i_j}) \\ &= \sum_{k=1}^{n+1} (-1)^{k-1} \sum_{1 \leq i_1 < \ldots < i_k = n+1} \mathbf{P}(\cap_{j=1}^k A_{i_j}),\end{aligned}$$

we obtain

$$\begin{aligned}\mathbf{P}(\cup_{k=1}^{n+1} A_k) &= \sum_{k=1}^n (-1)^{k-1} \sum_{1 \leq i_1 < \ldots < i_k \leq n} \mathbf{P}(\cap_{j=1}^k A_{i_j}) + \sum_{k=1}^{n+1} (-1)^{k-1} \sum_{1 \leq i_1 < \ldots < i_k = n+1} \mathbf{P}(\cap_{j=1}^k A_{i_j}) \\ &= \sum_{k=1}^{n+1} (-1)^{k-1} \sum_{1 \leq i_1 < \ldots < i_k \leq n+1} \mathbf{P}(\cap_{j=1}^k A_{i_j}),\end{aligned}$$

which concludes the proof. □

For three events $A, B, C \subseteq \Omega$, Theorem 2.9 simplifies to

$$\mathbf{P}(A \cup B \cup C) = \mathbf{P}(A) + \mathbf{P}(B) + \mathbf{P}(C) - \mathbf{P}(A \cap B) - \mathbf{P}(A \cap C) - \mathbf{P}(B \cap C) + \mathbf{P}(A \cap B \cap C).$$

2.2 Elementary combinatorics

Combinatorics is the branch of Mathematics where objects are counted. A standard problem in combinatorics is to determine the cardinality of a finite set, i.e., the number of its elements. This situation arises naturally in the context of a Laplace probability space $(\Omega, \mathbf{P})$, where

$$\mathbf{P}(A) = \frac{|A|}{|\Omega|}$$

for each $A \subseteq \Omega$. Recall that we denote by $|S|$ the cardinality of a set S, i.e., the number of elements of S. The computation of $|A|$ and $|\Omega|$ can be seen as a problem from combinatorics.

The focus of this section will be on urn models. Here an urn with M balls, which we denote by $1, \ldots, M$, is given. We choose n of these balls. Our goal is to determine how many choices we have to select n of the M balls.

First we note that there are different ways we can choose the balls:

- *With or without replacement.* After drawing a ball we can put it back into the urn or keep it.
- *Ordered or unordered sample.* One can take the order in which the balls were drawn into account (ordered sample) or ignore the order (unordered sample).

Thus, we will consider four different situations in the sequel:

Ordered samples with replacement. We are interested in the cardinality of

$$\Omega_1 := \{(a_1, \ldots, a_n) : a_1, \ldots, a_n \in \{1, \ldots, M\}\} = \{1, \ldots, M\}^n.$$

Here, we can think of a_i as the i-th ball we select. A short computation shows that

$$|\Omega_1| = M^n \tag{2.2}$$

since for each of the n balls we draw we have M choices.

Example 2.10. For $\ell, m \in \mathbb{N}$ we consider the set $F_{\ell,m} := \{f : \{1, \ldots, \ell\} \to \{1, \ldots, m\}\}$ of all functions from $\{1, \ldots, \ell\}$ to $\{1, \ldots, m\}$. Then, it follows from (2.2) that

$$|F_{\ell,m}| = m^\ell.$$

Ordered samples without replacement. Here the possible choices are given by

$$\Omega_2 := \{(a_1, \ldots, a_n) : a_1, \ldots, a_n \in \{1, \ldots, M\}, a_i \neq a_j \text{ for } i, j \in \{1, \ldots, n\}, i \neq j\}.$$

We obtain that

$$|\Omega_2| = M \cdot (M-1) \cdot \ldots \cdot (M-n+1) \tag{2.3}$$

since there are M choices for the first ball, $M-1$ choices for the second ball (because we have already drawn the first ball), ... and finally $M-n+1$ choices for the last ball. Note that this formula even holds for $n > M$ since in this case we have $\Omega_2 = \emptyset$.

For $m \in \mathbb{N}$ we define

$$m! := m \cdot (m-1) \cdot \ldots \cdot 2 \cdot 1,$$

which we call the factorial of m. Moreover, we use the convention $0! := 1$. This allows us to rewrite (2.3) as

$$|\Omega_2| = \frac{M!}{(M-n)!}$$

if $M \geq n$.

Example 2.11. For $\ell, m \in \mathbb{N}$ with $\ell \leq m$,

$$\overline{F}_{\ell,m} := \{g : \{1, \ldots, \ell\} \to \{1, \ldots, m\} : g(i) \neq g(j) \text{ for } i, j \in \{1, \ldots, \ell\} \text{ with } i \neq j\}$$

is the set of all injective functions from $\{1, \ldots, \ell\}$ to $\{1, \ldots, m\}$. Recall that a function is injective if distinct arguments are mapped to different values. From (2.3) we deduce that

$$|\overline{F}_{\ell,m}| = \frac{m!}{(m-\ell)!}.$$

For $\ell = m$, the functions of $\overline{F}_{\ell,m}$ are the permutations. A permutation is a bijective map from a finite set to itself. Permutations can be interpreted as orders of the objects of a finite set. For Π_ℓ, the set of permutations of $\{1, \ldots, \ell\}$, we have that

$$|\Pi_\ell| = \ell!.$$

In order to deal with unordered samples we introduce $[a_1, \ldots, a_n]$ as notation for an unordered n-tuple, i.e., the order of $a_1, \ldots, a_n$ does not play a role. This is in contrast to $(a_1, \ldots, a_n)$, where the order of the entries is taken into account. For example, one has $[1, 2] = [2, 1]$, while $(1, 2) \neq (2, 1)$.

Unordered samples without replacement. We consider

$$\Omega_3 := \{[a_1, \ldots, a_n] : a_1, \ldots, a_n \in \{1, \ldots, M\}, a_i \neq a_j \text{ for } i, j \in \{1, \ldots, n\}, i \neq j\}.$$

This means that Ω_3 describes the situation that we choose n distinct objects from M objects. So Ω_3 can be interpreted as the set of all subsets of size n of a set with M elements.

Theorem 2.12. One has

$$|\Omega_3| = \frac{M \cdot (M-1) \cdot \ldots \cdot (M-n+1)}{n!}. \tag{2.4}$$

Proof. Every unordered sample $[a_1, \ldots, a_n]$ can be ordered in $n!$ ways so that

$$|\Omega_2| = |\Omega_3| n!$$

and, by (2.3),

$$|\Omega_3| = \frac{|\Omega_2|}{n!} = \frac{M \cdot (M-1) \cdot \ldots \cdot (M-n+1)}{n!},$$

which is the desired statement. □

Note that (2.4) can be written as

$$|\Omega_3| = \frac{M!}{n!(M-n)!}$$

for $M \geq n$. Since ratios as the one on the right-hand side shall be often used in the sequel, we make the following definition.

Definition 2.13. For $m, k \in \mathbb{N}_0$ we define

$$\binom{m}{k} := \begin{cases} \frac{m!}{k!(m-k)!}, & k \leq m, \\ 0, & k > m, \end{cases}$$

which we call binomial coefficient. The symbol $\binom{m}{k}$ reads as 'm choose k'.

This allows us to write (2.4) as

$$|\Omega_3| = \binom{M}{n}.$$

Lemma 2.14. For $m \in \mathbb{N}$ and $k \in \mathbb{N}$ such that $k < m$,

$$\binom{m}{k} = \binom{m-1}{k-1} + \binom{m-1}{k}.$$

Proof. Here, we provide a combinatorial and an algebraic proof:

Combinatorial proof. As discussed above $\binom{m}{k}$ is the number of subsets with k elements of $\{1, \ldots, m\}$. Note that a subset of $\{1, \ldots, m\}$ can either contain m or not. Hence, we obtain all subsets with k elements of $\{1, \ldots, m\}$ by choosing $k-1$ elements from $\{1, \ldots, m-1\}$ if m belongs to the subset or by choosing k elements from $\{1, \ldots, m-1\}$ if m is not part of the subset. As $\{1, \ldots, m-1\}$ has $\binom{m-1}{k-1}$ subsets of size $k-1$ and $\binom{m-1}{k}$ subsets of size k, we see that

$$\binom{m}{k} = \binom{m-1}{k-1} + \binom{m-1}{k}.$$

Algebraic proof. We have that

$$\begin{aligned}\binom{m-1}{k-1}+\binom{m-1}{k} &= \frac{(m-1)!}{(k-1)!(m-1-(k-1))!}+\frac{(m-1)!}{k!(m-1-k)!}\\ &= \frac{(m-1)!}{k!(m-k)!}(k+m-k)=\frac{m!}{k!(m-k)!}=\binom{m}{k},\end{aligned}$$

which is the desired identity. □

Theorem 2.15. For $m \in \mathbb{N}$ and $a, b \in \mathbb{R}$,

$$(a+b)^m = \sum_{i=0}^{m} \binom{m}{i} a^i b^{m-i}.$$

Proof. We have

$$(a+b)^m = \underbrace{(a+b)\cdot \ldots \cdot (a+b)}_{m \text{ factors}}$$

so that we need to multiply the right-hand side out. This leads to products of the form $a^i b^{m-i}$ with $i \in \{0, 1, \ldots, m\}$. We obtain $a^i b^{m-i}$ when we choose the a in i of the factors and the b in the remaining ones. By Theorem 2.12, there are $\binom{m}{i}$ possibilities to choose i out of m factors, which concludes the proof. □

Unordered samples with replacement. We define

$$\Omega_4 := \{[a_1, \ldots, a_n] : a_1, \ldots, a_n \in \{1, \ldots, M\}\}.$$

Theorem 2.16. It holds that

$$|\Omega_4| = \binom{M+n-1}{n}.$$

Proof. By $K(M, n)$ we denote the number of possibilities to choose an unordered sample of n numbers from $\{1, \ldots, M\}$ with replacement, i.e., $|\Omega_4| = K(M, n)$. We prove now by induction over n that

$$K(M, n) = \binom{M+n-1}{n} \tag{2.5}$$

for all $M \in \mathbb{N}$.

For the base case $n = 1$ we have

$$K(M, n) = M = \binom{M}{1}$$

for all $M \in \mathbb{N}$ so that (2.5) is satisfied in this situation.

For the induction step we assume that (2.5) is true for n, which is our induction hypothesis. We show that the induction hypothesis implies the statement for $n+1$. Let $M \in \mathbb{N}$ be fixed. Because the order of $a_1, \ldots, a_{n+1}$ does not play a role, we have that

$$\begin{aligned} K(M, n+1) &= \big|\{[a_1, \ldots, a_{n+1}] : a_1, \ldots, a_{n+1} \in \{1, \ldots, M\}\}\big| \\ &= \big|\{[a_1, \ldots, a_{n+1}] : a_1 \leq \ldots \leq a_{n+1}, a_1, \ldots, a_{n+1} \in \{1, \ldots, M\}\}\big|. \end{aligned}$$

Considering the different values for a_{n+1}, we obtain

$$\begin{aligned} K(M, n+1) &= \big| \cup_{\ell=1}^{M} \{[a_1, \ldots, a_{n+1}] : a_1 \leq \ldots \leq a_{n+1} = \ell, a_1, \ldots, a_n \in \{1, \ldots, M\}\}\big| \\ &= \sum_{\ell=1}^{M} \big|\{[a_1, \ldots, a_{n+1}] : a_1 \leq \ldots \leq a_{n+1} = \ell, a_1, \ldots, a_n \in \{1, \ldots, M\}\}\big|. \end{aligned}$$

If $a_1 \leq \ldots \leq a_{n+1} = \ell$, we just have to choose n numbers from $\{1, \ldots, \ell\}$, whence

$$\big|\{[a_1, \ldots, a_{n+1}] : a_1 \leq \ldots \leq a_{n+1} = \ell, a_1, \ldots, a_n \in \{1, \ldots, M\}\}\big| = K(\ell, n)$$

and

$$K(M, n+1) = \sum_{\ell=1}^{M} K(\ell, n).$$

Now it follows from the induction hypothesis (2.5) that

$$K(M, n+1) = \sum_{\ell=1}^{M} \binom{\ell + n - 1}{n} = \sum_{j=n}^{M+n-1} \binom{j}{n} = 1 + \sum_{j=n+1}^{M+n-1} \binom{j}{n}.$$

From Lemma 2.14 we obtain

$$K(M, n+1) = 1 + \sum_{j=n+1}^{M+n-1} \binom{j+1}{n+1} - \binom{j}{n+1}.$$

Since we have a telescopic sum on the right-hand side, the formula simplifies to

$$K(M, n+1) = 1 + \binom{M+n}{n+1} - \binom{n+1}{n+1} = \binom{M + (n+1) - 1}{n+1},$$

which concludes the proof. □

Note that Ω_4 is the sample space for the random experiment that we draw n balls from an urn with M balls with replacement and do not take their order into account. This random experiment is not described by a Laplace probability space since the elementary events have different probabilities as can be seen from the following example. Assume that we draw two balls from an urn with the balls $1, 2$ with replacement. In this situation each of the ordered pairs $(1,1), (1,2), (2,1), (2,2)$ has probability $\frac{1}{4}$. Then the probability of the unordered pair $[1,2]$ is $\frac{1}{2}$ since it corresponds to the ordered pairs $(1,2)$ and $(2,1)$,

while $[1,1]$ and $[2,2]$ correspond only to the ordered pairs $(1,1)$ and $(2,2)$, respectively, and, thus, have each probability $\frac{1}{4}$.

A further combinatorial problem is to distribute balls into bins. Here, we assume that n balls and M bins are given. Our goal is to compute how many choices we have to put the n balls into the M bins. For this problem there are different situations:

- *At most one ball per bin or no restriction on the number of balls per bin.* We may put at most one ball in each bin or there is no such restriction.

- *Distinguishable or indistinguishable balls.* In the first case, we take into account which ball is in which bin, while in the second case we just count the number of balls in each bin.

The crucial observation is now that this problem can be reformulated in terms of urn models as well. For this we put our M bins into an urn. Then we draw n bins from the urn. The first bin is the one for the first ball, the second one for the second ball and so on. If we may put at most one ball into each bin, we select the bins without replacement, if there may be more than one ball in each bin, we draw the bins with replacement. For distinguishable ball we must take the order of the chosen bins into account, while for indistinguishable balls the order of the bins is irrelevant.

Example 2.17 (Number of subsets)**.** Let A be a non-empty finite set with n elements. Recall that $\mathcal{P}(A)$ is the power set of A, i.e., the set of all subsets of A. Then, we have

$$|\mathcal{P}(A)| = 2^n.$$

This formula follows from the observation that we can decide for each of the n elements of A whether it belongs to a subset or not. The problem can be reformulated as distributing n distinguishable balls (i.e., the elements of A) into two bins (i.e., the subset and its complement), which have no restrictions on the number of balls they can take.

Example 2.18 (Lotto)**.** We consider the lottery where 6 out of 49 numbers are drawn without replacement. This random experiment can be described by the probability space $(\Omega, \mathbf{P})$ with

$$\Omega = \{[a_1, \ldots, a_6] : a_1, \ldots, a_6 \in \{1, \ldots, 49\}, a_i \neq a_j, i, j \in \{1, \ldots, 6\}, i \neq j\}$$

and

$$\mathbf{P}(A) = \frac{|A|}{|\Omega|} \quad \text{for} \quad A \subseteq \Omega.$$

Since we have the situation of an unordered sample without replacement, it follows from Theorem 2.12 that

$$|\Omega| = \binom{49}{6}.$$

Now we assume that a player has predicted six numbers (e.g. $1, \dots, 6$). For $i \in \{0, 1, \dots, 6\}$ we are interested in the event A_i that i of the predicted numbers are matched. We obtain

$$|A_i| = \binom{6}{i}\binom{43}{6-i}.$$

The first factor is the number of choices for the i numbers out of the predicted ones that are matched, while the second factor provides the number of ways to select the remaining $6 - i$ numbers from the 43 not predicted numbers. Thus, we derive that

$$\mathbf{P}(A_i) = \frac{\binom{6}{i}\binom{43}{6-i}}{\binom{49}{6}}$$

for $i \in \{0, 1, \dots, 6\}$.

2.3 Discrete probability distributions

In this section, we construct for a given at most countable non-empty set Ω a probability $\mathbf{P}$. In the following, we often refer to $\mathbf{P}$ as a probability distribution.

For a discrete probability space $(\Omega, \mathbf{P})$ and an event $A \subseteq \Omega$ we have

$$\mathbf{P}(A) = \mathbf{P}(\cup_{\omega \in A}\{\omega\}) = \sum_{\omega \in A} \mathbf{P}(\{\omega\}). \tag{2.6}$$

Since $\{\omega_1\} \cap \{\omega_2\} = \emptyset$ for $\omega_1, \omega_2 \in \Omega$ with $\omega_1 \neq \omega_2$, for a finite set A the last equality follows from Lemma 2.8 (i). If A is an infinite subset of Ω, the identity follows from the σ-additivity in the definition of a discrete probability space. From (2.6) we see that we can compute the probabilities of all events if the probabilities of the elementary events are given. In the next result, we use the probabilities of the elementary events to construct discrete probability spaces.

Theorem 2.19. Let Ω be an at most countable non-empty set and let $(p_\omega)_{\omega \in \Omega}$ be a collection of non-negative real numbers such that

$$\sum_{\omega \in \Omega} p_\omega = 1.$$

Then, $(\Omega, \mathbf{P})$ with

$$\mathbf{P}(A) = \sum_{\omega \in A} p_\omega \quad \text{for} \quad A \subseteq \Omega$$

is a discrete probability space.

Proof. In the sequel we verify that $\mathbf{P}$ satisfies the definition of a probability (see Definition 2.1). Since $p_\omega \geq 0$ for all $\omega \in \Omega$, we have that, for $A \subseteq \Omega$,

$$\mathbf{P}(A) = \sum_{\omega \in A} p_\omega \geq 0 \quad \text{and} \quad \mathbf{P}(A) = \sum_{\omega \in A} p_\omega \leq \sum_{\omega \in \Omega} p_\omega = 1,$$

whence $\mathbf{P}$ is a function from $\mathcal{P}(\Omega)$ to $[0,1]$. Moreover, we obtain

$$\mathbf{P}(\Omega) = \sum_{\omega\in\Omega} p_\omega = 1.$$

Let $A_1, A_2, \ldots \subseteq \Omega$ be a sequence of disjoint sets. From the disjointness of the sets (and the absolute convergence of $\sum_{\omega\in\Omega} p_\omega$ if $|\Omega| = \infty$) it follows that

$$\mathbf{P}(\cup_{i=1}^\infty A_i) = \sum_{\omega\in\cup_{i=1}^\infty A_i} p_\omega = \sum_{i=1}^\infty \sum_{\omega\in A_i} p_\omega = \sum_{i=1}^\infty \mathbf{P}(A_i),$$

which shows the σ-additivity of $\mathbf{P}$. □

Example 2.20 (Laplace distribution)**.** Let Ω be a finite non-empty set. Define

$$p_\omega := \frac{1}{|\Omega|} \quad \text{for} \quad \omega \in \Omega$$

so that $p_\omega \geq 0$ for $\omega \in \Omega$ and

$$\sum_{\omega\in\Omega} p_\omega = \sum_{\omega\in\Omega} \frac{1}{|\Omega|} = \frac{|\Omega|}{|\Omega|} = 1.$$

Then, by Theorem 2.19, $(\Omega, \mathbf{P})$ with

$$\mathbf{P}(A) = \sum_{\omega\in A} \frac{1}{|\Omega|} \quad \text{for} \quad A \subseteq \Omega$$

is a discrete probability space. Because the previous equation can be rewritten as

$$\mathbf{P}(A) = \frac{|A|}{|\Omega|},$$

this is a Laplace probability space as we have studied before. Here, we denote $\mathbf{P}$ as Laplace distribution or uniform distribution because each elementary event has the same probability.

Example 2.21 (Dirac distribution)**.** Let Ω be an at most countable non-empty set and let $\omega_0 \in \Omega$ be fixed. Define

$$p_\omega := \begin{cases} 1, & \omega = \omega_0, \\ 0, & \omega \neq \omega_0, \end{cases} \qquad \text{for} \quad \omega \in \Omega.$$

By Theorem 2.19, $(\Omega, \mathbf{P})$ with

$$\mathbf{P}(A) = \sum_{\omega\in A} p_\omega \quad \text{for} \quad A \subseteq \Omega$$

is a discrete probability space. Note that

$$\mathbf{P}(A) = \begin{cases} 1, & \omega_0 \in A, \\ 0, & \omega_0 \notin A. \end{cases}$$

We call the probability $\mathbf{P}$ the Dirac distribution concentrated at ω_0.

Theorem 2.22. For $m \in \mathbb{N}$ let $(\Omega_1, \mathbf{P}_1), \ldots, (\Omega_m, \mathbf{P}_m)$ be discrete probability spaces and define $\Omega := \times_{i=1}^m \Omega_i$ and

$$\mathbf{P}(A) := \sum_{(\omega_1,\ldots,\omega_m)\in A} \prod_{i=1}^m \mathbf{P}_i(\{\omega_i\}) \quad \text{for} \quad A \subseteq \Omega.$$

Then, $(\Omega, \mathbf{P})$ is a discrete probability space.

Proof. Define

$$p_{(\omega_1,\ldots,\omega_m)} := \prod_{i=1}^m \mathbf{P}_i(\{\omega_i\}) \quad \text{for} \quad (\omega_1, \ldots, \omega_m) \in \Omega$$

so that $p_{(\omega_1,\ldots,\omega_m)} \geq 0$ for $(\omega_1, \ldots, \omega_m) \in \Omega$ and

$$\sum_{(\omega_1,\ldots,\omega_m)\in\Omega} \prod_{i=1}^m \mathbf{P}_i(\{\omega_i\}) = \prod_{i=1}^m \sum_{\omega_i\in\Omega_i} \mathbf{P}_i(\{\omega_i\}) = \prod_{i=1}^m \mathbf{P}_i(\Omega_i) = 1.$$

Thus, $(\Omega, \mathbf{P})$ is a discrete probability space by Theorem 2.19. □

Definition 2.23 (Product probability space)**.** The probability space $(\Omega, \mathbf{P})$ constructed in Theorem 2.22 is called the (discrete) product probability space of $(\Omega_1, \mathbf{P}_1), \ldots, (\Omega_m, \mathbf{P}_m)$.

Remark 2.24. Let $\mathbf{P}$ be the probability in Theorem 2.22.

- Sometimes one uses for $\mathbf{P}$ the notation $\mathbf{P}_1 \otimes \ldots \otimes \mathbf{P}_m$.
- For $A_1 \subseteq \Omega_1, \ldots, A_m \subseteq \Omega_m$, one has

$$\mathbf{P}(A_1 \times \ldots \times A_m) = \prod_{i=1}^m \mathbf{P}_i(A_i).$$

Example 2.25 (Rolling a die twice)**.** Let $\Omega_D = \{1, \ldots, 6\}$ and $\mathbf{P}_D(A) = \frac{|A|}{6}$ for $A \subseteq \Omega_D$, i.e., $(\Omega_D, \mathbf{P}_D)$ is the discrete probability space that describes the random experiment of throwing a die once. The product probability space $(\Omega, \mathbf{P})$ of $(\Omega_D, \mathbf{P}_D)$ with itself is given by

$$\Omega = \Omega_D \times \Omega_D = \{1, \ldots, 6\}^2$$

and

$$\mathbf{P}(A) = \sum_{(\omega_1,\omega_2)\in A} \mathbf{P}_D(\{\omega_1\})\mathbf{P}_D(\{\omega_2\}) = \sum_{(\omega_1,\omega_2)\in A} \frac{1}{36} \quad \text{for} \quad A \subseteq \{1,\dots,6\}^2.$$

Because of

$$\mathbf{P}(A) = \sum_{(\omega_1,\omega_2)\in A} \frac{1}{36} = \frac{|A|}{36},$$

we see that $\mathbf{P}$ is the Laplace distribution on $\{1,\dots,6\}^2$.

Note that the probability space in the previous example is the canonical probability space for the random experiment that we throw a die twice. More generally, the n-fold product probability space of $(\Omega_D, \mathbf{P}_D)$ is the probability space for the random experiment that we throw a die n times.

Assume that we have several random experiments which we conduct in such a way that they do not affect each other. The combination of these random experiments can be seen as a new single random experiment. For this large random experiment we obtain a suitable probability space by taking the product probability space of the probability spaces of the underlying random experiments.

Example 2.26 (Tossing an unfair coin n times)**.** Let $(\Omega_C, \mathbf{P}_C)$ be the discrete probability space such that $\Omega_C = \{0,1\}$ and $\mathbf{P}_C(\{0\}) = 1-p$ and $\mathbf{P}_C(\{1\}) = p$ with $p \in [0,1]$, which describes the random experiment of tossing an unfair coin once. For $n \in \mathbb{N}$ the n-fold product probability space $(\Omega, \mathbf{P})$ of $(\Omega_C, \mathbf{P}_C)$ is given by

$$\Omega = \{0,1\}^n$$

and

$$\mathbf{P}(A) = \sum_{(\omega_1,\dots,\omega_n)\in A} \prod_{i=1}^n \mathbf{P}_C(\{\omega_i\}) = \sum_{(\omega_1,\dots,\omega_n)\in A} \prod_{i=1}^n p^{\omega_i}(1-p)^{1-\omega_i} \quad \text{for} \quad A \subseteq \{0,1\}^n.$$

Now $(\Omega, \mathbf{P})$ is a suitable probability space for the random experiment that we toss an unfair coin n times.

Moreover, the probability space $(\Omega, \mathbf{P})$ describes the random experiment that a random experiment with the outcomes success (denoted by 1) and failure (denoted by 0) and with success probability p is repeated n times. For $k \in \{0,1,\dots,n\}$ the probability of k successes and $n-k$ failures in a fixed order is

$$p^k(1-p)^{n-k}.$$

Example 2.27 (Binomial distribution)**.** Let $n \in \mathbb{N}$ and $p \in [0,1]$ be fixed. We define

$$p_k := \binom{n}{k} p^k(1-p)^{n-k} \quad \text{for} \quad k \in \{0,1,\dots,n\}.$$

We have that $p_k \geq 0$ for $k \in \{0, 1, \ldots, n\}$. It follows from Theorem 2.15 that

$$\sum_{k=0}^{n} p_k = \sum_{k=0}^{n} \binom{n}{k} p^k (1-p)^{n-k} = (p + 1 - p)^n = 1.$$

Thus, Theorem 2.19 yields that $(\Omega, \mathbf{P})$ with $\Omega = \{0, 1, \ldots, n\}$ and

$$\mathbf{P}(A) = \sum_{k \in A} \binom{n}{k} p^k (1-p)^{n-k} \quad \text{for} \quad A \subseteq \{0, 1, \ldots, n\}$$

is a probability space. We denote $\mathbf{P}$ as binomial distribution.

Assume that a random experiment with the outcomes success and failure and with success probability p is repeated n times. From Example 2.26 and the interpretation of the binomial coefficient it follows that

$$\binom{n}{k} p^k (1-p)^{n-k}$$

is the probability that we observe exactly k successes.

In the following two examples, we construct probability distributions on $\Omega = \mathbb{N}_0$.

Example 2.28 (Poisson distribution)**.** Let $\lambda \in (0, \infty)$ be fixed. For $k \in \mathbb{N}_0$ we define

$$p_k := \frac{\lambda^k}{k!} e^{-\lambda}.$$

Then, we have $p_k \geq 0$ for $k \in \mathbb{N}_0$ and by the definition of the exponential series

$$\sum_{k=0}^{\infty} p_k = \sum_{k=0}^{\infty} \frac{\lambda^k}{k!} e^{-\lambda} = e^{-\lambda} \sum_{k=0}^{\infty} \frac{\lambda^k}{k!} = e^{-\lambda} e^{\lambda} = 1.$$

Now it follows from Theorem 2.19 that

$$\mathbf{P}(A) = \sum_{k \in A} \frac{\lambda^k}{k!} e^{-\lambda} \quad \text{for} \quad A \subseteq \mathbb{N}_0$$

is a probability distribution on $\mathbb{N}_0$, which is called Poisson distribution. The Poisson distribution is often used when rare events are counted.

Example 2.29 (Geometric distribution)**.** Let $p \in (0, 1)$ be fixed and define

$$p_k := p^k (1-p) \quad \text{for} \quad k \in \mathbb{N}_0.$$

Note that $p_k \geq 0$ for $k \in \mathbb{N}_0$ and that by the formula for the geometric series

$$\sum_{k=0}^{\infty} p_k = \sum_{k=0}^{\infty} p^k (1-p) = (1-p) \sum_{k=0}^{\infty} p^k = (1-p) \frac{1}{1-p} = 1.$$

By Theorem 2.19,

$$\mathbf{P}(A) = \sum_{k \in A} p^k (1-p) \quad \text{for} \quad A \subseteq \mathbb{N}_0$$

is a probability distribution on $\mathbb{N}_0$, which is called geometric distribution.

Let $k \in \mathbb{N}_0$ and assume that a random experiment with the outcomes success and failure and with success probability $1-p$ is repeated $k+1$ times. From Example 2.26 we know that

$$p^k(1-p)$$

is the probability that we have at first k failures and then a success in the $k+1$-st trial. If we repeat our random experiment until we observe the first success, the geometric distribution describes the number of failures we get before the first success.

Example 2.30 (Fixpoints of random permutations)**.** Recall that Π_n is the set of all permutations $\pi : \{1, \dots, n\} \to \{1, \dots, n\}$ for $n \in \mathbb{N}$. In the sequel we consider the Laplace probability space $(\Omega, \mathbf{P})$ with $\Omega = \Pi_n$ and

$$\mathbf{P}(A) = \frac{|A|}{|\Omega|} = \frac{|A|}{n!} \quad \text{for} \quad A \subseteq \Pi_n,$$

where we used Example 2.11 in the last step. We say that $i \in \{1, \dots, n\}$ is a fixpoint of $\pi \in \Pi_n$ if $\pi(i) = i$. In the following we compute the probability

$$p_0 := \mathbf{P}(\{\pi \in \Pi_n : \pi \text{ has no fixpoint}\}).$$

We define $A_i := \{\pi \in \Pi_n : \pi(i) = i\}$ for $i \in \{1, \dots, n\}$ so that

$$p_0 = 1 - \mathbf{P}(\cup_{i=1}^n A_i).$$

By the inclusion-exclusion principle (see Theorem 2.9), we obtain

$$\mathbf{P}(\cup_{i=1}^n A_i) = \sum_{k=1}^n (-1)^{k-1} \sum_{1 \le i_1 < \dots < i_k \le n} \mathbf{P}(\cap_{j=1}^k A_{i_j}). \tag{2.7}$$

For the probabilities on the right-hand side we have

$$\mathbf{P}(\cap_{j=1}^k A_{i_j}) = \frac{|\{\pi \in \Pi_n : \pi(i_j) = i_j, j = 1, \dots, k\}|}{n!} = \frac{(n-k)!}{n!}.$$

Together with the observation that there are $\binom{n}{k}$ summands in the inner sum in (2.7) this leads to

$$\mathbf{P}(\cup_{i=1}^n A_i) = \sum_{k=1}^n (-1)^{k-1} \binom{n}{k} \frac{(n-k)!}{n!} = \sum_{k=1}^n \frac{(-1)^{k-1}}{k!}$$

so that

$$p_0 = 1 - \sum_{k=1}^n \frac{(-1)^{k-1}}{k!} = \sum_{k=0}^n \frac{(-1)^k}{k!}. \tag{2.8}$$

For $\ell \in \{1, \dots, n\}$ we consider

$$p_\ell := \mathbf{P}(\{\pi \in \Pi_n : \ \pi \text{ has exactly } \ell \text{ fixpoints}\}) = \frac{|\{\pi \in \Pi_n : \ \pi \text{ has exactly } \ell \text{ fixpoints}\}|}{n!}.$$

Since we can choose ℓ fixpoints in $\binom{n}{\ell}$ different ways and we have a permutation of $n-\ell$ elements after removing the fixpoints, it follows from (2.8) that

$$p_\ell = \frac{1}{n!}\binom{n}{\ell}|\{\pi \in \Pi_{n-\ell} : \ \pi \text{ has no fixpoint}\}| = \frac{1}{n!}\binom{n}{\ell}(n-\ell)!\sum_{k=0}^{n-\ell}\frac{(-1)^k}{k!} = \frac{1}{\ell!}\sum_{k=0}^{n-\ell}\frac{(-1)^k}{k!}.$$

From (2.8) we see that this formula holds for $\ell = 0$ as well. For $\ell \in \mathbb{N}_0$ it follows from the series representation of the exponential function that

$$\lim_{n\to\infty} p_\ell = \lim_{n\to\infty}\frac{1}{\ell!}\sum_{k=0}^{n-\ell}\frac{(-1)^k}{k!} = \frac{1}{\ell!}\sum_{k=0}^{\infty}\frac{(-1)^k}{k!} = \frac{1}{\ell!}e^{-1}.$$

The term on the right-hand side is the probability of ℓ for a Poisson distribution with parameter $\lambda = 1$.

2.4 Conditional probabilities and independence

Let a random experiment be given and let us assume that we are interested in the probability of an event A. The probability of A can be seen as a measure for the chance that the event A occurs. Now let us assume that we know that an event B has occurred in our random experiment since, for example, the random experiment was already conducted, but we have only limited knowledge about the outcome. Our goal is to understand how the information that B occurs affects the chance and, thus, the probability of A. We will formalise this with conditional probabilities.

Let $(\Omega, \mathbf{P})$ be a Laplace probability space with $|\Omega| = n$, i.e., $\mathbf{P}(M) = \frac{|M|}{n}$ for $M \subseteq \Omega$. Suppose that $A, B \subseteq \Omega$ are such that $|A| = k$, $|B| = m \geq 1$ and $|A \cap B| = \ell$ and that it is known that B occurs. Given that B occurs all m elementary events from B have the probability $1/m$. For ℓ of these elementary events A occurs as well. Thus, the new probability for A under the information that B occurs is

$$\frac{\ell}{m} = \frac{\ell/n}{m/n} = \frac{\mathbf{P}(A \cap B)}{\mathbf{P}(B)}.$$

This motivates the following definition.

Definition 2.31 (Conditional probability)**.** Let $(\Omega, \mathbf{P})$ be a discrete probability space and let $A, B \subseteq \Omega$ with $\mathbf{P}(B) > 0$. The conditional probability of A given B is defined as

$$\mathbf{P}(A|B) := \frac{\mathbf{P}(A \cap B)}{\mathbf{P}(B)}.$$

Example 2.32 (Rolling a die once)**.** Let $\Omega = \{1, \dots, 6\}$ and $\mathbf{P}(A) = \frac{|A|}{6}$ for $A \subseteq \{1, \dots, 6\}$. For $E = \{2, 4, 6\}$ and $S = \{1, 2, 3\}$, one has

$$\mathbf{P}(E|S) = \frac{\mathbf{P}(E \cap S)}{\mathbf{P}(S)} = \frac{\mathbf{P}(\{2\})}{\mathbf{P}(\{1,2,3\})} = \frac{\frac{1}{6}}{\frac{3}{6}} = \frac{1}{3}.$$

Note that the conditional probability is different from $\mathbf{P}(E) = \frac{1}{2}$.

Theorem 2.33. Let $(\Omega, \mathbf{P})$ be a discrete probability space and let $B \subseteq \Omega$ with $\mathbf{P}(B) > 0$. Then,

$$\mathbf{P}(\cdot|B) : \mathcal{P}(\Omega) \ni A \mapsto \mathbf{P}(A|B)$$

is a probability distribution on Ω.

Proof. For $A \subseteq \Omega$ we obtain

$$\mathbf{P}(A|B) = \frac{\mathbf{P}(A \cap B)}{\mathbf{P}(B)} \geq 0 \quad \text{and} \quad \mathbf{P}(A|B) = \frac{\mathbf{P}(A \cap B)}{\mathbf{P}(B)} \leq \frac{\mathbf{P}(B)}{\mathbf{P}(B)} = 1,$$

whence $\mathbf{P}(\cdot|B) : \mathcal{P}(\Omega) \to [0, 1]$. Moreover, we have

$$\mathbf{P}(\Omega|B) = \frac{\mathbf{P}(\Omega \cap B)}{\mathbf{P}(B)} = \frac{\mathbf{P}(B)}{\mathbf{P}(B)} = 1.$$

For disjoint sets $A_1, A_2, \dots \subseteq \Omega$, the sets $A_1 \cap B, A_2 \cap B, \dots \subseteq \Omega$ are disjoint as well so that the σ-additivity of $\mathbf{P}$ leads to

$$\begin{aligned}\mathbf{P}(\cup_{i=1}^{\infty} A_i|B) &= \frac{\mathbf{P}(\cup_{i=1}^{\infty} A_i \cap B)}{\mathbf{P}(B)} = \frac{\mathbf{P}(\cup_{i=1}^{\infty}(A_i \cap B))}{\mathbf{P}(B)} \\ &= \frac{\sum_{i=1}^{\infty} \mathbf{P}(A_i \cap B)}{\mathbf{P}(B)} = \sum_{i=1}^{\infty} \frac{\mathbf{P}(A_i \cap B)}{\mathbf{P}(B)} = \sum_{i=1}^{\infty} \mathbf{P}(A_i|B).\end{aligned}$$

Thus, $\mathbf{P}(\cdot|B)$ satisfies all conditions that are required for a probability in Definition 2.1. □

Since, by Theorem 2.33, $\mathbf{P}(\cdot|B)$ is a probability, we can apply our results for probabilities such as Lemma 2.8 and Theorem 2.9 to $\mathbf{P}(\cdot|B)$. Note that $\mathbf{P}(B|\cdot)$ is not a probability.

In the following result we consider an at most countable collection of sets with an index set I. This means either that we have a finite number of sets (i.e., I is finite) or that we can index our sets by $\mathbb{N}$ (i.e., we can choose $I = \mathbb{N}$).

Theorem 2.34 (Law of total probability)**.** Let $(\Omega, \mathbf{P})$ be a discrete probability space and let $(B_i)_{i \in I}$ be an at most countable collection of disjoint subsets of Ω (i.e., $B_j \cap B_k = \emptyset$ for $j, k \in I$ with $j \neq k$) such that $\cup_{i \in I} B_i = \Omega$ and $\mathbf{P}(B_i) > 0$ for $i \in I$. Then, for $A \subseteq \Omega$,

$$\mathbf{P}(A) = \sum_{i \in I} \mathbf{P}(A|B_i)\mathbf{P}(B_i).$$

Proof. First note that

$$\mathbf{P}(A) = \mathbf{P}(A \cap \Omega) = \mathbf{P}(A \cap \cup_{i \in I} B_i) = \mathbf{P}(\cup_{i \in I} A \cap B_i) = \sum_{i \in I} \mathbf{P}(A \cap B_i),$$

where we used the disjointness of $A \cap B_i$ for $i \in I$ in the last step (together with Lemma 2.8 (i) for finite I and σ-additivity for infinite I). From the definition of the conditional probabilities, we obtain

$$\mathbf{P}(A) = \sum_{i \in I} \mathbf{P}(A \cap B_i) = \sum_{i \in I} \frac{\mathbf{P}(A \cap B_i)}{\mathbf{P}(B_i)} \mathbf{P}(B_i) = \sum_{i \in I} \mathbf{P}(A|B_i)\mathbf{P}(B_i),$$

which concludes the proof. □

Note that a collection of disjoint subsets $(B_i)_{i \in I}$ of Ω such that $\cup_{i \in I} B_i = \Omega$ is called a partition of Ω.

Example 2.35. A factory has two machines that produce the same part. One machine produces 60% of the output with a defect rate of 5%, while the other machine produces 40% of the items with a defect rate of 10%. What is the probability that a randomly selected part is defective?

Let A be the event that our randomly selected part is defective, while M_i is the event that it comes from machine $i \in \{1, 2\}$. We know that

$$\mathbf{P}(M_1) = 0.6, \quad \mathbf{P}(M_2) = 0.4, \quad \mathbf{P}(A|M_1) = 0.05 \quad \text{and} \quad \mathbf{P}(A|M_2) = 0.1.$$

Thus, it follows from Theorem 2.34 that

$$\mathbf{P}(A) = \mathbf{P}(A|M_1)\mathbf{P}(M_1) + \mathbf{P}(A|M_2)\mathbf{P}(M_2) = 0.05 \cdot 0.6 + 0.1 \cdot 0.4 = 0.03 + 0.04 = 0.07.$$

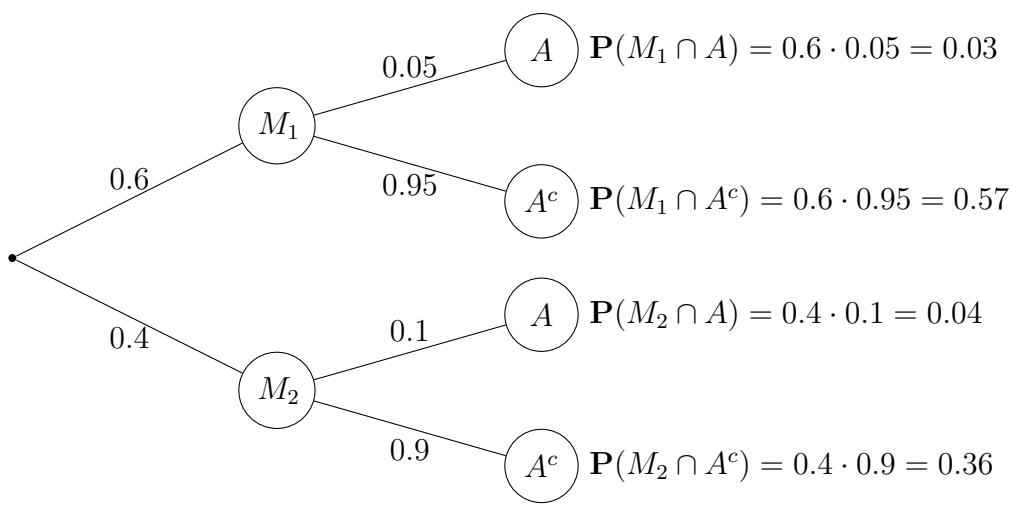
Situations as in the previous example can be visualised by so-called tree diagrams. We do not give a formal definition of tree diagrams and the underlying probability spaces as it is surprisingly technical. Instead, we provide two examples for tree diagrams.

Example 2.36. Consider the situation from Example 2.35. We can think of this as a two-step random experiment concerning a randomly selected part:

- First, we determine which machine produced the particular part. The respective probabilities are $\mathbf{P}(M_1) = 0.6$ and $\mathbf{P}(M_2) = 0.4$. This corresponds to the first layer in the tree diagram below.

- Then, we determine whether it is defective or not. Since we already know which machine produced it, we obtain conditional probabilities. If the first machine produced the part we have $\mathbf{P}(A|M_1) = 0.05$ and $\mathbf{P}(A^c|M_1) = 0.95$. For the second machine we get $\mathbf{P}(A|M_2) = 0.1$ as well as $\mathbf{P}(A^c|M_2) = 0.9$. These numbers lead to the second layer in the figure below.

One visualises this two-step random experiment as follows in a tree diagram, where the probabilities at the end of each path are calculated via $\mathbf{P}(C \cap D) = \mathbf{P}(C) \cdot \mathbf{P}(D|C)$ for any two events $C, D \subseteq \Omega$ with $\mathbf{P}(C) > 0$.

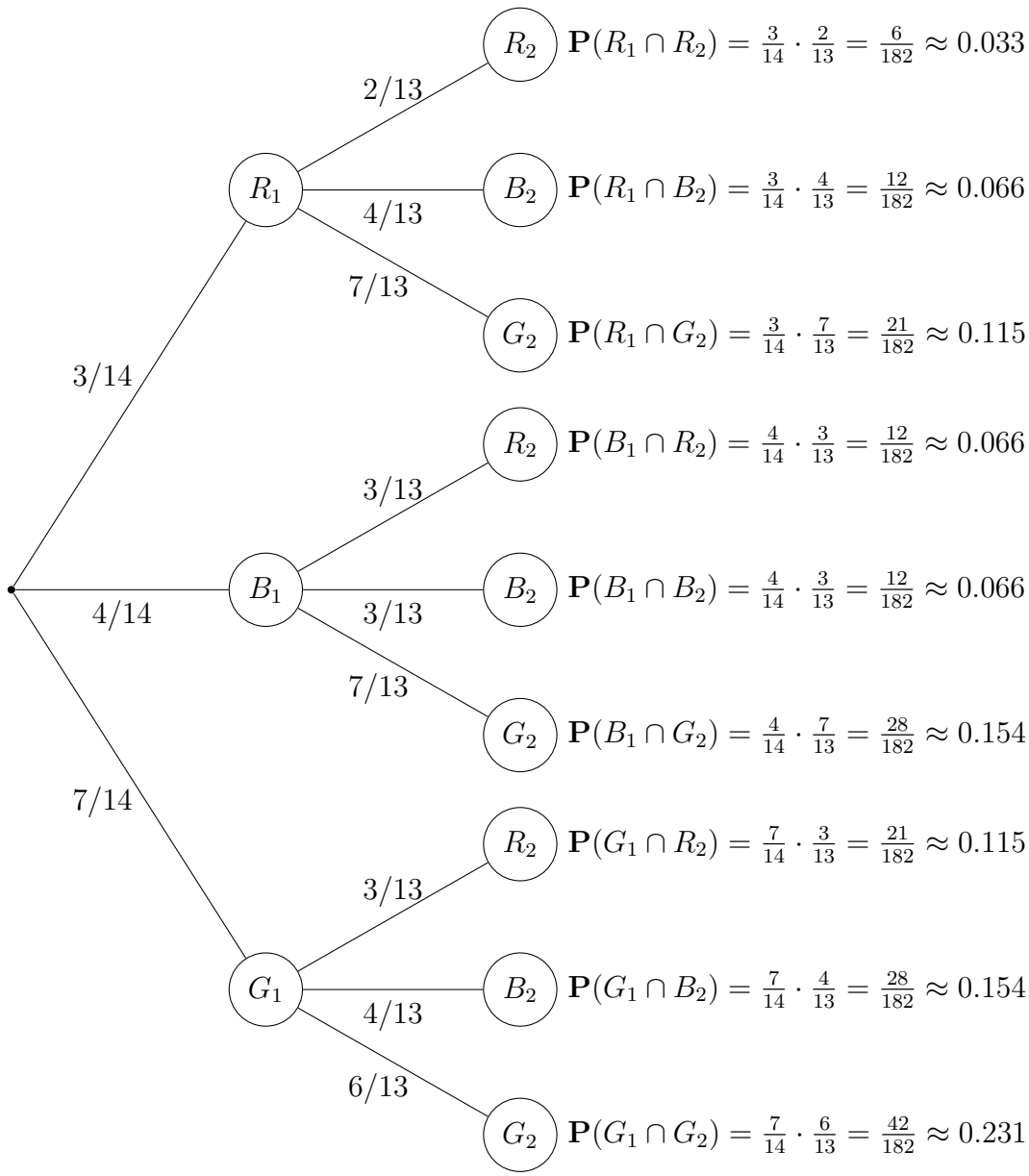


We calculate the probability of the part being defective by adding up the probabilities of all paths that lead to a defective part, i.e. to event A. As in Example 2.35 we obtain

$$\mathbf{P}(A) = 0.03 + 0.04 = 0.07.$$

In the previous example we considered a binary tree, i.e. in each step there were two possible outcomes: machine 1 or machine 2 in the first step and defective or not defective in the second step. Tree diagrams are also applicable when there are more possible outcomes in each step, as shown in the next example. One could also consider examples of tree diagrams with more than two layers but we do not provide an example here.

Example 2.37. Consider a bag that contains 3 red, 4 blue and 7 green balls. We draw 2 balls without replacement and are interested in the probability of drawing two balls of different colours. We write R_i, B_i and G_i to denote drawing a red, blue or green ball in the i-th step for $i = 1, 2$, respectively. The corresponding tree diagram is the following.



As we are interested in drawing two balls having different colours, let A denote this event. Then,

$$\mathbf{P}(A) \approx 0.066 + 0.115 + 0.066 + 0.154 + 0.115 + 0.154 = 0.67.$$

In general we do not have $\mathbf{P}(A|B) = \mathbf{P}(B|A)$. The following result allows us to compute $\mathbf{P}(B|A)$ from $\mathbf{P}(A|B)$.

Theorem 2.38 (Bayes formula)**.** Let $(\Omega, \mathbf{P})$ be a discrete probability space and let $(B_i)_{i \in I}$ be an at most countable collection of disjoint subsets of Ω (i.e., $B_j \cap B_k = \emptyset$ for $j, k \in I$ with $j \neq k$) such that $\cup_{i \in I} B_i = \Omega$ and $\mathbf{P}(B_i) > 0$ for $i \in I$. Then, for $A \subseteq \Omega$ with

$\mathbf{P}(A) > 0$,

$$\mathbf{P}(B_i|A) = \frac{\mathbf{P}(B_i)\mathbf{P}(A|B_i)}{\sum_{j\in I}\mathbf{P}(B_j)\mathbf{P}(A|B_j)} \quad \text{for} \quad i \in I.$$

Proof. From the definition of the conditional probability and Theorem 2.34 it follows that

$$\mathbf{P}(B_i|A) = \frac{\mathbf{P}(B_i \cap A)}{\mathbf{P}(A)} = \frac{\mathbf{P}(B_i)\mathbf{P}(A|B_i)}{\mathbf{P}(A)} = \frac{\mathbf{P}(B_i)\mathbf{P}(A|B_i)}{\sum_{j\in I}\mathbf{P}(A|B_j)\mathbf{P}(B_j)}$$

for $i \in I$. □

Example 2.39. We consider a fire detector. Let A denote the event that there is an alarm and let B be the event that a fire occurs. We assume that

$$\mathbf{P}(B) = 10^{-5}, \quad \mathbf{P}(A|B) = 0.99 \quad \text{and} \quad \mathbf{P}(A|B^c) = 10^{-4}.$$

In the following we compute $\mathbf{P}(B|A)$, i.e., the probability of a fire given that there is an alarm. Obviously, we have

$$\mathbf{P}(B^c) = 1 - \mathbf{P}(B) = 1 - 10^{-5}.$$

It follows from Theorem 2.38 that

$$\begin{aligned}\mathbf{P}(B|A) &= \frac{\mathbf{P}(B)\mathbf{P}(A|B)}{\mathbf{P}(B)\mathbf{P}(A|B) + \mathbf{P}(B^c)\mathbf{P}(A|B^c)} \\ &= \frac{10^{-5}\cdot 0.99}{10^{-5}\cdot 0.99 + (1-10^{-5})\cdot 10^{-4}} = \frac{0.99}{0.99 + 9.9999} \approx 0.09.\end{aligned}$$

This means that only for 9 percent of the alarms there is indeed a fire.

Example 2.40 (Spam filters)**.** Conditional probabilitites and the Bayes theorem can be used to detect spam e-mails. For a fixed word w let W be the event that an e-mail contains the word w, while S is the event that an e-mail is spam and H is the event that an e-mail is legitimate (i.e., no spam). Here, H stands for 'ham'. To decide if an e-mail is spam or not, we are interested in the conditional $\mathbf{P}(S|W)$. The idea is to classify an e-mail with the word w as spam if this probability is sufficiently large.

It follows from Theorem 2.38 that

$$\mathbf{P}(S|W) = \frac{\mathbf{P}(S)\mathbf{P}(W|S)}{\mathbf{P}(S)\mathbf{P}(W|S) + \mathbf{P}(H)\mathbf{P}(W|H)}.$$

Here, $\mathbf{P}(W|S)$ and $\mathbf{P}(W|H)$ can be estimated from some classified e-mails (for example, e-mails that were sorted by the user), while one can think of $\mathbf{P}(S) = 1-\mathbf{P}(H)$ as parameters that one can choose.

In order to decide, whether an e-mail is spam, one has to combine the spam probabilities for all words of the e-mail.

Definition 2.41 (Independent events)**.** Let $(\Omega, \mathbf{P})$ be a discrete probability space. Two events $A, B \subseteq \Omega$ are called independent if

$$\mathbf{P}(A \cap B) = \mathbf{P}(A)\mathbf{P}(B).$$

For $n \geq 2$ the events $A_1, \ldots, A_n \subseteq \Omega$ are called independent if for all $I \subseteq \{1, \ldots, n\}$ with $|I| \geq 2$,

$$\mathbf{P}(\cap_{i \in I} A_i) = \prod_{i \in I} \mathbf{P}(A_i).$$

Remark 2.42. For independent events $A, B \subseteq \Omega$ with $\mathbf{P}(B) > 0$ we have

$$\mathbf{P}(A|B) = \frac{\mathbf{P}(A \cap B)}{\mathbf{P}(B)} = \frac{\mathbf{P}(A)\mathbf{P}(B)}{\mathbf{P}(B)} = \mathbf{P}(A).$$

Analogously, $\mathbf{P}(A|B) = \mathbf{P}(A)$ implies that A and B are independent.

Example 2.43 (Rolling a die once)**.** Let $\Omega = \{1, \ldots, 6\}$ and $\mathbf{P}(A) = \frac{|A|}{6}$ for $A \subseteq \{1, \ldots, 6\}$ and

$$A = \{2, 4, 6\}, \quad B = \{1, 2\} \quad \text{and} \quad C = \{2\}.$$

This means that we consider for the random experiment that we throw a die once the events of an even number, of a number not larger than two and of a two. Note that

$$\mathbf{P}(A) = \frac{3}{6} = \frac{1}{2}, \quad \mathbf{P}(B) = \frac{2}{6} = \frac{1}{3} \quad \text{and} \quad \mathbf{P}(C) = \frac{1}{6}.$$

Because of

$$\mathbf{P}(A \cap B) = \mathbf{P}(\{2\}) = \frac{1}{6} \quad \text{and} \quad \mathbf{P}(A)\mathbf{P}(B) = \frac{1}{2} \cdot \frac{1}{3} = \frac{1}{6},$$

the events A and B are independent. On the other hand,

$$\mathbf{P}(A \cap C) = \mathbf{P}(\{2\}) = \frac{1}{6} \quad \text{and} \quad \mathbf{P}(A)\mathbf{P}(C) = \frac{1}{2} \cdot \frac{1}{6} = \frac{1}{12}$$

means that A and C are not independent. Consequently, the three events A, B and C are not independent.

Example 2.44. Let $(\Omega, \mathbf{P})$ be the product probability space of two discrete probability spaces $(\Omega_1, \mathbf{P}_1)$ and $(\Omega_2, \mathbf{P}_2)$ (i.e., $\Omega = \Omega_1 \times \Omega_2$). For $M_1 \subseteq \Omega_1$ and $M_2 \subseteq \Omega_2$ we define $E_1 := M_1 \times \Omega_2$ and $E_2 := \Omega_1 \times M_2$. This means that

$$E_1 \cap E_2 = (M_1 \times \Omega_2) \cap (\Omega_1 \times M_2) = M_1 \times M_2.$$

Recall that, by the properties of a product probability space, we have

$$\mathbf{P}(A \times B) = \mathbf{P}_1(A)\mathbf{P}_2(B)$$

for $A \subseteq \Omega_1$ and $B \subseteq \Omega_2$. Thus, we obtain that

$$\begin{aligned}\mathbf{P}(E_1 \cap E_2) &= \mathbf{P}(M_1 \times M_2) = \mathbf{P}_1(M_1)\mathbf{P}_2(M_2) = \mathbf{P}_1(M_1)\mathbf{P}_2(\Omega_2)\mathbf{P}_1(\Omega_1)\mathbf{P}_2(M_2) \\ &= \mathbf{P}(M_1 \times \Omega_2)\mathbf{P}(\Omega_1 \times M_2) = \mathbf{P}(E_1)\mathbf{P}(E_2),\end{aligned}$$

which implies that E_1 and E_2 are independent.

Remark 2.45. Let $(\Omega, \mathbf{P})$ be a discrete probability space and let $A_1, \ldots, A_n \subseteq \Omega$ with $n \in \mathbb{N}$ be independent events. Then we still have independence if we replace some of the sets by their complements. We demonstrate this by showing that $A_1^c, A_2, \ldots, A_n$ are independent. As in the definition of independence we choose $I \subseteq \{1, \ldots, n\}$ with $|I| \geq 2$. We only need to consider the case $1 \in I$. Then, we obtain

$$\mathbf{P}(A_1^c \cap \cap_{i \in I \setminus \{1\}} A_i) = \mathbf{P}(\cap_{i \in I \setminus \{1\}} A_i) - \mathbf{P}(A_1 \cap \cap_{i \in I \setminus \{1\}} A_i).$$

It follows from the independence of $A_1, \ldots, A_n$ that

$$\begin{aligned}\mathbf{P}(A_1^c \cap \cap_{i \in I \setminus \{1\}} A_i) &= \prod_{i \in I \setminus \{1\}} \mathbf{P}(A_i) - \prod_{i \in I} \mathbf{P}(A_i) \\ &= (1 - \mathbf{P}(A_1)) \prod_{i \in I \setminus \{1\}} \mathbf{P}(A_i) = \mathbf{P}(A_1^c) \prod_{i \in I \setminus \{1\}} \mathbf{P}(A_i).\end{aligned}$$

This proves the independence of $A_1^c, A_2, \ldots, A_n$. By the same argument we can successively replace sets by their complements without loosing independence.

Example 2.46 (Birthday problem)**.** Assume that we have a group of n people. We are interested in the probability that at least two of them have the same birthday.

For $k \in \{1, \ldots, n\}$ and $i \in \{1, \ldots, 365\}$ let $A_i^{(k)}$ denote the event that day i is the birthday of person k (we ignore leap years with 366 days). We define B as the event that there is at most one birthday on each day, i.e., we are interested in the probability of B^c. We have that

$$\mathbf{P}(B) = \sum_{\substack{i_1, \ldots, i_n = 1 \\ i_k \neq i_\ell, k \neq \ell}}^{365} \mathbf{P}(A_{i_1}^{(1)} \cap \ldots \cap A_{i_n}^{(n)}).$$

It is reasonable to believe that the events $A_{i_1}^{(1)}, \ldots, A_{i_n}^{(n)}$ are independent since the birthdays of the n persons should not be related (of course this is not the case if we have twins or siblings in our group). This leads to

$$\mathbf{P}(B) = \sum_{\substack{i_1, \ldots, i_n = 1 \\ i_k \neq i_\ell, k \neq \ell}}^{365} \mathbf{P}(A_{i_1}^{(1)}) \cdot \ldots \cdot \mathbf{P}(A_{i_n}^{(n)}).$$

Moreover, we assume that the birthdays are uniformly distributed, i.e.,

$$\mathbf{P}(A_i^{(k)}) = \frac{1}{365} \quad \text{for} \quad k \in \{1, \ldots, n\}, i \in \{1, \ldots, 365\}.$$

Together with determining the number of summands (which corresponds to the problem of an ordered sample without replacement), we obtain

$$\mathbf{P}(B) = \frac{365 \cdot 364 \cdot \ldots \cdot (365 - n + 1)}{365^n}.$$

Thus, the desired probability is

$$\mathbf{P}(B^c) = 1 - \frac{365 \cdot 364 \cdot \ldots \cdot (365 - n + 1)}{365^n}.$$

2.5 General probability spaces

In the previous sections we have studied discrete probability spaces where we assume that the sample space Ω is at most countable. The latter assumption excludes situations like $\Omega = \mathbb{R}$, $\Omega = [0, \infty)$ or $\Omega = [0, 1]$. Since there are many random experiments, where such sample spaces are necessary, we generalise our definition of a probability space in the following. When the cardinality of our sample space is larger than at most countable, it is in general not possible to assign a probability to every subset of Ω. Thus, we allow only some subsets of Ω as events. These families of sets are introduced in the following definition.

Definition 2.47 (σ-algebra)**.** Let Ω be a non-empty set. A family $\mathscr{F}$ of subsets of Ω is called a σ-algebra on Ω if the following properties are satisfied:

(i) $\Omega \in \mathscr{F}$

(ii) $A \in \mathscr{F} \quad \Longrightarrow \quad A^c \in \mathscr{F}$

(iii) $A_1, A_2, \ldots \in \mathscr{F} \quad \Longrightarrow \quad \cup_{i=1}^{\infty} A_i \in \mathscr{F}$

Remark 2.48. It follows from (i) and (ii) in Definition 2.47 that $\emptyset \in \mathscr{F}$. If we choose $A_i = \emptyset$ in (iii) for all sets except of finitely many, we see that finite unions of elements of $\mathscr{F}$ are again elements of $\mathscr{F}$. From this observation and (iii) together with (ii) we deduce that at most countable intersections of elements of $\mathscr{F}$ belong to $\mathscr{F}$ as well.

Example 2.49. Assume that Ω is a non-empty set.

1. Obviously, $\mathscr{F}_1 = \{\emptyset, \Omega\}$ is a σ-algebra. It is the smallest possible σ-algebra.

2. It is easy to see that $\mathscr{F}_2 = \mathcal{P}(\Omega)$ is a σ-algebra. Since $\mathscr{F}_2$ is the largest family of subsets of Ω, namely all of them, it is the largest possible σ-algebra.

Using the concept of a σ-Algebra we can introduce general probability spaces, where the sample space Ω is not required to be at most countable.

Definition 2.50 (Probability space)**.** A probability space $(\Omega, \mathscr{F}, \mathbf{P})$ consists of a non-empty set Ω, a σ-algebra $\mathscr{F}$ on Ω and a function $\mathbf{P} : \mathscr{F} \to [0, 1]$ such that

(i) $\mathbf{P}(\Omega) = 1$

(ii) for all disjoint $A_1, A_2, \ldots \in \mathscr{F}$ (i.e., $A_i \cap A_j = \emptyset$ for $i \neq j$),

$$\mathbf{P}(\cup_{k=1}^{\infty} A_k) = \sum_{k=1}^{\infty} \mathbf{P}(A_k). \tag{2.9}$$

The set Ω is denoted as sample space and $\mathbf{P}$ is called probability or probability measure.

A discrete probability space is only a special case of a probability space as introduced in Definition 2.50. This follows from the following example.

Example 2.51. Each discrete probability space $(\Omega, \mathbf{P})$ is a probability space according to Definition 2.50 if we choose $\mathscr{F} = \mathcal{P}(\Omega)$, which is a σ-algebra as discussed in Example 2.49.

Remark 2.52. Let $(\Omega, \mathscr{F}, \mathbf{P})$ be a probability space as in Definition 2.50.

- We call a set $A \in \mathscr{F}$ measurable. The probability measure $\mathbf{P}$ assigns probabilities only to measurable subsets of Ω.
- We refer to condition (2.9) again as σ-additivity as for discrete probability spaces.
- All our rules for the computation of probabilities from Section 2.1 prevail for general probability spaces. The definitions of conditional probabilities and of the independence of events and the corresponding results in Section 2.4 can be extended to general probability spaces as well.

Lemma 2.53. Let $(\Omega, \mathscr{F}, \mathbf{P})$ be a probability space.

1. For increasing events $A_1, A_2, \ldots \in \mathscr{F}$ (i.e., $A_1 \subseteq A_2 \subseteq \ldots$),
$$\lim_{n\to\infty} \mathbf{P}(A_n) = \mathbf{P}(\cup_{i=1}^{\infty} A_i).$$
2. For decreasing events $A_1, A_2, \ldots \in \mathscr{F}$ (i.e., $A_1 \supseteq A_2 \supseteq \ldots$),
$$\lim_{n\to\infty} \mathbf{P}(A_n) = \mathbf{P}(\cap_{i=1}^{\infty} A_i).$$

Proof. We start with the first statement and define $B_1 := A_1$ and $B_k := A_k \setminus A_{k-1}$ for $k \geq 2$ so that $B_1, B_2, \ldots \in \mathscr{F}$ are disjoint, $A_n = \cup_{i=1}^{n} B_i$ for $n \in \mathbb{N}$ and $\cup_{i=1}^{\infty} A_i = \cup_{i=1}^{\infty} B_i$. Then, we obtain

$$\lim_{n\to\infty} \mathbf{P}(A_n) = \lim_{n\to\infty} \mathbf{P}(\cup_{i=1}^{n} B_i) = \lim_{n\to\infty} \sum_{i=1}^{n} \mathbf{P}(B_i) = \sum_{i=1}^{\infty} \mathbf{P}(B_i) = \mathbf{P}(\cup_{i=1}^{\infty} B_i) = \mathbf{P}(\cup_{i=1}^{\infty} A_i),$$

where we used Lemma 2.8 (i) and the σ-additivity in (2.9).

For the second statement we use that for a decreasing sequence $A_1, A_2, \ldots \in \mathscr{F}$ the sequence $A_1^c, A_2^c, \ldots \in \mathscr{F}$ is increasing. Then, it follows from the first statement applied to the complements that

$$\lim_{n\to\infty} \mathbf{P}(A_n) = 1 - \lim_{n\to\infty} \mathbf{P}(A_n^c) = 1 - \mathbf{P}(\cup_{i=1}^{\infty} A_i^c) = \mathbf{P}((\cup_{i=1}^{\infty} A_i^c)^c) = \mathbf{P}(\cap_{i=1}^{\infty} A_i),$$

which completes the proof. $\square$

For each non-empty set Ω and a family $\mathscr{A}$ of subsets of Ω there exists a smallest σ-algebra on Ω that contains all sets from $\mathscr{A}$. We denote this σ-algebra by $\sigma(\mathscr{A})$ and call it the σ-algebra generated by A.

Example 2.54. Let $\Omega = \{1, 2, 3, 4\}$ and let $\mathscr{A} = \{\{1\}\}$. Then, the smallest σ-algebra on $\{1, 2, 3, 4\}$ that contains $\{1\}$ is

$$\sigma(\{\{1\}\}) = \{\emptyset, \{1\}, \{2, 3, 4\}, \{1, 2, 3, 4\}\}.$$

Definition 2.55 (Borel σ-algebra)**.** For $\Omega = \mathbb{R}$ and $\mathscr{A} = \{(a, b] : a, b \in \mathbb{R}, a \leq b\}$ we denote

$$\mathcal{B}(\mathbb{R}) := \sigma(\mathscr{A})$$

as Borel σ-algebra on $\mathbb{R}$. For $\Omega = \mathbb{R}^d$, $d \geq 2$, and $\mathscr{A} = \{\times_{i=1}^d (a_i, b_i] : a_i, b_i \in \mathbb{R}, a_i \leq b_i, i \in \{1, \ldots, d\}\}$ we call

$$\mathcal{B}(\mathbb{R}^d) := \sigma(\mathscr{A})$$

the Borel σ-algebra on $\mathbb{R}^d$.

Example 2.56. In measure theory, an important result is the construction of the so-called d-dimensional Lebesgue measure λ_d on $\mathbb{R}^d$. It is the unique map $\lambda_d : \mathcal{B}(\mathbb{R}^d) \to [0, \infty]$ such that

$$\lambda_d(\cup_{i=1}^\infty A_i) = \sum_{i=1}^\infty \lambda_d(A_i) \tag{2.10}$$

for disjoint $A_1, A_2, \ldots \in \mathcal{B}(\mathbb{R}^d)$ and

$$\lambda_d(\times_{i=1}^d (a_i, b_i]) = \prod_{i=1}^d (b_i - a_i) \tag{2.11}$$

for $a_i, b_i \in \mathbb{R}$ with $a_i \leq b_i$ for $i \in \{1, \ldots, d\}$. Since in (2.11) the volume of a rectangular solid is computed, the d-dimensional Lebesgue measure can be interpreted as the volume of sets in $\mathbb{R}^d$.

We use the Lebesgue measure to construct a probability on $\mathbb{R}$. For $\Omega = \mathbb{R}$ and $\mathbf{P}(\cdot) = \lambda_1(\cdot \cap (0, 1])$, $(\Omega, \mathcal{B}(\mathbb{R}), \mathbf{P})$ is a probability space. The σ-additivity of $\mathbf{P}$ follows from the σ-additivity of the Lebesgue measure in (2.10) and $\mathbf{P}(\Omega) = \lambda_1(\mathbb{R} \cap (0, 1]) = \lambda_1((0, 1]) = 1$ is implied by (2.11).

Chapter 3

Random variables and expectation

3.1 Discrete random variables

Often we are not interested in the complete outcome of a random experiment but only in some real-valued quantity. For example, when we roll two dice, it can be the case that we are not interested in the exact numbers of both dice, but only in their eye sum. In order order to describe such situations, we introduce random variables.

Definition 3.1 (Random variable)**.** Let $(\Omega, \mathbf{P})$ be a discrete probability space. A map $X : \Omega \to \mathbb{R}$ is called a random variable.

Example 3.2 (Tossing a fair coin twice)**.** Let $\Omega = \{0,1\}^2$ and $\mathbf{P}(A) = \frac{|A|}{4}$ for $A \subseteq \Omega$. Then,

$$X : \Omega \to \mathbb{R}, \quad \omega \mapsto X(\omega) := \begin{cases} 2, & \omega = (1,1), \\ 1, & \omega = (0,1) \text{ or } \omega = (1,0), \\ 0, & \omega = (0,0), \end{cases}$$

is a random variable. We can think of X as the number of heads when we toss a fair coin twice.

Recall that for a random variable $X : \Omega \to \mathbb{R}$ and $A \subseteq \mathbb{R}$,

$$X(\Omega) = \{X(\omega) : \omega \in \Omega\} \quad \text{and} \quad X^{-1}(A) = \{\omega \in \Omega : X(\omega) \in A\}.$$

Theorem 3.3. For a discrete probability space $(\Omega, \mathbf{P})$ and a random variable $X : \Omega \to \mathbb{R}$ define $\mathbf{P}_X : \mathcal{P}(X(\Omega)) \to [0,1]$ by

$$\mathbf{P}_X(A) := \mathbf{P}(X \in A) := \mathbf{P}(X^{-1}(A)) = \mathbf{P}(\{\omega \in \Omega : X(\omega) \in A\}) \quad \text{for} \quad A \subseteq X(\Omega).$$

Then, $(X(\Omega), \mathbf{P}_X)$ is a discrete probability space.

Proof. Since $\mathbf{P}(M) \in [0,1]$ for all $M \subseteq \Omega$, $\mathbf{P}_X$ is a map from $X(\Omega)$ to $[0,1]$. We have that

$$\mathbf{P}_X(X(\Omega)) = \mathbf{P}(X^{-1}(X(\Omega))) = \mathbf{P}(\Omega) = 1.$$

For disjoint sets $A_1, A_2, \ldots \subseteq X(\Omega)$, the sets $X^{-1}(A_1), X^{-1}(A_2), \ldots \subseteq \Omega$ are disjoint as well. Thus, the σ-additivity of $\mathbf{P}$ implies that

$$\mathbf{P}_X(\cup_{i=1}^{\infty} A_i) = \mathbf{P}(X^{-1}(\cup_{i=1}^{\infty} A_i)) = \mathbf{P}(\cup_{i=1}^{\infty} X^{-1}(A_i)) = \sum_{i=1}^{\infty} \mathbf{P}(X^{-1}(A_i)) = \sum_{i=1}^{\infty} \mathbf{P}_X(A_i),$$

which proves the σ-additivity of $\mathbf{P}_X$. □

Definition 3.4. The probability $\mathbf{P}_X$ in Theorem 3.17 is called the distribution of the random variable X.

One often has the situation that the underlying probability space is not explicitly given and that one knows only the distributions of random variables.

Example 3.5 (Tossing a coin twice)**.** The random variable X from Example 3.2 has the distribution

$$\mathbf{P}_X : \mathcal{P}(\{0,1,2\}) \to [0,1] \quad \text{with} \quad \mathbf{P}_X(\{0\}) = \frac{1}{4}, \quad \mathbf{P}_X(\{1\}) = \frac{1}{2} \quad \text{and} \quad \mathbf{P}_X(\{2\}) = \frac{1}{4}.$$

Example 3.6 (Indicator random variable)**.** Let $(\Omega, \mathbf{P})$ be a discrete probability space. The indicator function of an event $A \subseteq \Omega$ is defined by

$$\mathbf{1}_A : \Omega \ni \omega \mapsto \mathbf{1}_A(\omega) := \begin{cases} 1, & \omega \in A, \\ 0, & \omega \notin A. \end{cases}$$

The indicator function $\mathbf{1}_A$ is a random variable whose distribution is given by

$$\mathbf{P}(\mathbf{1}_A = 1) = \mathbf{P}(A) \quad \text{and} \quad \mathbf{P}(\mathbf{1}_A = 0) = \mathbf{P}(A^c) = 1 - \mathbf{P}(A),$$

which is a binomial distribution with the parameters $n = 1$ and $p = \mathbf{P}(A)$. For $n = 1$ the binomial distribution is called Bernoulli distribution.

Example 3.7 (Binomial distribution)**.** A random experiment with the possible outcomes success and failure and with success probability $p \in [0,1]$ is repeated n times. Let the random variable X count the number of successes. For $i \in \{1, \ldots, n\}$ we denote by A_i the event that we have a success in the i-th trial. Since A_i and A_j are independent for $i, j \in \{1, \ldots, n\}$ with $i \neq j$, we see that

$$\mathbf{P}(A_1 \cap \ldots \cap A_k \cap A_{k+1}^c \cap \ldots \cap A_n^c) = \mathbf{P}(A_1) \cdot \ldots \cdot \mathbf{P}(A_k) \cdot \mathbf{P}(A_{k+1}^c) \cdot \ldots \cdot \mathbf{P}(A_n^c) = p^k (1-p)^{n-k}$$

for $k \in \{0, 1, \ldots, n\}$. Since we get the same probability for all $\binom{n}{k}$ possible orders of k successes and $n - k$ failures, we obtain

$$\mathbf{P}(X = k) = \binom{n}{k} p^k (1-p)^{n-k}.$$

Thus, we conclude that X follows a binomial distribution with parameters n and p as introduced in Example 2.27.

Example 3.8 (Poisson distribution)**.** We say that a random variable Y follows a Poisson distribution with parameter $\lambda \in [0, \infty)$ if

$$\mathbf{P}(Y = k) = \frac{\lambda^k}{k!} e^{-\lambda} \quad \text{for} \quad k \in \mathbb{N}_0.$$

In Example 2.28, we established that the probabilities on the right-hand side indeed define a probability distribution. Let X be a random variable with a binomial distribution with parameters n and $p = \frac{\lambda}{n}$ such that $\lambda \leq n$, i.e.,

$$\mathbf{P}(X = k) = \binom{n}{k} \left(\frac{\lambda}{n}\right)^k \left(1 - \frac{\lambda}{n}\right)^{n-k} \quad \text{for} \quad k \in \mathbb{N}_0.$$

Letting $n \to \infty$ yields

$$\begin{aligned} \lim_{n\to\infty} \binom{n}{k} \left(\frac{\lambda}{n}\right)^k \left(1 - \frac{\lambda}{n}\right)^{n-k} &= \lim_{n\to\infty} \frac{n \cdot (n-1) \cdot \ldots \cdot (n-k+1)}{n^k} \frac{\lambda^k}{k!} \left(1 - \frac{\lambda}{n}\right)^{n-k} \\ &= \frac{\lambda^k}{k!} \lim_{n\to\infty} \left(1 - \frac{\lambda}{n}\right)^n = \frac{\lambda^k}{k!} e^{-\lambda}, \end{aligned}$$

where we used a property of the exponential function in the last step. Thus, we can approximate a binomial distribution with parameters n and p by a Poisson distribution with parameter $\lambda = np$ if n is sufficiently large.

Example 3.9.[1] A vaccine is given to 1.6 million people. In seven cases a rare severe disease is diagnosed after the vaccination. In order to decide whether the disease is caused by the vaccine, one has to compare the number of cases in the vaccinated group with the number of cases one would have observed without the vaccination. Assume that the probability to get the disease is $p = 1/1600000$ for each person. Thus, the total number of cases for a group of the same size without vaccination follows a binomial distribution with the parameters $n = 1600000$ and $p = 1/1600000$. We are interested in $\mathbf{P}(X \geq 7)$, i.e., in the event that one observes at least as many cases as in the group of vaccinated people.

Since n is very large, we can approximation the distribution of X by that of a Poisson distributed random variable Y with parameter $\lambda = np = 1$. We obtain

$$\mathbf{P}(Y \leq 6) = \sum_{k=0}^{6} \frac{1^k}{k!} e^{-1} = e^{-1} + \frac{e^{-1}}{1!} + \frac{e^{-1}}{2!} + \frac{e^{-1}}{3!} + \frac{e^{-1}}{4!} + \frac{e^{-1}}{5!} + \frac{e^{-1}}{6!} = \frac{1957}{720} e^{-1}$$

so that

$$\mathbf{P}(X \geq 7) \approx \mathbf{P}(Y \geq 7) = 1 - \mathbf{P}(Y \leq 6) = 1 - \frac{1957}{720} e^{-1} \approx 0.000083.$$

This means that the number of cases in the vaccinated group is significantly higher than one would have observed without the vaccination. This indicates that the disease is caused by the vaccine.

[1] This example uses data reported in `https://www.pei.de/SharedDocs/Downloads/EN/newsroom-en/hp-news/faq-temporary-suspension-astrazeneca.pdf?__blob=publicationFile&v=5`.

Example 3.10 (Hypergeometric distribution)**.** We consider an urn with b blue and r red balls, from which we draw $n \leq b + r$ balls without replacement. Let the random variable X describe the number of drawn blue balls. For $k \in \{0, 1, \ldots, n\}$ we obtain

$$\mathbf{P}(X = k) = \frac{\binom{b}{k}\binom{r}{n-k}}{\binom{b+r}{n}}.$$

The distribution of X is called hypergeometric distribution.

3.2 General random variables

In this section we extend the concept of random variables to general probability spaces.

Definition 3.11 (Random variable)**.** Let $(\Omega, \mathscr{F}, \mathbf{P})$ be a probability space. A map $X : \Omega \to \mathbb{R}$ is called a random variable if

$$\{\omega \in \Omega : X(\omega) \in A\} \in \mathscr{F} \quad \text{for all} \quad A \in \mathcal{B}(\mathbb{R}). \tag{3.1}$$

One says that a function $X : \Omega \to \mathbb{R}$ satisfying (3.1) is measurable. Next we consider how this definition is related to our definition of random variables on discrete probability spaces.

Example 3.12. Let X be a random variable on a discrete probability space $(\Omega, \mathbf{P})$, i.e., a map $X : \Omega \to \mathbb{R}$. Then, for any $A \subseteq \mathbb{R}$, one has

$$\{\omega \in \Omega : X(\omega) \in A\} \in \mathcal{P}(\Omega).$$

Thus, X is a random variable on the probability space $(\Omega, \mathcal{P}(\Omega), \mathbf{P})$ in the sense of Definition 3.11.

In the following we will often denote the underlying probability space of a random variable by $(\Omega, \mathscr{F}, \mathbf{P})$ without mentioning it explicitly. For a random variable X we use, similarly as in the case of discrete random variables, the notation

$$\mathbf{P}_X(A) := \mathbf{P}(X \in A) := \mathbf{P}(X^{-1}(A)) = \mathbf{P}(\{\omega \in \Omega : X(\omega) \in A\}) \quad \text{for} \quad A \in \mathcal{B}(\mathbb{R})$$

for the distribution of X. By condition (3.1) in the definition of a random variable, the probability on the right-hand side is well defined. As in the proof of Theorem 3.17, one can show that $(\mathbb{R}, \mathcal{B}(\mathbb{R}), \mathbf{P}_X)$ is a probability space.

Definition 3.13 (Distribution function)**.** Let X be a random variable. The function

$$F_X : \mathbb{R} \to [0, 1], \quad y \mapsto \mathbf{P}(X \leq y)$$

is called the distribution function of X.

Note that $X \leq y$ can be rewritten as $X \in (-\infty, y]$. Because of $(-\infty, y] \in \mathcal{B}(\mathbb{R})$ the probability in the definition of the distribution function is well defined.

Example 3.14 (Bernoulli distribution)**.** Let X be a Bernoulli distributed random variable with parameter $p \in (0,1)$, i.e., $\mathbf{P}(X = 1) = p$ and $\mathbf{P}(X = 0) = 1 - p$. The distribution function of X is

$$F_X(y) = \mathbf{P}(X \leq y) = \begin{cases} 0, & y < 0, \\ 1 - p, & y \in [0,1), \\ 1, & y \geq 1. \end{cases}$$

The following result collects some properties of distribution functions. We denote by $\lim_{z \to y+}$ the right-sided limit in $y \in \mathbb{R}$.

Theorem 3.15. Let F_X be the distribution function of a random variable X. Then,

1. F_X is non-decreasing, i.e., $F_X(y_1) \leq F_X(y_2)$ for $y_1, y_2 \in \mathbb{R}$ with $y_1 \leq y_2$.
2. $\lim_{y \to -\infty} F_X(y) = 0$ and $\lim_{y \to \infty} F_X(y) = 1$
3. F_X is right-continuous, i.e.,

$$\lim_{z \to y+} F_X(z) = F_X(y) \quad \text{for} \quad y \in \mathbb{R}.$$

Proof. It follows from Lemma 2.8 (iv) that

$$F_X(y_1) = \mathbf{P}(X \leq y_1) \leq \mathbf{P}(X \leq y_2) = F_X(y_2)$$

for $y_1, y_2 \in \mathbb{R}$ with $y_1 \leq y_2$. In the rest of the proof, the existence of all limits follows from the facts that F is non-decreasing and bounded. Lemma 2.53 yields that

$$\begin{aligned} \lim_{y \to -\infty} F_X(y) &= \lim_{y \to -\infty} \mathbf{P}(X \leq y) = \lim_{n \to \infty} \mathbf{P}(X \in (-\infty, -n]) \\ &= \mathbf{P}(\cap_{n=1}^{\infty} \{X \in (-\infty, -n]\}) = \mathbf{P}(\emptyset) = 0 \end{aligned}$$

and

$$\begin{aligned} \lim_{y \to \infty} F_X(y) &= \lim_{y \to \infty} \mathbf{P}(X \leq y) = \lim_{n \to \infty} \mathbf{P}(X \in (-\infty, n]) \\ &= \mathbf{P}(\cup_{n=1}^{\infty} \{X \in (-\infty, n]\}) = \mathbf{P}(\Omega) = 1. \end{aligned}$$

A further application of Lemma 2.53 leads to

$$\begin{aligned} \lim_{z \to y+} F_X(z) &= \lim_{n \to \infty} F_X(y + 1/n) = \lim_{n \to \infty} \mathbf{P}(X \leq y + 1/n) \\ &= \mathbf{P}(\cap_{n=1}^{\infty} \{X \leq y + 1/n\}) = \mathbf{P}(X \leq y) = F_X(y) \end{aligned}$$

for $y \in \mathbb{R}$. $\square$

Remark 3.16. For a distribution function F_X of a random variable X and $y \in \mathbb{R}$ we define $F_X(y-) := \lim_{z \to y-} F_X(z)$, i.e., $F_X(y-)$ is the left-sided limit of F_X in y. Since F_X is non-decreasing, this limit always exists. From Lemma 2.53 it follows that

$$\begin{aligned} F_X(y-) &= \lim_{n\to\infty} F_X(y-1/n) = \lim_{n\to\infty} \mathbf{P}(X \le y - 1/n) \\ &= \mathbf{P}(\cup_{n\in\mathbb{N}}\{X \in (-\infty, y-1/n]\}) = \mathbf{P}(X \in (-\infty, y)) = \mathbf{P}(X < y). \end{aligned}$$

This implies that

$$\mathbf{P}(X = y) = \mathbf{P}(X \le y) - \mathbf{P}(X < y) = F_X(y) - F_X(y-).$$

Obviously, we have $\mathbf{P}(X = y) = 0$ if y is a continuity point of F_X. Thus, continuity of F_X implies that $\mathbf{P}(X = z) = 0$ for all $z \in \mathbb{R}$.

For $a, b \in \mathbb{R}$ with $a < b$ we obtain

$$\begin{aligned} \mathbf{P}(a < X \le b) &= F(b) - F(a) \\ \mathbf{P}(a \le X \le b) &= F(b) - F(a-) \\ \mathbf{P}(a < X < b) &= F(b-) - F(a) \\ \mathbf{P}(a \le X < b) &= F(b-) - F(a-). \end{aligned}$$

The next result, whose proof is outside the scope of this class, says that the opposite direction in Theorem 3.15 holds as well.

Theorem 3.17. Let $F : \mathbb{R} \to [0,1]$ satisfy the three properties of a distribution function in Theorem 3.15. Then, there exists a probability space $(\Omega, \mathscr{F}, \mathbf{P})$ with a random variable X such that F is the distribution function of X, i.e., $\mathbf{P}(X \le y) = F(y)$ for $y \in \mathbb{R}$.

We call a function F as in Theorem 3.17 a distribution function since we know that there exists a random variable with F as distribution function. In this case, where the random variable is not further specified, F has no random variable as index. Theorem 3.17 ensures that we obtain probability distributions by providing distribution functions. In order to construct distribution functions, we use so-called densities.

Definition 3.18. A function $f : \mathbb{R} \to \mathbb{R}$ is called a density if

$$f(x) \ge 0 \quad \text{for} \quad x \in \mathbb{R} \quad \text{and} \quad \int_{-\infty}^{\infty} f(x)\,\mathrm{d}x = 1.$$

In the previous definition we implicitly assume that f is such that the integral is defined. The next result implies that each density leads to a distribution function.

Theorem 3.19. For a density $f : \mathbb{R} \to \mathbb{R}$ define

$$F(y) := \int_{-\infty}^{y} f(x)\,\mathrm{d}x \quad \text{for} \quad y \in \mathbb{R}.$$

Then, F is a distribution function.

Proof. Because of $f(x) \geq 0$ for $x \in \mathbb{R}$, F is non-decreasing. The identity $\int_{-\infty}^{\infty} f(x)\,\mathrm{d}x = 1$ implies that

$$\lim_{y\to-\infty} F(y) = 0 \quad \text{and} \quad \lim_{y\to\infty} F(y) = 1.$$

Since F is defined as an integral, it is continuous. Thus, F satisfies all assumptions of a distribution function. □

We denote a distribution with a density as absolutely continuous. Distributions with continuous distribution functions are called continuous. Note that all absolutely continuous distributions are continuous.

Remark 3.20. For a random variable X whose distribution function F_X has a density f_X one can show that

$$\mathbf{P}(X \in A) = \int_A f_X(x)\,\mathrm{d}x$$

for all $A \in \mathcal{B}(\mathbb{R})$. This statement requires a suitable definition of the integral which is not introduced in this class.

Example 3.21 (Uniform distribution)**.** For $a, b \in \mathbb{R}$ with $a < b$ we define $F : \mathbb{R} \to \mathbb{R}$ by

$$F(y) = \begin{cases} 0, & y < a, \\ \frac{y-a}{b-a}, & a \leq y \leq b, \\ 1, & y > b. \end{cases}$$

Since F is non-decreasing and continuous and

$$\lim_{y\to-\infty} F(y) = 0 \quad \text{and} \quad \lim_{y\to\infty} F(y) = 1,$$

F is obviously a distribution function. The corresponding density is

$$f(y) = F'(y) = \begin{cases} 0, & y < a, \\ \frac{1}{b-a}, & a \leq y \leq b, \\ 0, & y > b. \end{cases}$$

Note that F is not differentiable in a and b, but for these arguments one can choose an arbitrary value for f. Here, we took $1/(b-a)$, but one can choose 0 as well. We say that a random variable with distribution function F is uniformly distributed on $[a, b]$.

Example 3.22 (Pareto distribution)**.** For $\alpha > 0$ let $f : \mathbb{R} \to \mathbb{R}$ be given by

$$f(x) = \begin{cases} 0, & x < 1, \\ \alpha x^{-\alpha-1}, & x \geq 1. \end{cases}$$

Since $f(x) \geq 0$ for $x \in \mathbb{R}$ and

$$\int_{-\infty}^{\infty} f(x)\,\mathrm{d}x = \int_1^{\infty} \alpha x^{-\alpha-1}\,\mathrm{d}x = [-x^{-\alpha}]_1^{\infty} = 0 - (-1) = 1,$$

f is a density. Next we compute its distribution function. For $y < 1$ we have

$$\int_{-\infty}^{y} f(x)\,\mathrm{d}x = 0,$$

while for $y \geq 1$ we obtain

$$\int_{-\infty}^{y} f(x)\,\mathrm{d}x = \int_{1}^{y} \alpha x^{-\alpha-1}\,\mathrm{d}x = [-x^{-\alpha}]_1^y = 1 - y^{-\alpha}.$$

Thus, f has the distribution function

$$F(y) = \begin{cases} 0, & y < 1, \\ 1 - y^{-\alpha}, & y \geq 1. \end{cases}$$

The corresponding distribution is called Pareto distribution.

Example 3.23 (Exponential distribution)**.** For $\lambda > 0$ define $f : \mathbb{R} \to \mathbb{R}$ by

$$f(x) = \begin{cases} 0, & x < 0, \\ \lambda e^{-\lambda x}, & x \geq 0. \end{cases}$$

Since $f(x) \geq 0$ for $x \in \mathbb{R}$ and

$$\int_{-\infty}^{\infty} f(x)\,\mathrm{d}x = \int_{0}^{\infty} \lambda e^{-\lambda x}\,\mathrm{d}x = \big[-e^{-\lambda x}\big]_0^{\infty} = 0 - (-1) = 1,$$

f is a density. Next we compute its distribution function. For $y < 0$ we have

$$\int_{-\infty}^{y} f(x)\,\mathrm{d}x = 0,$$

while for $y \geq 0$ we obtain

$$\int_{-\infty}^{y} f(x)\,\mathrm{d}x = \int_{0}^{y} \lambda e^{-\lambda x}\,\mathrm{d}x = \big[-e^{-\lambda x}\big]_0^{y} = -e^{\lambda y} - (-1) = 1 - e^{-\lambda y}.$$

Thus, we have shown that the density f has the distribution function

$$F(y) = \begin{cases} 0, & y < 0, \\ 1 - e^{-\lambda y}, & y \geq 0. \end{cases}$$

We say that a random variable with distribution function F is exponentially distributed with parameter λ.

Let the random variable X be exponentially distributed with parameter $\lambda > 0$. For $t, h > 0$ we have that

$$\mathbf{P}(X > t+h | X > t) = \frac{\mathbf{P}(X > t+h)}{\mathbf{P}(X > t)} = \frac{1 - \mathbf{P}(X \leq t+h)}{1 - \mathbf{P}(X \leq t)} = \frac{e^{-\lambda(t+h)}}{e^{-\lambda t}} = e^{-\lambda h} = \mathbf{P}(X > h).$$

Thus, an exponentially distributed random variable X satisfies

$$\mathbf{P}(X > t + h | X > t) = \mathbf{P}(X > h)$$

for $t, h > 0$. This property is called memorylessness. One can show that the exponential distribution is the only continuous distribution that is memoryless.

Example 3.24 (Normal distribution)**.** Let the function $\varphi : \mathbb{R} \to \mathbb{R}$ be given by

$$\varphi(x) = \frac{1}{\sqrt{2\pi}} e^{-\frac{x^2}{2}} \quad \text{for} \quad x \in \mathbb{R}.$$

In the following we show that φ is a density. Obviously, we have $\varphi(x) \geq 0$ for $x \in \mathbb{R}$. For

$$I := \int_{-\infty}^{\infty} \varphi(x) \, \mathrm{d}x$$

we obtain

$$\begin{aligned} I^2 &= \left(\int_{-\infty}^{\infty} \varphi(x) \, \mathrm{d}x \right)^2 = \frac{1}{2\pi} \int_{-\infty}^{\infty} e^{-\frac{x^2}{2}} \, \mathrm{d}x \int_{-\infty}^{\infty} e^{-\frac{y^2}{2}} \, \mathrm{d}y \\ &= \frac{1}{2\pi} \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} e^{-\frac{x^2+y^2}{2}} \, \mathrm{d}x \, \mathrm{d}y = \frac{1}{2\pi} \int_{\mathbb{R}^2} e^{-\frac{z_1^2+z_2^2}{2}} \, \mathrm{d}(z_1, z_2). \end{aligned}$$

For two-dimensional integrals of an integrable function $h : \mathbb{R}^2 \to \mathbb{R}$ one knows from analysis that

$$\int_{\mathbb{R}^2} h(z_1, z_2) \, \mathrm{d}(z_1, z_2) = \int_0^{\infty} \int_0^{2\pi} h(r \cos \alpha, r \sin \alpha) \, r \, \mathrm{d}\alpha \, \mathrm{d}r,$$

which is called transformation to polar coordinates. This formula leads to

$$I^2 = \frac{1}{2\pi} \int_0^{\infty} \int_0^{2\pi} e^{-\frac{(r \cos \alpha)^2 + (r \sin \alpha)^2}{2}} \, r \, \mathrm{d}\alpha \, \mathrm{d}r.$$

Together with $(\cos \alpha)^2 + (\sin \alpha)^2 = 1$ for $\alpha \in [0, 2\pi]$, it follows that

$$\begin{aligned} I^2 &= \frac{1}{2\pi} \int_0^{\infty} \int_0^{2\pi} e^{-\frac{r^2}{2}} \, r \, \mathrm{d}\alpha \, \mathrm{d}r = \frac{1}{2\pi} 2\pi \int_0^{\infty} e^{-\frac{r^2}{2}} \, r \, \mathrm{d}r \\ &= \int_0^{\infty} e^{-\frac{r^2}{2}} \, r \, \mathrm{d}r = \left[-e^{-\frac{r^2}{2}} \right]_0^{\infty} = 0 - (-1) = 1. \end{aligned}$$

Thus, we have $I = 1$ and φ is a density. The distribution function of φ is

$$\Phi : \mathbb{R} \to [0, 1], \quad y \mapsto \Phi(y) := \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{y} e^{-\frac{x^2}{2}} \, \mathrm{d}x.$$

Note that the integral in the definition of Φ cannot be further computed. We say that a random variable N with distribution function Φ follows a standard normal distribution or a standard Gaussian distribution.

For $\mu \in \mathbb{R}$ and $\sigma \in (0, \infty)$ one can define the random variable $X = \mu + \sigma N$, which has the distribution function

$$F_X(y) = \mathbf{P}(X \leq y) = \mathbf{P}(\mu + \sigma N \leq y) = \mathbf{P}\left(N \leq \frac{y-\mu}{\sigma}\right) = \Phi\left(\frac{y-\mu}{\sigma}\right)$$

for $y \in \mathbb{R}$ and the density

$$f_X(x) = \frac{d}{dx}\left(\Phi\left(\frac{x-\mu}{\sigma}\right)\right) = \Phi'\left(\frac{x-\mu}{\sigma}\right)\frac{1}{\sigma} = \varphi\left(\frac{x-\mu}{\sigma}\right)\frac{1}{\sigma} = \frac{1}{\sqrt{2\pi}\sigma}e^{-\frac{(x-\mu)^2}{2\sigma^2}}$$

for $x \in \mathbb{R}$. We denote the distribution of X as normal distribution or Gaussian distribution with parameters μ and σ^2.

Next we introduce independence of random variables, a very important property, which shall be often used throughout this class.

Definition 3.25 (Independence of random variables)**.** Random variables $X_1, \ldots, X_n$ are called independent if

$$\mathbf{P}(X_1 \in A_1, \ldots, X_n \in A_n) = \prod_{i=1}^{n} \mathbf{P}(X_i \in A_i) \tag{3.2}$$

for all $A_1, \ldots, A_n \in \mathcal{B}(\mathbb{R})$.

Remark 3.26. In Definition 2.41, we introduced the independence of events. One has that the random variables $X_1, \ldots, X_n$ are independent if and only if the events

$$\{X_1 \in A_1\}, \ldots, \{X_n \in A_n\}$$

are independent for all $A_1, \ldots, A_n \in \mathcal{B}(\mathbb{R})$. For $I \subseteq \{1, \ldots, n\}$ it follows from the choice $A_i = \mathbb{R}$ for $i \in \{1, \ldots, n\} \setminus I$ in (3.2) that

$$\mathbf{P}(X_i \in A_i \text{ for } i \in I) = \prod_{i \in I} \mathbf{P}(X_i \in A_i).$$

Thus, any subset of independent random variables is again independent.

Since the condition in (3.2) involves a large class of sets $A_1, \ldots, A_n$, we often use the following characterisations of independence, whose proofs are beyond the scope of this class.

Theorem 3.27. Let $X_1, \ldots, X_n$ be random variables.

1) Then, $X_1, \dots, X_n$ are independent if and only if

$$\mathbf{P}(X_1 \le y_1, \dots, X_n \le y_n) = \prod_{i=1}^{n} \mathbf{P}(X_i \le y_i)$$

for all $y_1, \dots, y_n \in \mathbb{R}$.

2) Assume that $X_1(\Omega), \dots, X_n(\Omega)$ are at most countable. Then, $X_1, \dots, X_n$ are independent if and only if

$$\mathbf{P}(X_1 = y_1, \dots, X_n = y_n) = \prod_{i=1}^{n} \mathbf{P}(X_i = y_i)$$

for all $y_1 \in X_1(\Omega), \dots, y_n \in X_n(\Omega)$.

Example 3.28 (Rolling a die twice)**.** Let $\Omega = \{1, \dots, 6\}^2$ and $\mathbf{P}(A) = \frac{|A|}{36}$ for $A \subseteq \Omega$. Define $X_1 : \Omega \ni (\omega_1, \omega_2) \mapsto \omega_1$ and $X_2 : \Omega \ni (\omega_1, \omega_2) \mapsto \omega_2$. One can interpret X_1 and X_2 as the numbers displayed by a die that is rolled twice. Obviously, we have $X_1(\Omega) = X_2(\Omega) = \{1, \dots, 6\}$. For $i_1, i_2 \in \{1, \dots, 6\}$, we obtain

$$\begin{aligned}
\mathbf{P}(X_1 = i_1, X_2 = i_2) = \mathbf{P}(\{(i_1, i_2)\}) &= \frac{1}{36} \\
\mathbf{P}(X_1 = i_1) = \mathbf{P}(\{(i_1, 1), \dots, (i_1, 6)\}) &= \frac{6}{36} = \frac{1}{6} \\
\mathbf{P}(X_1 = i_2) = \mathbf{P}(\{(1, i_2), \dots, (6, i_2)\}) &= \frac{6}{36} = \frac{1}{6}
\end{aligned}$$

so that

$$\mathbf{P}(X_1 = i_1, X_2 = i_2) = \frac{1}{36} = \frac{1}{6} \cdot \frac{1}{6} = \mathbf{P}(X_1 = i_1)\mathbf{P}(X_2 = i_2).$$

Thus, by Theorem 3.27, the random variables X_1 and X_2 are independent. This means that the outcomes of two die rolls are independent.

In the following it will often be postulated that some given random variables are independent.

Some of the concepts that we introduced in this section can be extended to several random variables. For random variables $X_1, \dots, X_n$ we call

$$\mathbf{P}_{X_1,\dots,X_n}(A) := \mathbf{P}((X_1, \dots, X_n) \in A) = \mathbf{P}(\{\omega \in \Omega : (X_1(\omega), \dots, X_n(\omega)) \in A\})$$

for $A \in \mathcal{B}(\mathbb{R}^n)$ the distribution of $X_1, \dots, X_n$. Note that, similarly to the univariate case, $(\mathbb{R}^n, \mathcal{B}(\mathbb{R}^n), \mathbf{P}_{X_1,\dots,X_n})$ is a probability space. The distribution function $F_{X_1,\dots,X_n}$ of $X_1, \dots, X_n$ is defined by

$$F_{X_1,\dots,X_n}(y_1, \dots, y_n) = \mathbf{P}(X_1 \le y_1, \dots, X_n \le y_n)$$

for $y_1, \dots, y_n \in \mathbb{R}$. For $i \in \{1, \dots, n\}$ we denote the distribution of X_i as marginal distribution.

Example 3.29. Let the distribution of two random variables X and Y be given by

$$\mathbf{P}(X=1,Y=1)=0.1, \quad \mathbf{P}(X=1,Y=2)=0.2, \quad \mathbf{P}(X=2,Y=1)=0.3$$

and $\mathbf{P}(X=2,Y=2)=0.4$. Then, X has the marginal distribution

$$\mathbf{P}(X=1)=\mathbf{P}(X=1,Y=1)+\mathbf{P}(X=1,Y=2)=0.1+0.2=0.3$$

and

$$\mathbf{P}(X=2)=\mathbf{P}(X=2,Y=1)+\mathbf{P}(X=2,Y=2)=0.3+0.4=0.7,$$

while the marginal distribution of Y is

$$\mathbf{P}(Y=1)=0.1+0.3=0.4 \quad \text{and} \quad \mathbf{P}(Y=2)=0.2+0.4=0.6.$$

The random variables X and Y are not independent since

$$\mathbf{P}(X=1,Y=1)=0.1\neq 0.3\cdot 0.4=\mathbf{P}(X=1)\mathbf{P}(Y=1).$$

Note that the distribution of X and Y is not determined by the marginal distributions of X and Y since the random variables $\widetilde{X}$ and $\widetilde{Y}$ with

$$\mathbf{P}(\widetilde{X}=1,\widetilde{Y}=1)=0.2, \quad \mathbf{P}(\widetilde{X}=1,\widetilde{Y}=2)=0.1, \quad \mathbf{P}(\widetilde{X}=2,\widetilde{Y}=1)=0.2$$

and $\mathbf{P}(\widetilde{X}=2,\widetilde{Y}=2)=0.5$ have the same marginal distributions.

In the following we write $X\sim \mathrm{D}$ to denote that a random variable X follows a distribution D. We use the following abbreviations for distributions:

$\mathrm{Ber}(p)$	Bernoulli distribution with parameter p
$\mathrm{Binom}(n,p)$	Binomial distribution with parameters n and p
$\mathrm{Poisson}(\lambda)$	Poisson distribution with parameter λ
$\mathrm{Geometric}(p)$	Geometric distribution with parameter p
$\mathrm{Normal}(\mu,\sigma^2)$	Normal distribution with parameters μ and σ^2
$\mathrm{Exp}(\lambda)$	Exponential distribution with parameter λ
$\mathrm{Pareto}(\alpha)$	Pareto distribution with parameter α
$\mathrm{Uniform}([a,b])$	Uniform distribution on $[a,b]$

3.3 Expectation of discrete random variables

In the following we introduce the expectation of a random variable, which can be interpreted as the 'average' value of the random variable. We start with the case of discrete probability spaces.

Definition 3.30 (Expectation)**.** Let $(\Omega,\mathbf{P})$ be a discrete probability space with a random variable X. The expectation $\mathbf{E}[X]$ of X is defined by

$$\mathbf{E}[X]=\sum_{\omega\in\Omega}X(\omega)\mathbf{P}(\{\omega\}) \tag{3.3}$$

if the sum on the right-hand side is absolutely convergent.

Sometimes $\mathbf{E}[X]$ is also denoted as expected value of X. If we repeat a random experiment that is described by X many times, then the average over the realisations of X tends to the expectation $\mathbf{E}[X]$.

Remark 3.31. The requirement in Definition 3.30 that the sum on the right-hand side is absolutely convergent means that

$$\sum_{\omega\in\Omega} |X(\omega)|\mathbf{P}(\{\omega\}) < \infty,$$

which is equivalent to $\mathbf{E}[|X|] < \infty$. This assumption is only relevant if Ω is a countably infinite set. The expectation of the random variable X is not defined if the right-hand side of (3.3) is not absolutely convergent. If $X(\omega) \geq 0$ ($X(\omega) \leq 0$) for all $\omega \in \Omega$ and the right-hand side of (3.3) diverges to ∞ (to $-\infty$), we say that $\mathbf{E}[X] = \infty$ ($\mathbf{E}[X] = -\infty$).

Example 3.32 (Expectation for rolling a die once)**.** For $\Omega = \{1, \ldots, 6\}$, $\mathbf{P}(A) = \frac{|A|}{6}$ for $A \subseteq \Omega$ and $X : \Omega \ni \omega \mapsto \omega$ we have

$$\mathbf{E}[X] = \sum_{\omega\in\Omega} X(\omega)\mathbf{P}(\{\omega\}) = \sum_{i=1}^{6} \frac{i}{6} = \frac{1+\ldots+6}{6} = \frac{21}{6} = \frac{7}{2}.$$

A motivation to study expectations comes from gambling, where a player pays a stake and gets a random payoff. Such a game is called fair if the stake equals the expected gain since then the player has neither an advantage nor a disadvantage in the long run. In case that a player gets the number displayed by a die, the fair stake would be $\frac{7}{2}$.

Theorem 3.33. For a random variable X with $\mathbf{E}[|X|] < \infty$, we have

$$\mathbf{E}[X] = \sum_{y\in X(\Omega)} y\mathbf{P}(X = y).$$

Proof. Since $\cup_{y\in X(\Omega)}\{\omega \in \Omega : X(\omega) = y\} = \Omega$ and the sets $\{\omega \in \Omega : X(\omega) = y\}$, $y \in X(\Omega)$, are disjoint, we obtain

$$\begin{aligned}\mathbf{E}[X] &= \sum_{\omega\in\Omega} X(\omega)\mathbf{P}(\{\omega\}) = \sum_{y\in X(\Omega)} \sum_{\omega\in\Omega, X(\omega)=y} X(\omega)\mathbf{P}(\{\omega\}) \\ &= \sum_{y\in X(\Omega)} y \sum_{\omega\in\Omega, X(\omega)=y} \mathbf{P}(\{\omega\}) = \sum_{y\in X(\Omega)} y\mathbf{P}(X = y),\end{aligned}$$

which is the desired identity. □

Theorem 3.33 implies that the expectation of a random variable depends only on the distribution of the random variable.

Example 3.34 (Expectation of an indicator random variable)**.** Let $X = \mathbf{1}_A$ for some $A \subseteq \Omega$ (see Example 3.6 for the definition of an indicator random variable). Because of $X(\Omega) = \{0, 1\}$ it follows from Theorem 3.33 that

$$\mathbf{E}[X] = 0 \cdot \mathbf{P}(X = 0) + 1 \cdot \mathbf{P}(X = 1) = \mathbf{P}(\mathbf{1}_A = 1) = \mathbf{P}(A).$$

Example 3.35 (Expectation of the Poisson distribution)**.** For $X \sim \text{Poisson}(\lambda)$ with $\lambda > 0$ we have, by Theorem 3.33,

$$\begin{aligned}\mathbf{E}[X] &= \sum_{k=0}^{\infty} k\mathbf{P}(X=k) = \sum_{k=0}^{\infty} k\frac{\lambda^k}{k!}e^{-\lambda} = \sum_{k=1}^{\infty} \frac{\lambda^k}{(k-1)!}e^{-\lambda} = \sum_{k=0}^{\infty} \frac{\lambda^{k+1}}{k!}e^{-\lambda} \\ &= \lambda e^{-\lambda}\sum_{k=0}^{\infty}\frac{\lambda^k}{k!} = \lambda e^{-\lambda}e^{\lambda} = \lambda.\end{aligned}$$

In the next theorem we collect several important properties of expectations. For random variables X and Y we write $X \geq 0$ if $X(\omega) \geq 0$ for all $\omega \in \Omega$ and $X \geq Y$ if $X(\omega) \geq Y(\omega)$ for all $\omega \in \Omega$.

Theorem 3.36. Let X and Y be random variables such that $\mathbf{E}[|X|] < \infty$ and $\mathbf{E}[|Y|] < \infty$. Then, the following statements hold:

(i) $\mathbf{E}[X+Y] = \mathbf{E}[X] + \mathbf{E}[Y]$

(ii) For $a \in \mathbb{R}$, $\mathbf{E}[aX] = a\mathbf{E}[X]$.

(iii) If $X = c$ with $c \in \mathbb{R}$, then $\mathbf{E}[X] = c$.

(iv) If $X \geq 0$, then $\mathbf{E}[X] \geq 0$.

(v) If $X \geq Y$, then $\mathbf{E}[X] \geq \mathbf{E}[Y]$.

Proof. By the definition of the expectation we have

$$\begin{aligned}\mathbf{E}[X+Y] &= \sum_{\omega\in\Omega}(X+Y)(\omega)\mathbf{P}(\{\omega\}) = \sum_{\omega\in\Omega}(X(\omega)+Y(\omega))\mathbf{P}(\{\omega\}) \\ &= \sum_{\omega\in\Omega}X(\omega)\mathbf{P}(\{\omega\}) + \sum_{\omega\in\Omega}Y(\omega)\mathbf{P}(\{\omega\}) = \mathbf{E}[X]+\mathbf{E}[Y].\end{aligned}$$

Here, by the assumptions $\mathbf{E}[|X|] < \infty$ and $\mathbf{E}[|Y|] < \infty$ all sums are absolutely convergent. Similarly, we obtain

$$\mathbf{E}[aX] = \sum_{\omega\in\Omega}(aX)(\omega)\mathbf{P}(\{\omega\}) = \sum_{\omega\in\Omega}aX(\omega)\mathbf{P}(\{\omega\}) = a\sum_{\omega\in\Omega}X(\omega)\mathbf{P}(\{\omega\}) = a\mathbf{E}[X].$$

For $X = c$ we derive

$$\mathbf{E}[X] = \sum_{\omega\in\Omega}X(\omega)\mathbf{P}(\{\omega\}) = \sum_{\omega\in\Omega}c\mathbf{P}(\{\omega\}) = c\sum_{\omega\in\Omega}\mathbf{P}(\{\omega\}) = c\mathbf{P}(\Omega) = c.$$

Combining $X \geq 0$ with the fact that probabilities are non-negative, we deduce

$$\mathbf{E}[X] = \sum_{\omega\in\Omega}X(\omega)\mathbf{P}(\{\omega\}) \geq 0.$$

Since $Z := X - Y \geq 0$, it follows from (iv), (i) and (ii) that

$$0 \leq \mathbf{E}[Z] = \mathbf{E}[X-Y] = \mathbf{E}[X] + \mathbf{E}[-Y] = \mathbf{E}[X] - \mathbf{E}[Y],$$

which yields $\mathbf{E}[X] \geq \mathbf{E}[Y]$. □

Theorem 3.36 (i) and (ii) mean that the expectation is a linear map from the space of all random variables with finite expectation to the real numbers.

Example 3.37 (Expectation of the binomial distribution)**.** A random experiment with the two possible outcomes success and failure and success probability $p \in [0,1]$ is repeated n times. Let X denote the total number of successes. We can write the random variable X as

$$X = \sum_{i=1}^{n} \mathbf{1}_{A_i},$$

where A_i is the event of a success in the i-th repetition for $i \in \{1, \dots, n\}$. From Example 3.34 we know that $\mathbf{E}[\mathbf{1}_{A_i}] = \mathbf{P}(A_i) = p$ for $i \in \{1, \dots, n\}$. Thus it follow from Theorem 3.36 (i) that

$$\mathbf{E}[X] = \mathbf{E}\left[\sum_{i=1}^{n} \mathbf{1}_{A_i}\right] = \sum_{i=1}^{n} \mathbf{E}[\mathbf{1}_{A_i}] = \sum_{i=1}^{n} p = np.$$

Together with the observation next to Theorem 3.33 that the expectation of a random variable depends only on its distribution, we obtain for any random variable Y that is binomially distributed with the parameters n and p that $\mathbf{E}[Y] = np$.

Example 3.38 (Expected number of fixpoints of a random permutation)**.** Recall that Π_n is the set of all permutations of $\{1, \dots, n\}$ and let $\Omega = \Pi_n$ and $\mathbf{P}(A) = \frac{|A|}{n!}$ for $A \subseteq \Omega$, i.e., we consider a uniformly chosen random permutation, which we denote by π. Let X be the number of fixpoints of π, which can be written as

$$X = \sum_{i=1}^{n} \mathbf{1}_{\{\pi(i)=i\}}.$$

Since

$$\mathbf{E}[\mathbf{1}_{\{\pi(i)=i\}}] = \mathbf{P}(\pi(i) = i) = \frac{(n-1)!}{n!} = \frac{1}{n}$$

for $i \in \{1, \dots, n\}$, it follows from Theorem 3.36 (i) that

$$\mathbf{E}[X] = \mathbf{E}\left[\sum_{i=1}^{n} \mathbf{1}_{\{\pi(i)=i\}}\right] = \sum_{i=1}^{n} \mathbf{E}[\mathbf{1}_{\{\pi(i)=i\}}] = \sum_{i=1}^{n} \frac{1}{n} = \frac{n}{n} = 1.$$

Theorem 3.39. Let X and Y be independent random variables such that $\mathbf{E}[|X|] < \infty$ and $\mathbf{E}[|Y|] < \infty$. Then, $\mathbf{E}[|XY|] < \infty$ and

$$\mathbf{E}[XY] = \mathbf{E}[X]\mathbf{E}[Y].$$

Proof. By the assumptions $\mathbf{E}[|X|] < \infty$ and $\mathbf{E}[|Y|] < \infty$, the sums

$$\sum_{x \in X(\Omega)} x\mathbf{P}(X = x) \quad \text{and} \quad \sum_{y \in Y(\Omega)} y\mathbf{P}(Y = y)$$

are absolutely convergent. Together with the independence of X and Y, we obtain that

$$\begin{aligned}\mathbf{E}[X]\mathbf{E}[Y] &= \sum_{x\in X(\Omega)} x\mathbf{P}(X=x) \sum_{y\in Y(\Omega)} y\mathbf{P}(Y=y) \\ &= \sum_{x\in X(\Omega), y\in Y(\Omega)} xy\mathbf{P}(X=x)\mathbf{P}(Y=y) = \sum_{x\in X(\Omega), y\in Y(\Omega)} xy\mathbf{P}(X=x, Y=y) \\ &= \sum_{z\in (XY)(\Omega)} z\mathbf{P}(XY=z) = \mathbf{E}[XY].\end{aligned}$$

Since all sums are absolutely convergent, we see that $\mathbf{E}[|XY|] < \infty$. □

The statement of Theorem 3.39 can be extended to products of more than two independent random variables.

3.4 Expectation of general random variables

In the following we construct the expectation of random variables on a general probability space $(\Omega, \mathscr{F}, \mathbf{P})$ in an informal way without proofs. For an indicator random variable $\mathbf{1}_A$ with $A \in \mathscr{F}$ we define

$$\mathbf{E}[\mathbf{1}_A] = \mathbf{P}(A), \tag{3.4}$$

which is motivated by Example 3.34. For a random variable $X = \sum_{i=1}^m c_i \mathbf{1}_{A_i}$ with $c_1, \ldots, c_m > 0$ and $A_1, \ldots, A_m \in \mathscr{F}$ (such random variables are called simple in the following) let

$$\mathbf{E}[X] = \sum_{i=1}^m c_i \mathbf{P}(A_i). \tag{3.5}$$

This definition follows naturally from (3.4) and the desired property that the expectation is linear as in the case a of a discrete probability space (see Theorem 3.36 (i) and (ii)).

Next assume that X is a non-negative random variable. We define a sequence of simple random variables $(X_n)_{n\in\mathbb{N}}$ by

$$X_n(\omega) = \begin{cases} \frac{k}{10^n}, & \frac{k}{10^n} \le X(\omega) \le \frac{k+1}{10^n} \quad \text{for some} \quad k \in \{0, 1, \ldots, 10^n - 1\}, \\ 10^n, & X(\omega) > 10^n, \end{cases} \quad \text{for} \quad \omega \in \Omega.$$

Note that the sequence $(X_n)_{n\in\mathbb{N}}$ approximates X since $\lim_{n\to\infty} X_n(\omega) = X(\omega)$ for all $\omega \in \Omega$ and that $\mathbf{E}[X_n]$ for $n \in \mathbb{N}$ is given by (3.5) because X_n is a simple random variable. Since, for each $\omega \in \Omega$, $(X_n(\omega))_{n\in\mathbb{N}}$ is a monotonously non-decreasing sequence, the sequence $(\mathbf{E}[X_n])_{n\in\mathbb{N}}$ is monotonously non-decreasing as well. Thus, we can define the expectation of X as

$$\mathbf{E}[X] = \lim_{n\to\infty} \mathbf{E}[X_n]. \tag{3.6}$$

Note that different choices of the approximating sequence $(X_n)_{n\in\mathbb{N}}$ lead to the same value for $\mathbf{E}[X]$. In (3.6), we allow $\mathbf{E}[X] = \infty$.

For a general random variable X let

$$X_+(\omega) = \begin{cases} X(\omega), & X(\omega) \geq 0, \\ 0, & X(\omega) < 0, \end{cases} \quad \text{and} \quad X_-(\omega) = \begin{cases} 0, & X(\omega) \geq 0, \\ -X(\omega), & X(\omega) < 0, \end{cases} \quad \text{for} \quad \omega \in \Omega$$

so that $X = X_+ - X_-$. Since X_+ and X_- are non-negative random variables, their expectations exist and we can define

$$\mathbf{E}[X] = \mathbf{E}[X_+] - \mathbf{E}[X_-]$$

if $\mathbf{E}[X_+] < \infty$ or $\mathbf{E}[X_-] < \infty$. The latter assumption is required since for $\mathbf{E}[X_+] = \infty$ and $\mathbf{E}[X_-] = \infty$ the difference $\mathbf{E}[X_+] - \mathbf{E}[X_-]$ is not well defined. We call X integrable if $\mathbf{E}[X_+] < \infty$ and $\mathbf{E}[X_-] < \infty$, which is equivalent to $\mathbf{E}[|X|] < \infty$. A random variable is integrable if and only if it has a finite expectation.

The statements of Theorem 3.36 and Theorem 3.39 carry over to random variables on general probability spaces. From the construction of the expectation above one can deduce the following formula for the expectation of a random variable with an absolutely continuous distribution, which we will not prove.

Theorem 3.40. For a random variable X with a density f we have

$$\mathbf{E}[X] = \int_{-\infty}^{\infty} x f(x) \, \mathrm{d}x$$

if the condition

$$-\int_{-\infty}^{0} x f(x) \, \mathrm{d}x < \infty \quad \text{or} \quad \int_{0}^{\infty} x f(x) \, \mathrm{d}x < \infty \tag{3.7}$$

is satisfied.

Remark 3.41. If both integrals in (3.7) are finite, which is equivalent to

$$\int_{-\infty}^{\infty} |x| f(x) \, \mathrm{d}x < \infty,$$

we call X integrable and $\mathbf{E}[X]$ is finite. If only the first (second) integral is finite, we obtain $\mathbf{E}[X] = \infty$ ($\mathbf{E}[X] = -\infty$).

Example 3.42 (Expectation of the uniform distribution)**.** For a random variable $X \sim \mathrm{Uniform}([a, b])$ with $a, b \in \mathbb{R}$ such that $a < b$, we obtain

$$\mathbf{E}[X] = \int_a^b \frac{x}{b-a} \, \mathrm{d}x = \left[\frac{x^2}{2(b-a)} \right]_a^b = \frac{b^2 - a^2}{2(b-a)} = \frac{(a+b)(b-a)}{2(b-a)} = \frac{a+b}{2}.$$

Example 3.43 (Expectation of the exponential distribution)**.** For $X \sim \mathrm{Exponential}(\lambda)$ with $\lambda > 0$, integration by parts leads to

$$\begin{aligned} \mathbf{E}[X] &= \int_0^\infty x \lambda e^{-\lambda x} \, \mathrm{d}x = \big[-x e^{-\lambda x} \big]_0^\infty - \int_0^\infty -e^{-\lambda x} \, \mathrm{d}x \\ &= 0 + \int_0^\infty e^{-\lambda x} \, \mathrm{d}x = \left[-\frac{e^{-\lambda x}}{\lambda} \right]_0^\infty = 0 - \left(-\frac{1}{\lambda} \right) = \frac{1}{\lambda}. \end{aligned}$$

Example 3.44 (Expectation of the normal distribution)**.** Let $N \sim \text{Normal}(0,1)$. Since

$$\int_0^\infty \frac{x}{\sqrt{2\pi}} e^{-\frac{x^2}{2}}\,\mathrm{d}x = \frac{1}{\sqrt{2\pi}} \int_0^\infty x e^{-\frac{x^2}{2}}\,\mathrm{d}x = \frac{1}{\sqrt{2\pi}} \big[-e^{-\frac{x^2}{2}} \big]_0^\infty = \frac{0-(-1)}{\sqrt{2\pi}} = \frac{1}{\sqrt{2\pi}} < \infty,$$

we obtain

$$\begin{aligned}\mathbf{E}[N] &= \int_{-\infty}^\infty \frac{x}{\sqrt{2\pi}} e^{-\frac{x^2}{2}}\,\mathrm{d}x = \int_{-\infty}^0 \frac{x}{\sqrt{2\pi}} e^{-\frac{x^2}{2}}\,\mathrm{d}x + \int_0^\infty \frac{x}{\sqrt{2\pi}} e^{-\frac{x^2}{2}}\,\mathrm{d}x \\ &= -\int_0^\infty \frac{x}{\sqrt{2\pi}} e^{-\frac{x^2}{2}}\,\mathrm{d}x + \int_0^\infty \frac{x}{\sqrt{2\pi}} e^{-\frac{x^2}{2}}\,\mathrm{d}x = 0,\end{aligned}$$

where we used the symmetry of $x \mapsto e^{-\frac{x^2}{2}}$ in the second last step. A random variable $Y \sim \text{Normal}(\mu, \sigma^2)$ with $\mu \in \mathbb{R}$ and $\sigma \in (0,\infty)$ has the same distribution as $\mu + \sigma N$ with $N \sim \text{Normal}(0,1)$. Thus, it follows from Theorem 3.36 that

$$\mathbf{E}[Y] = \mathbf{E}[\mu + \sigma N] = \mu + \sigma \mathbf{E}[N] = \mu + \sigma \cdot 0 = \mu.$$

This means that the parameter μ of a $\text{Normal}(\mu, \sigma^2)$-distributed random variable is its expectation.

3.5 Functions of random variables

Let X be a random variable on a probability space $(\Omega, \mathscr{F}, \mathbf{P})$. Our goal is to construct a new random variable by applying a function $h : \mathbb{R} \to \mathbb{R}$ to X. By Definition 3.11, $h(X)$ is a random variable if and only if

$$\{\omega \in \Omega : h(X(\omega)) \in A\} \in \mathscr{F} \quad \text{for all} \quad A \in \mathcal{B}(\mathbb{R}). \tag{3.8}$$

We say that a function $h : \mathbb{R} \to \mathbb{R}$ is measurable if

$$h^{-1}(M) \in \mathcal{B}(\mathbb{R}) \quad \text{for all} \quad M \in \mathcal{B}(\mathbb{R}).$$

For a measurable function $h : \mathbb{R} \to \mathbb{R}$, the assumption (3.8) is satisfied since, for all $A \in \mathcal{B}(\mathbb{R})$,

$$\{\omega \in \Omega : h(X(\omega)) \in A\} = \{\omega \in \Omega : X(\omega) \in \underbrace{h^{-1}(A)}_{\in \mathcal{B}(\mathbb{R})}\} \in \mathscr{F}.$$

Thus $Y = h(X)$ is a random variable. Throughout this class we will usually not verify measurability assumptions as all functions we deal with in examples will be measurable. For example, all continuous functions are measurable.

The following result allows us to compute the expectation of a transformation of a random variable.

Theorem 3.45. For a random variable X on a probability space $(\Omega, \mathscr{F}, \mathbf{P})$ and a measurable map $h : \mathbb{R} \to \mathbb{R}$ let $Y = h(X)$.

(i) Let $X(\Omega)$ be at most countable. Then

$$\mathbf{E}[Y] = \sum_{x \in X(\Omega)} h(x)\mathbf{P}(X = x)$$

if

$$\sum_{x \in X(\Omega)} |h(x)|\mathbf{P}(X = x) < \infty. \tag{3.9}$$

(ii) Let X have an absolutely continuous distribution with density f. Then

$$\mathbf{E}[Y] = \int_{-\infty}^{\infty} h(x)f(x)\,\mathrm{d}x$$

if

$$\int_{-\infty}^{\infty} |h(x)|f(x)\,\mathrm{d}x < \infty. \tag{3.10}$$

Proof. We proof only (i) since the proof of (ii) is beyond the scope of this class. If we have a random variable such that $X(\Omega)$ is at most countable, we can assume without loss of generality that we have an underlying discrete probability space $(\Omega, \mathbf{P})$. Then, we obtain

$$\begin{aligned}\mathbf{E}[Y] &= \sum_{\omega \in \Omega} Y(\omega)\mathbf{P}(\{\omega\}) = \sum_{\omega \in \Omega} h(X(\omega))\mathbf{P}(\{\omega\}) = \sum_{x \in X(\Omega)} \sum_{\omega \in \Omega, X(\omega)=x} h(X(\omega))\mathbf{P}(\{\omega\}) \\ &= \sum_{x \in X(\Omega)} h(x) \sum_{\omega \in \Omega, X(\omega)=x} \mathbf{P}(\{\omega\}) = \sum_{x \in X(\Omega)} h(x)\mathbf{P}(X = x),\end{aligned}$$

where all sums are absolutely convergent by (3.9). □

Remark 3.46. The conditions (3.9) and (3.10) ensure that the expectation of Y is finite. If we replace them by the weaker assumptions

$$\sum_{x \in X(\Omega)} \max\{0, h(x)\}\mathbf{P}(X = x) < \infty \quad \text{or} \quad \sum_{x \in X(\Omega)} \max\{0, -h(x)\}\mathbf{P}(X = x) < \infty$$

and

$$\int_{-\infty}^{\infty} \max\{0, h(x)\}f(x)\,\mathrm{d}x < \infty \quad \text{or} \quad \int_{-\infty}^{\infty} \max\{0, -h(x)\}f(x)\,\mathrm{d}x < \infty,$$

respectively, then we allow also the cases $\mathbf{E}[Y] = \infty$ and $\mathbf{E}[Y] = -\infty$.

Example 3.47. For $X \sim \text{Poisson}(\lambda)$ with $\lambda > 0$, it follows from Theorem 3.45 (i) that

$$\begin{aligned}\mathbf{E}[X(X-1)] &= \sum_{k=0}^{\infty} k(k-1)\mathbf{P}(X = k) = \sum_{k=0}^{\infty} k(k-1)\frac{\lambda^k}{k!}e^{-\lambda} = \sum_{k=2}^{\infty} \frac{\lambda^k}{(k-2)!}e^{-\lambda} \\ &= \sum_{k=0}^{\infty} \frac{\lambda^{k+2}}{k!}e^{-\lambda} = \lambda^2 e^{-\lambda} \sum_{k=0}^{\infty} \frac{\lambda^k}{k!} = \lambda^2 e^{-\lambda} e^{\lambda} = \lambda^2,\end{aligned}$$

where we used the exponential series in the second last step.

Example 3.48. For $X \sim \text{Uniform}([a,b])$ with $a, b \in \mathbb{R}$, $a < b$, (see Example 3.21) Theorem 3.45 (ii) leads to

$$\mathbf{E}[X^2] = \int_a^b \frac{x^2}{b-a}\,\mathrm{d}x = \left[\frac{x^3}{3(b-a)}\right]_a^b = \frac{b^3-a^3}{3(b-a)} = \frac{a^2+ab+b^2}{3}.$$

For $m \in \mathbb{N}$ and a random variable X such that $\mathbf{E}[|X|^m] < \infty$ we call $\mathbf{E}[X^m]$ the m-th moment of X.

3.6 Variance and covariance

In the following we introduce the variance which measures the dispersion of a random variable around its expectation. The variance is beside the expectation one of the key characteristics of random variables. We denote the underlying probability space by $(\Omega, \mathscr{F}, \mathbf{P})$ throughout this section.

Definition 3.49 (Variance)**.** Let X be a random variable such that $\mathbf{E}[X^2] < \infty$. The variance of X is defined as

$$\mathbf{Var}[X] = \mathbf{E}[(X - \mathbf{E}[X])^2].$$

Note that the variance of a random variable depends - as the expectation - only on the distribution of the random variable. For $m \in \mathbb{N}$ and a random variable X such that $\mathbf{E}[|X|^m] < \infty$ we denote $\mathbf{E}[(X - \mathbf{E}[X])^m]$ as the m-th centred moment of X. Thus, the variance of X is the second centred moment of X.

Example 3.50 (Variance for rolling a die once)**.** Let the random variable X describe the random experiment of rolling a die once, i.e., $\mathbf{P}(X = i) = \frac{1}{6}$ for $i \in \{1, \dots, 6\}$. Since $\mathbf{E}[X] = \frac{7}{2}$ (see Example 3.32), we obtain

$$\mathbf{Var}[X] = \sum_{i=1}^{6} \left(i - \frac{7}{2}\right)^2 \frac{1}{6} = \frac{1}{6}\left(\frac{25}{4} + \frac{9}{4} + \frac{1}{4} + \frac{1}{4} + \frac{9}{4} + \frac{25}{4}\right) = \frac{35}{12}.$$

Instead of the definition, we often use the following formula for the variance.

Lemma 3.51. Let X be a random variable such that $\mathbf{E}[X^2] < \infty$. Then

$$\mathbf{Var}[X] = \mathbf{E}[X^2] - \mathbf{E}[X]^2.$$

Proof. It follows from Definition 3.49 and Theorem 3.36 that

$$\begin{aligned}\mathbf{Var}[X] &= \mathbf{E}[(X - \mathbf{E}[X])^2] = \mathbf{E}[X^2 - 2X\mathbf{E}[X] + \mathbf{E}[X]^2] = \mathbf{E}[X^2] - 2\mathbf{E}[X\mathbf{E}[X]] + \mathbf{E}[X]^2 \\ &= \mathbf{E}[X^2] - 2\mathbf{E}[X]\mathbf{E}[X] + \mathbf{E}[X]^2 = \mathbf{E}[X^2] - \mathbf{E}[X]^2.\end{aligned}$$

Here, we used that $\mathbf{E}[X\mathbf{E}[X]] = \mathbf{E}[X]\mathbf{E}[X]$ since $\mathbf{E}[X]$ is a constant that we can take out of the expectation. $\square$

Example 3.52 (Variance of the Poisson distribution)**.** For $X \sim \text{Poisson}(\lambda)$ with $\lambda > 0$, we obtain from Lemma 3.51 and the Examples 3.35 and 3.47 that

$$\begin{aligned}\mathbf{Var}[X] &= \mathbf{E}[X^2] - \mathbf{E}[X]^2 = \mathbf{E}[X(X-1)+X] - \mathbf{E}[X]^2 \\ &= \mathbf{E}[X(X-1)] + \mathbf{E}[X] - \mathbf{E}[X]^2 = \lambda^2 + \lambda - \lambda^2 = \lambda.\end{aligned}$$

Example 3.53 (Variance of the uniform distribution)**.** For $X \sim \text{Uniform}([a,b])$ with $a, b \in \mathbb{R}$, $a < b$, it follows from Lemma 3.51 and the Examples 3.42 and 3.48 that

$$\begin{aligned}\mathbf{Var}[X] &= \mathbf{E}[X^2] - \mathbf{E}[X]^2 = \frac{b^2+ab+a^2}{3} - \frac{(a+b)^2}{4} \\ &= \frac{4b^2+4ab+4a^2-3b^2-6ab-3a^2}{12} = \frac{b^2+a^2-2ab}{12} = \frac{(b-a)^2}{12}.\end{aligned}$$

Lemma 3.54. For a random variable X such that $\mathbf{E}[X^2] < \infty$ and $a, c \in \mathbb{R}$, we have

$$\mathbf{Var}[aX] = a^2\mathbf{Var}[X] \quad \text{and} \quad \mathbf{Var}[X+c] = \mathbf{Var}[X].$$

Proof. It follows from Lemma 3.51 and Theorem 3.36 (ii) that

$$\mathbf{Var}[aX] = \mathbf{E}[(aX)^2] - \mathbf{E}[aX]^2 = a^2\mathbf{E}[X^2] - a^2\mathbf{E}[X]^2 = a^2(\mathbf{E}[X^2] - \mathbf{E}[X]^2) = a^2\mathbf{Var}[X].$$

By Definition 3.49 and Theorem 3.36, we have

$$\mathbf{Var}[X+c] = \mathbf{E}[(X+c-\mathbf{E}[X+c])^2] = \mathbf{E}[(X+c-\mathbf{E}[X]-c)^2] = \mathbf{E}[(X-\mathbf{E}[X])^2] = \mathbf{Var}[X],$$

which completes the proof. □

Example 3.55 (Variance of the normal distribution)**.** For $N \sim \text{Normal}(0,1)$, we have

$$\mathbf{E}[N^2] = \int_{-\infty}^{\infty} \frac{x^2}{\sqrt{2\pi}} e^{-\frac{x^2}{2}} \,\mathrm{d}x.$$

The integration by parts formula

$$\int_{-\infty}^{\infty} g(x)h'(x)\,\mathrm{d}x = \big[g(x)h(x)\big]_{-\infty}^{\infty} - \int_{-\infty}^{\infty} g'(x)h(x)\,\mathrm{d}x$$

with $g(x) = x$ and $h(x) = -\frac{1}{\sqrt{2\pi}} e^{-\frac{x^2}{2}}$ leads to

$$\mathbf{E}[N^2] = \left[-\frac{x}{\sqrt{2\pi}} e^{-\frac{x^2}{2}}\right]_{-\infty}^{\infty} + \int_{-\infty}^{\infty} \frac{1}{\sqrt{2\pi}} e^{-\frac{x^2}{2}} \,\mathrm{d}x.$$

Since the first term on the right-hand side vanishes and the second term equals one as integral over the density of N (see Example 3.24), we obtain that $\mathbf{E}[N^2] = 1$. This implies together with $\mathbf{E}[N] = 0$ (see Example 3.44) that

$$\mathbf{Var}[N] = \mathbf{E}[N^2] - \mathbf{E}[N]^2 = 1 - 0 = 1.$$

Since a random variable $Y \sim \text{Normal}(\mu, \sigma^2)$ with $\mu \in \mathbb{R}$ and $\sigma \in (0, \infty)$ has the same distribution as $\mu + \sigma N$ with $N \sim \text{Normal}(0,1)$, it follows from Lemma 3.54 that

$$\mathbf{Var}[Y] = \mathbf{Var}[\mu + \sigma N] = \mathbf{Var}[\sigma N] = \sigma^2\mathbf{Var}[N] = \sigma^2.$$

Remark 3.56. For a random variable X with $\mathbf{E}[X^2] < \infty$, we denote $\sqrt{\mathbf{Var}[X]}$ as the standard deviation of X. This means that the parameter σ of the normal distribution is its standard deviation.

In order to describe the joint behaviour of two random variables, we introduce their covariance.

Definition 3.57 (Covariance)**.** Let X and Y be random variables such that $\mathbf{E}[X^2] < \infty$ and $\mathbf{E}[Y^2] < \infty$. The covariance of X and Y is defined as

$$\mathbf{Cov}(X, Y) = \mathbf{E}[(X - \mathbf{E}[X])(Y - \mathbf{E}[Y])].$$

Note that $\mathbf{Cov}(X, X) = \mathbf{Var}[X]$ for a random variable X satisfying $\mathbf{E}[X^2] < \infty$. The covariance of two random variables has the following important properties.

Lemma 3.58. Let X and Y be random variables such that $\mathbf{E}[X^2] < \infty$ and $\mathbf{E}[Y^2] < \infty$.

(i) Then,

$$\mathbf{Cov}(X, Y) = \mathbf{E}[XY] - \mathbf{E}[X]\mathbf{E}[Y].$$

(ii) If X and Y are independent, then

$$\mathbf{Cov}(X, Y) = 0.$$

Proof. By the same arguments as in the proof of Lemma 3.51, we obtain that

$$\begin{aligned}\mathbf{Cov}(X, Y) &= \mathbf{E}[(X - \mathbf{E}[X])(Y - \mathbf{E}[Y])] = \mathbf{E}[XY] - \mathbf{E}[X\mathbf{E}[Y]] - \mathbf{E}[\mathbf{E}[X]Y] + \mathbf{E}[X]\mathbf{E}[Y] \\ &= \mathbf{E}[XY] - \mathbf{E}[X]\mathbf{E}[Y] - \mathbf{E}[X]\mathbf{E}[Y] + \mathbf{E}[X]\mathbf{E}[Y] = \mathbf{E}[XY] - \mathbf{E}[X]\mathbf{E}[Y].\end{aligned}$$

It follows from (i) and the independence of X and Y (see Theorem 3.39) that

$$\mathbf{Cov}(X, Y) = \mathbf{E}[XY] - \mathbf{E}[X]\mathbf{E}[Y] = \mathbf{E}[X]\mathbf{E}[Y] - \mathbf{E}[X]\mathbf{E}[Y] = 0,$$

which completes the proof. □

It follows from Lemma 3.58 (i) that the covariance is bilinear in the sense that

$$\begin{aligned}&\mathbf{Cov}(a_1X_1 + a_2X_2, b_1Y_1 + b_2Y_2) \\ &= a_1b_1\mathbf{Cov}(X_1, Y_1) + a_2b_1\mathbf{Cov}(X_2, Y_1) + a_1b_2\mathbf{Cov}(X_1, Y_2) + a_2b_2\mathbf{Cov}(X_2, Y_2)\end{aligned}$$

for all $a_1, a_2, b_1, b_2 \in \mathbb{R}$ and all random variables X_1, X_2, Y_1 and Y_2 such that $\mathbf{E}[X_1^2] < \infty$, $\mathbf{E}[X_2^2] < \infty$, $\mathbf{E}[Y_1^2] < \infty$ and $\mathbf{E}[Y_2^2] < \infty$.

Note that the reverse direction of Lemma 3.58 (ii) does not hold since there exist random variables with covariance zero that are not independent. This is illustrated by the following example.

Example 3.59. Let X be a random variable such that

$$\mathbf{P}(X=1)=\mathbf{P}(X=-1)=\mathbf{P}(X=2)=\mathbf{P}(X=-2)=\frac{1}{4}$$

and let $Y=X^2$. Then, we have

$$\mathbf{E}[X]=\frac{-2-1+1+2}{4}=0,\quad \mathbf{E}[Y]=\mathbf{E}[X^2]=\frac{(-2)^2+(-1)^2+1^2+2^2}{4}=\frac{10}{4}=\frac{5}{2}$$

and

$$\mathbf{E}[XY]=\mathbf{E}[X^3]=\frac{(-2)^3+(-1)^3+1^3+2^3}{4}=0$$

so that

$$\mathbf{Cov}(X,Y)=\mathbf{E}[XY]-\mathbf{E}[X]\mathbf{E}[Y]=0-0\cdot\frac{5}{2}=0.$$

On the other hand, we have that

$$\mathbf{P}(X=1,Y=4)=\mathbf{P}(X=1,X\in\{-2,2\})=\mathbf{P}(\emptyset)=0$$

and

$$\mathbf{P}(X=1)\mathbf{P}(Y=4)=\mathbf{P}(X=1)\mathbf{P}(X\in\{-2,2\})=\frac{1}{4}\cdot\frac{1}{2}=\frac{1}{8},$$

whence X and Y are not independent.

Definition 3.60 (Correlation coefficient)**.** For random variables X and Y such that $\mathbf{E}[X^2]<\infty$, $\mathbf{E}[Y^2]<\infty$, $\mathbf{Var}[X]>0$ and $\mathbf{Var}[Y]>0$, the correlation coefficient $\varrho_{X,Y}$ of X and Y is defined by

$$\varrho_{X,Y}=\frac{\mathbf{Cov}(X,Y)}{\sqrt{\mathbf{Var}[X]}\sqrt{\mathbf{Var}[Y]}}.$$

We say that X and Y are uncorrelated if $\varrho_{X,Y}=0$.

The advantage of the correlation coefficient compared with the covariance is that it does not change if we multiply X and Y with positive constants.

Remark 3.61. Let X and Y be as in Definition 3.60.

(i) The so-called Cauchy-Schwarz inequality, which we do not prove, implies that

$$|\mathbf{Cov}(X,Y)|\le\sqrt{\mathbf{Var}[X]}\sqrt{\mathbf{Var}[Y]}.$$

This yields that $\varrho_{X,Y}\in[-1,1]$.

(ii) If X and Y are independent, then we have

$$\varrho_{X,Y}=\frac{\mathbf{Cov}(X,Y)}{\sqrt{\mathbf{Var}[X]}\sqrt{\mathbf{Var}[Y]}}=\frac{0}{\sqrt{\mathbf{Var}[X]}\sqrt{\mathbf{Var}[Y]}}=0$$

by Lemma 3.58 (ii). Obviously, $\mathbf{Cov}(X,Y)=0$ implies $\varrho_{X,Y}=0$. Thus, not all uncorrelated random variables are independent (see Example 3.59).

(iii) One can show that $\varrho_{X,Y} \in \{-1, 1\}$ if and only if $Y = aX + c$ with $a, c \in \mathbb{R}$. In particular, one has $\varrho_{X,Y} = 1$ if $a > 0$ and $\varrho_{X,Y} = -1$ if $a < 0$. Thus, we can think or the correlation coefficient as a measure for the linear dependence between X and Y.

Covariances allow us to compute the variance of sums of random variables.

Theorem 3.62 (Variance of sums of random variables)**.** Let $X_1, \dots, X_n$ be random variables satisfying $\mathbf{E}[X_1^2] < \infty, \dots, \mathbf{E}[X_n^2] < \infty$. Then,

$$\mathbf{Var}\left[\sum_{i=1}^n X_i\right] = \sum_{i=1}^n \mathbf{Var}[X_i] + 2 \sum_{1 \le i < j \le n} \mathbf{Cov}(X_i, X_j). \tag{3.11}$$

Proof. We have

$$\begin{aligned}
\mathbf{Var}\left[\sum_{i=1}^n X_i\right] &= \mathbf{E}\left[\left(\sum_{i=1}^n X_i - \mathbf{E}\left[\sum_{i=1}^n X_i\right]\right)^2\right] = \mathbf{E}\left[\left(\sum_{i=1}^n X_i - \sum_{i=1}^n \mathbf{E}[X_i]\right)^2\right] \\
&= \mathbf{E}\left[\left(\sum_{i=1}^n (X_i - \mathbf{E}[X_i])\right)^2\right] = \mathbf{E}\left[\sum_{i,j=1}^n (X_i - \mathbf{E}[X_i])(X_j - \mathbf{E}[X_j])\right] \\
&= \sum_{i,j=1}^n \mathbf{E}[(X_i - \mathbf{E}[X_i])(X_j - \mathbf{E}[X_j])] \\
&= \sum_{i=1}^n \mathbf{E}[(X_i - \mathbf{E}[X_i])^2] + \sum_{i,j=1, i \ne j}^n \mathbf{E}[(X_i - \mathbf{E}[X_i])(X_j - \mathbf{E}[X_j])] \\
&= \sum_{i=1}^n \mathbf{Var}[X_i] + 2 \sum_{1 \le i < j \le n} \mathbf{Cov}(X_i, X_j),
\end{aligned}$$

which is the desired identity. □

Remark 3.63. If the random variables $X_1, \dots, X_n$ in Theorem 3.62 are pairwise uncorrelated or even independent, the formula in (3.11) simplifies to

$$\mathbf{Var}\left[\sum_{i=1}^n X_i\right] = \sum_{i=1}^n \mathbf{Var}[X_i].$$

Example 3.64 (Variance of the binomial distribution)**.** A random experiment with the two possible outcomes success and failure and success probability $p \in [0, 1]$ is repeated n times. The total number of successes X, which follows a binomial distribution with parameters n and p, can be written as

$$X = \sum_{i=1}^n \mathbf{1}_{A_i},$$

where A_i is the event that one observes a success in the i-th repetition for $i \in \{1, \dots, n\}$. For $i, j \in \{1, \dots, n\}$ with $i \neq j$, the events A_i and A_j are independent. Thus, $\mathbf{1}_{A_i}$ and $\mathbf{1}_{A_j}$ are independent and it follows from Lemma 3.58 (ii) that $\mathbf{Cov}(\mathbf{1}_{A_i}, \mathbf{1}_{A_j}) = 0$.

For $i \in \{1, \dots, n\}$ we obtain from Example 3.34 that

$$\mathbf{Var}[\mathbf{1}_{A_i}] = \mathbf{E}[\mathbf{1}_{A_i}^2] - \mathbf{E}[\mathbf{1}_{A_i}]^2 = \mathbf{E}[\mathbf{1}_{A_i}] - \mathbf{E}[\mathbf{1}_{A_i}]^2 = \mathbf{P}(A_i) - \mathbf{P}(A_i)^2 = p - p^2 = p(1-p).$$

Then, Theorem 3.62 yields

$$\begin{aligned}\mathbf{Var}[X] &= \mathbf{Var}\left[\sum_{i=1}^n \mathbf{1}_{A_i}\right] = \sum_{i=1}^n \mathbf{Var}[\mathbf{1}_{A_i}] + 2 \sum_{1 \le i < j \le n} \mathbf{Cov}(\mathbf{1}_{A_i}, \mathbf{1}_{A_j}) \\ &= \sum_{i=1}^n \mathbf{Var}[\mathbf{1}_{A_i}] = \sum_{i=1}^n p(1-p) = np(1-p).\end{aligned}$$

This implies that for any random variable $Y \sim \mathrm{Binom}(n, p)$ the variance is given by

$$\mathbf{Var}[Y] = np(1-p).$$

Chapter 4

Sums of independent random variables

4.1 Convolution of probability distributions

In this section we compute the distribution of the sum of two independent random variables. This distribution is called the convolution of the distributions of the two random variables. We start with the case of integer valued random variables.

Theorem 4.1 (Convolution of integer valued probability distributions)**.** Let X and Y be independent integer valued random variables. Then, for $m \in \mathbb{Z}$,

$$\mathbf{P}(X+Y=m) = \sum_{k=-\infty}^{\infty} \mathbf{P}(X=m-k)\mathbf{P}(Y=k).$$

Proof. We have that

$$\mathbf{P}(X+Y=m) = \mathbf{P}(\cup_{k=-\infty}^{\infty}\{X=m-k, Y=k\}).$$

Now it follows from σ-additivity and independence of X and Y that

$$\mathbf{P}(X+Y=m) = \sum_{k=-\infty}^{\infty} \mathbf{P}(X=m-k, Y=k) = \sum_{k=-\infty}^{\infty} \mathbf{P}(X=m-k)\mathbf{P}(Y=k),$$

which completes the proof. □

As example for Theorem 4.1 we consider the convolution of two Poisson distributions.

Example 4.2 (Sum of two independent Poisson distributed random variables)**.** For independent random variables $X \sim \text{Poisson}(\lambda)$ and $Y \sim \text{Poisson}(\mu)$ with $\lambda, \mu > 0$, it follows from Theorem 4.1 that, for $m \in \mathbb{Z}$,

$$\mathbf{P}(X+Y=m) = \sum_{k=-\infty}^{\infty} \mathbf{P}(X=m-k)\mathbf{P}(Y=k).$$

Note that the probabilities on the right-hand side become zero if $m-k<0$ or $k<0$. For $m<0$, this implies that

$$\mathbf{P}(X+Y=m)=0.$$

For $m\geq 0$ we obtain

$$\begin{aligned}\mathbf{P}(X+Y=m)&=\sum_{k=0}^{m}\mathbf{P}(X=m-k)\mathbf{P}(Y=k)=\sum_{k=0}^{m}\frac{\lambda^{m-k}}{(m-k)!}e^{-\lambda}\frac{\mu^k}{k!}e^{-\mu}\\&=\frac{e^{-(\lambda+\mu)}}{m!}\sum_{k=0}^{m}\frac{m!}{k!(m-k)!}\lambda^{m-k}\mu^k=\frac{e^{-(\lambda+\mu)}}{m!}\sum_{k=0}^{m}\binom{m}{k}\lambda^{m-k}\mu^k\\&=\frac{e^{-(\lambda+\mu)}}{m!}(\lambda+\mu)^m=\frac{(\lambda+\mu)^m}{m!}e^{-(\lambda+\mu)},\end{aligned}$$

where we used Theorem 2.15 in the second last step. Thus, we have shown that $X+Y\sim$ Poisson$(\lambda+\mu)$.

In order two study the sum of two independent random variables with absolutely continuous distributions, we introduce the convolution of densities.

Definition 4.3 (Convolution of densities)**.** For two densities $f:\mathbb{R}\to\mathbb{R}$ and $g:\mathbb{R}\to\mathbb{R}$ we define $f*g:\mathbb{R}\to\mathbb{R}$ as

$$(f*g)(t)=\int_{-\infty}^{\infty}f(t-s)g(s)\,\mathrm{d}s$$

for $t\in\mathbb{R}$. We call $f*g$ the convolution of f and g.

For the density of the sum of two independent random variables with densities we have the following result, which we will not prove.

Theorem 4.4 (Convolution of random variables with absolutely continuous distributions)**.** Let X and Y be independent random variables with densities f_X and f_Y, respectively. Then, the density f_{X+Y} of $X+Y$ is given by

$$f_{X+Y}=f_X*f_Y.$$

Note that Theorem 4.4 says that the density of $X+Y$ is the convolution of the densities of X and Y. As example, we study the convolution of two standard normal distributions.

Example 4.5 (Sum of two independent random variables with standard normal distributions)**.** Let $N_1\sim$ Normal$(0,1)$ and $N_2\sim$ Normal$(0,1)$ be independent. It follows from Theorem 4.4 that N_1+N_2 has the density

$$\begin{aligned}f_{N_1+N_2}(t)=(f_{N_1}*f_{N_2})(t)&=\int_{-\infty}^{\infty}f_{N_1}(t-s)f_{N_2}(s)\,\mathrm{d}s=\frac{1}{2\pi}\int_{-\infty}^{\infty}e^{-\frac{(t-s)^2}{2}}e^{-\frac{s^2}{2}}\,\mathrm{d}s\\&=\frac{1}{2\pi}\int_{-\infty}^{\infty}e^{-\frac{t^2-2st+2s^2}{2}}\,\mathrm{d}s=\frac{1}{2\pi}e^{-\frac{t^2}{4}}\int_{-\infty}^{\infty}e^{-\frac{t^2/2-2st+2s^2}{2}}\,\mathrm{d}s\\&=\frac{1}{2\pi}e^{-\frac{t^2}{4}}\int_{-\infty}^{\infty}e^{-\frac{(t/\sqrt{2}-\sqrt{2}s)^2}{2}}\,\mathrm{d}s\end{aligned}$$

for $t \in \mathbb{R}$. By the substitution $u = t/\sqrt{2} - \sqrt{2}s$, we obtain

$$f_{N_1+N_2}(t) = \frac{1}{2\pi} e^{-\frac{t^2}{4}} \frac{1}{\sqrt{2}} \int_{-\infty}^{\infty} e^{-\frac{u^2}{2}} \, \mathrm{d}u = \frac{1}{\sqrt{2\pi}\sqrt{2}} e^{-\frac{t^2}{4}} \int_{-\infty}^{\infty} \frac{1}{\sqrt{2\pi}} e^{-\frac{u^2}{2}} \, \mathrm{d}u = \frac{1}{\sqrt{2\pi}\sqrt{2}} e^{-\frac{t^2}{4}},$$

where we used that the integral of the density of a standard normal distribution equals one. Since $f_{N_1+N_2}$ is the density of a normal distribution with expectation 0 and variance 2, we see that $N_1 + N_2 \sim \text{Normal}(0, 2)$.

Remark 4.6. The convolution of the distributions $\mathbf{P}_X$ and $\mathbf{P}_Y$ of two random variables X and Y is often denoted by $\mathbf{P}_X * \mathbf{P}_Y$.

4.2 Moment generating functions

In this section we introduce moment generating functions. They can be also used to determine the distributions of sums of independent random variables, whence they provide an alternative to the convolution formulas from Section 4.1.

Definition 4.7 (Moment generating function)**.** The moment generating function M_X of a random variable X is defined as

$$M_X(t) = \mathbf{E}[e^{tX}]$$

for $t \in \mathbb{R}$ such that $\mathbf{E}[e^{tX}] < \infty$.

We say that the moment generating function exists for some $t \in \mathbb{R}$ if $\mathbf{E}[e^{tX}] < \infty$, i.e., if the expectation of e^{tX} is finite. Note that the expectation of e^{tX} is either finite or equal to infinity as e^{tX} is a non-negative random variable.

Example 4.8 (Moment generating function of the Poisson distribution)**.** For $X \sim \text{Poisson}(\lambda)$ with $\lambda > 0$, we have that

$$\begin{aligned} M_X(t) = \mathbf{E}[e^{tX}] &= \sum_{k=0}^{\infty} e^{kt} \mathbf{P}(X = k) = \sum_{k=0}^{\infty} e^{kt} \frac{\lambda^k}{k!} e^{-\lambda} \\ &= e^{-\lambda} \sum_{k=0}^{\infty} \frac{(\lambda e^t)^k}{k!} = e^{-\lambda} e^{\lambda e^t} = e^{\lambda(e^t - 1)} \end{aligned}$$

for all $t \in \mathbb{R}$.

Example 4.9 (Moment generating function of the normal distribution)**.** For $N \sim \text{Normal}(0, 1)$ we obtain

$$\begin{aligned} M_N(t) = \mathbf{E}[e^{tN}] &= \int_{-\infty}^{\infty} e^{tx} \frac{1}{\sqrt{2\pi}} e^{-\frac{x^2}{2}} \, \mathrm{d}x = \int_{-\infty}^{\infty} \frac{1}{\sqrt{2\pi}} e^{-\frac{x^2 - 2tx}{2}} \, \mathrm{d}x = e^{\frac{t^2}{2}} \int_{-\infty}^{\infty} \frac{1}{\sqrt{2\pi}} e^{-\frac{x^2 - 2tx + t^2}{2}} \, \mathrm{d}x \\ &= e^{\frac{t^2}{2}} \int_{-\infty}^{\infty} \frac{1}{\sqrt{2\pi}} e^{-\frac{(x-t)^2}{2}} \, \mathrm{d}x = e^{\frac{t^2}{2}} \int_{-\infty}^{\infty} \frac{1}{\sqrt{2\pi}} e^{-\frac{y^2}{2}} \, \mathrm{d}y = e^{\frac{t^2}{2}} \end{aligned}$$

for $t \in \mathbb{R}$. In the last two steps we used substitution and that the integral of the density of N is one.

Let $Y \sim \text{Normal}(\mu, \sigma^2)$ with $\mu \in \mathbb{R}$ and $\sigma \in (0, \infty)$. Since Y has the same distribution as $\mu + \sigma N$ with $N \sim \text{Normal}(0, 1)$, we deduce that, for $t \in \mathbb{R}$,

$$M_Y(t) = \mathbf{E}[e^{tY}] = \mathbf{E}[e^{t(\mu+\sigma N)}] = e^{t\mu}\mathbf{E}[e^{(t\sigma)N}] = e^{t\mu}M_N(\sigma t).$$

This yields that

$$M_Y(t) = e^{t\mu}e^{\frac{(\sigma t)^2}{2}} = e^{\mu t + \frac{\sigma^2 t^2}{2}}$$

for $t \in \mathbb{R}$.

Example 4.10. For $X \sim \text{Exponential}(\lambda)$ with $\lambda > 0$, we have

$$M_X(t) = \mathbf{E}[e^{tX}] = \int_0^\infty e^{tx}\lambda e^{-\lambda x}\,\mathrm{d}x = \int_0^\infty \lambda e^{(t-\lambda)x}\,\mathrm{d}x$$

for $t \in \mathbb{R}$. Since the integral on the right-hand side becomes infinity for $t \geq \lambda$, the moment generating function of X exists only for $t < \lambda$. In this case, we obtain

$$M_X(t) = \int_0^\infty \lambda e^{(t-\lambda)x}\,\mathrm{d}x = \left[\frac{\lambda e^{(t-\lambda)x}}{t-\lambda}\right]_0^\infty = 0 - \frac{\lambda}{t-\lambda} = \frac{\lambda}{\lambda - t}.$$

In the next result we collect some properties of moment generating functions, which we will not prove. We denote by $g^{(m)}$ the m-th derivative of a function g.

Theorem 4.11. Let X and Y be two random variables.

(i) If there is an $\varepsilon > 0$ such that $M_X(t)$ and $M_Y(t)$ exist for all $t \in (-\varepsilon, \varepsilon)$ and such that $M_X(t) = M_Y(t)$ for all $t \in (-\varepsilon, \varepsilon)$, then X and Y have the same distribution.

(ii) Let $\varepsilon > 0$ be such that $M_X(t)$ exists for all $t \in (-\varepsilon, \varepsilon)$. Then, for all $m \in \mathbb{N}$, $\mathbf{E}[|X|^m] < \infty$ and

$$\mathbf{E}[X^m] = M_X^{(m)}(0).$$

Theorem 4.11 (i) implies that the moment generating function (if it exists in a neighbourhood of the origin) uniquely determines the distribution of a random variable. Recall that we denote $\mathbf{E}[X^m]$ as the m-th moment of a random variable. Theorem 4.11 (ii) allows us to compute the moments of a random variable from its moment generating function.

Example 4.12 (Moments of the Poisson distribution)**.** Let $X \sim \text{Poisson}(\lambda)$ with $\lambda > 0$. Recall that

$$M_X(t) = e^{\lambda(e^t - 1)}$$

for $t \in \mathbb{R}$ so that

$$\begin{aligned}
M_X'(t) &= e^{\lambda(e^t-1)}\lambda e^t \\
M_X''(t) &= e^{\lambda(e^t-1)}\lambda^2 e^{2t} + e^{\lambda(e^t-1)}\lambda e^t = e^{\lambda(e^t-1)}(\lambda^2 e^{2t} + \lambda e^t) \\
M_X'''(t) &= e^{\lambda(e^t-1)}(\lambda^3 e^{3t} + \lambda^2 e^{2t}) + e^{\lambda(e^t-1)}(2\lambda^2 e^{2t} + \lambda e^t) = e^{\lambda(e^t-1)}(\lambda^3 e^{3t} + 3\lambda^2 e^{2t} + \lambda e^t).
\end{aligned}$$

Now Theorem 4.11 (ii) leads to

$$\begin{aligned}
\mathbf{E}[X] &= M_X'(0) = e^{\lambda \cdot 0}\lambda e^0 = \lambda \\
\mathbf{E}[X^2] &= M_X''(0) = e^0(\lambda^2 e^0 + \lambda e^0) = \lambda^2 + \lambda \\
\mathbf{E}[X^3] &= M_X'''(0) = e^0(\lambda^3 e^0 + 3\lambda^2 e^0 + \lambda e^0) = \lambda^3 + 3\lambda^2 + \lambda.
\end{aligned}$$

Example 4.13 (Moments of the normal distribution)**.** For $N \sim \text{Normal}(0,1)$ we have

$$M_N(t) = e^{\frac{t^2}{2}}$$

for $t \in \mathbb{R}$ so that

$$\begin{aligned}
M_N'(t) &= te^{\frac{t^2}{2}} \\
M_N''(t) &= e^{\frac{t^2}{2}} + t^2 e^{\frac{t^2}{2}} \\
M_N'''(t) &= te^{\frac{t^2}{2}} + 2te^{\frac{t^2}{2}} + t^3 e^{\frac{t^2}{2}} = 3te^{\frac{t^2}{2}} + t^3 e^{\frac{t^2}{2}} \\
M_N''''(t) &= 3e^{\frac{t^2}{2}} + 3t^2 e^{\frac{t^2}{2}} + 3t^2 e^{\frac{t^2}{2}} + t^4 e^{\frac{t^2}{2}} = 3e^{\frac{t^2}{2}} + 6t^2 e^{\frac{t^2}{2}} + t^4 e^{\frac{t^2}{2}}.
\end{aligned}$$

It follows from Theorem 4.11 (ii) that

$$\begin{aligned}
\mathbf{E}[N] &= M_N'(0) = 0 \cdot 1 = 0 \\
\mathbf{E}[N^2] &= M_N''(0) = 1 + 0 \cdot 1 = 1 \\
\mathbf{E}[N^3] &= M_N'''(0) = 3 \cdot 0 \cdot 1 + 0 \cdot 1 = 0 \\
\mathbf{E}[N^4] &= M_N''''(0) = 3 \cdot 1 + 6 \cdot 0 \cdot 1 + 0 \cdot 1 = 3.
\end{aligned}$$

The moment generating function can be also used to compute the distribution of the sum of independent random variables.

Theorem 4.14. Let X and Y be independent random variables. Then,

$$M_{X+Y}(t) = M_X(t)M_Y(t)$$

for all $t \in \mathbb{R}$ such that $M_X(t)$ and $M_Y(t)$ exist.

Proof. Let $t \in \mathbb{R}$ be such that $M_X(t) = \mathbf{E}[e^{tX}] < \infty$ and $M_Y(t) = \mathbf{E}[e^{tY}] < \infty$. Since e^{tX} and e^{tY} are independent, which is a consequence of the independence of X and Y, it follows from Theorem 3.39 that

$$\mathbf{E}[e^{tX}e^{tY}] < \infty$$

and that

$$M_{X+Y}(t) = \mathbf{E}[e^{t(X+Y)}] = \mathbf{E}[e^{tX}e^{tY)}] = \mathbf{E}[e^{tX}]\mathbf{E}[e^{tY}] = M_X(t)M_Y(t),$$

which completes the proof. □

Note that Theorem 4.14 can be extended to sums of finitely many independent random variables.

The Theorems 4.11 and 4.14 allow us to determine the distribution of $X + Y$ for two independent random variables X and Y in the following way. First we compute the moment generating function M_{X+Y} via Theorem 4.14. If we can find a random variable Z such that M_Z coincides with M_{X+Y} in some neighbourhood of the origin, then we know by Theorem 4.11 (i) that Z and $X+Y$ have the same distribution. This is an alternative approach to the one from Section 4.1 for the computation of convolutions of probability distributions. We illustrate this by considering the same examples as in Section 4.1.

Example 4.15 (Sum of two independent Poisson distributed random variables)**.** Let $X \sim \text{Poisson}(\lambda)$ and $Y \sim \text{Poisson}(\mu)$ be independent Poisson distributed random variables with $\lambda, \mu > 0$. It follows from Theorem 4.14 and Example 4.8 that

$$M_{X+Y}(t) = M_X(t)M_Y(t) = e^{\lambda(e^t-1)}e^{\mu(e^t-1)} = e^{(\lambda+\mu)(e^t-1)}$$

for $t \in \mathbb{R}$. Thus, $X+Y$ has the same moment generating function as a Poisson distributed random variable with parameter $\lambda + \mu$. By Theorem 4.11 (i), we have that $X + Y \sim \text{Poisson}(\lambda + \mu)$. This was also shown in Example 4.2 by using Theorem 4.1.

Example 4.16 (Sum of two independent normal distributed random variables)**.** Let $X_1 \sim \text{Normal}(\mu_1, \sigma_1^2)$ and $X_2 \sim \text{Normal}(\mu_2, \sigma_2^2)$ be independent random variables with $\mu_1, \mu_2 \in \mathbb{R}$ and $\sigma_1, \sigma_2 \in (0, \infty)$. It follows from Theorem 4.14 and Example 4.9 that

$$M_{X_1+X_2}(t) = M_{X_1}(t)M_{X_2}(t) = e^{\mu_1 t + \frac{\sigma_1^2 t^2}{2}} e^{\mu_2 t + \frac{\sigma_2^2 t^2}{2}} = e^{(\mu_1+\mu_2)t + \frac{(\sigma_1^2+\sigma_2^2)t^2}{2}}$$

for $t \in \mathbb{R}$. Since the expression on the right-hand side is the moment generating function of a normal distribution with expectation $\mu_1 + \mu_2$ and variance $\sigma_1^2 + \sigma_2^2$, we see that $X_1 + X_2 \sim \text{Normal}(\mu_1 + \mu_2, \sigma_1^2 + \sigma_2^2)$. This means that the sum of two independent normal distributed random variables follows again a normal distribution. We obtain its expectation and variance by taking the sums of the expectations and variances.

This example generalises Example 4.5, where Theorem 4.4 was used, from the special situation of standard normal distributions to general normal distributions.

Remark 4.17. Instead of moment generating functions one can study so-called characteristic functions. The characteristic function of a random variable X is defined as

$$\varphi_X(t) = \mathbf{E}[e^{\mathbf{i}tX}]$$

for $t \in \mathbb{R}$, where $\mathbf{i}$ is the imaginary unit. In contrast to the moment generating function, the characteristic function is defined for all random variables and for all values of t. The characteristic function uniquely determines the distribution of a random variable and for independent random variables X and Y one has

$$\varphi_{X+Y}(t) = \varphi_X(t)\varphi_Y(t)$$

for $t \in \mathbb{R}$.

4.3 Markov inequality and Chebyshev inequality

In this section we introduce so-called concentration inequalities, which bound probabilities that random variables deviate from some values like zero or their expectation by more than a certain amount, i.e., they control how concentrated random variables are around such values.

Recall that a function $h : I \to \mathbb{R}$ with $I \subseteq \mathbb{R}$ is called monotonously non-decreasing if $h(a) \leq h(b)$ for all $a, b \in I$ such that $a \leq b$.

Theorem 4.18 (Markov inequality)**.** Let X be a random variable and let $h : [0, \infty) \to [0, \infty)$ be a monotonously non-decreasing function such that $\mathbf{E}[h(|X|)] < \infty$. Then, for any $u > 0$ with $h(u) > 0$,

$$\mathbf{P}(|X| \geq u) \leq \frac{\mathbf{E}[h(|X|)]}{h(u)}.$$

Proof. Since h is non-negative and monotonously non-decreasing, we have

$$h(|X|) \geq \mathbf{1}\{|X| \geq u\}h(|X|) \geq \mathbf{1}\{|X| \geq u\}h(u),$$

where $\mathbf{1}\{|X| \geq u\}$ stands for the indicator random variable of the event $\{|X| \geq u\}$. It follows from Theorem 3.36 (v) that

$$\mathbf{E}[h(|X|)] \geq \mathbf{E}[\mathbf{1}\{|X| \geq u\}h(|X|)] \geq \mathbf{E}[\mathbf{1}\{|X| \geq u\}h(u)].$$

From Theorem 3.36 (ii) and Example 3.34 we obtain

$$\mathbf{E}[h(|X|)] \geq h(u)\mathbf{E}[\mathbf{1}\{|X| \geq u\}] = h(u)\mathbf{P}(|X| \geq u),$$

which proves the assertion. □

The Markov inequality allows us to bound the probability that the absolute value of a random variable exceeds a given threshold. It is often used to control the probabilities $\mathbf{P}(X \geq u)$ or $\mathbf{P}(X \leq -u)$ for large $u \in \mathbb{R}$, which are of interest for many applications since they are the probabilities of extreme events.

An important consequence of the Markov inequality is the Chebyshev inequality.

Theorem 4.19 (Chebyshev inequality)**.** Let X be a random variable such that $\mathbf{E}[X^2] < \infty$. Then, for all $u > 0$,

$$\mathbf{P}(|X - \mathbf{E}[X]| \geq u) \leq \frac{\mathbf{Var}[X]}{u^2}.$$

Proof. Applying the Markov inequality Theorem 4.18 with $h : [0, \infty) \ni x \mapsto x^2$ to the random variable $Y := X - \mathbf{E}[X]$, we obtain

$$\mathbf{P}(|X - \mathbf{E}[X]| \geq u) = \mathbf{P}(|Y| \geq u) \leq \frac{\mathbf{E}[h(|Y|)]}{h(u)} = \frac{\mathbf{E}[|X - \mathbf{E}[X]|^2]}{u^2} = \frac{\mathbf{Var}[X]}{u^2},$$

which completes the proof. □

The Chebyshev inequality provides a bound for the concentration of a random variable around its expectation. We illustrate this by the following example.

Example 4.20. For $N \sim \text{Normal}(0,1)$ and $u > 0$, it follows from the Chebyshev inequality Theorem 4.19 that

$$\mathbf{P}(|N| \geq u) = \mathbf{P}(|N - \mathbf{E}[N]| \geq u) \leq \frac{\mathbf{Var}[N]}{u^2} = \frac{1}{u^2}.$$

Here, we used $\mathbf{E}[N] = 0$ and $\mathbf{Var}[N] = 1$ (see Example 3.44 and Example 3.55).

4.4 Law of large numbers

In the following we derive the so-called weak law of large numbers, which describes the asymptotic behaviour of averages of independent and identically distributed random variables. In order to formulate the weak law of large numbers, we need the concept of convergence in probability.

Definition 4.21 (Convergence in probability)**.** Let $(X_n)_{n\in\mathbb{N}}$ be a sequence of random variables and let X be a random variable. We say that the sequence $(X_n)_{n\in\mathbb{N}}$ converges in probability to X if

$$\lim_{n\to\infty} \mathbf{P}(|X_n - X| \geq \varepsilon) = 0$$

for all $\varepsilon > 0$. We write $X_n \xrightarrow{\mathbf{P}} X$ as $n \to \infty$ for convergence in probability.

In Definition 3.25 we introduced independence for finite collections of random variables. We call an arbitrary collection of random variables independent if any finite subset of them is independent according to Definition 3.25. We denote a family of random variables as independent and identically distributed if they are independent and follow the same distribution. We abbreviate independent and identically distributed by i.i.d. in the following.

Theorem 4.22 (Weak law of large numbers)**.** Let $(X_i)_{i\in\mathbb{N}}$ be a sequence of i.i.d. random variables such that $\mathbf{E}[X_1^2] < \infty$ and define $S_n = \sum_{i=1}^n X_i$ for $n \in \mathbb{N}$. Then,

$$\frac{S_n}{n} \xrightarrow{\mathbf{P}} \mathbf{E}[X_1] \quad \text{as} \quad n \to \infty. \tag{4.1}$$

Proof. Let $\varepsilon > 0$ and $n \in \mathbb{N}$. By the linearity of expectation (see Theorem 3.36 (i) and (ii)) we have

$$\mathbf{E}\left[\frac{S_n}{n}\right] = \frac{1}{n}\mathbf{E}\left[\sum_{i=1}^n X_i\right] = \frac{1}{n}\sum_{i=1}^n \mathbf{E}[X_i] = \frac{n\mathbf{E}[X_1]}{n} = \mathbf{E}[X_1],$$

while Lemma 3.54 and Theorem 3.62 with Remark 3.63 lead to

$$\mathbf{Var}\left[\frac{S_n}{n}\right] = \frac{\mathbf{Var}[\sum_{i=1}^n X_i]}{n^2} = \frac{1}{n^2}\sum_{i=1}^n \mathbf{Var}[X_i] = \frac{n\mathbf{Var}[X_1]}{n^2} = \frac{\mathbf{Var}[X_1]}{n}.$$

Now it follows from the Chebyshev inequality (see Theorem 4.19) that

$$\mathbf{P}\left(\left|\frac{S_n}{n} - \mathbf{E}[X_1]\right| \geq \varepsilon\right) = \mathbf{P}\left(\left|\frac{S_n}{n} - \mathbf{E}\left[\frac{S_n}{n}\right]\right| \geq \varepsilon\right) \leq \frac{\mathbf{Var}[S_n/n]}{\varepsilon^2} = \frac{\mathbf{Var}[X_1]}{\varepsilon^2 n}. \tag{4.2}$$

Since the right-hand side vanishes as $n \to \infty$, this implies

$$\lim_{n\to\infty} \mathbf{P}\left(\left|\frac{S_n}{n} - \mathbf{E}[X_1]\right| \geq \varepsilon\right) = 0,$$

which proves the assertion. □

Remark 4.23. Let the assumptions of Theorem 4.22 be satisfied. Then, it follows from (4.2) in the proof of Theorem 4.22 that for all $n \in \mathbb{N}$ and $\varepsilon > 0$,

$$\mathbf{P}\left(\left|\frac{S_n}{n} - \mathbf{E}[X_1]\right| \geq \varepsilon\right) \leq \frac{\mathbf{Var}[X_1]}{\varepsilon^2 n}.$$

Roughly speaking, the weak law of large numbers means that the average of a large number of i.i.d. random variables will be close to the expectation of the random variables. We consider two applications of the weak law of large numbers.

Example 4.24 (Convergence of relative frequencies)**.** Let $(X_i)_{i\in\mathbb{N}}$ be a sequence of independent $\mathrm{Ber}(p)$-distributed random variables with $p \in [0,1]$. Since $\mathbf{E}[X_1^2] = p < \infty$ and $\mathbf{E}[X_1] = p$, it follows from the weak law of large numbers (see Theorem 4.22) that

$$\frac{1}{n}\sum_{i=1}^{n} X_i \xrightarrow{\mathbf{P}} p \quad \text{as} \quad n \to \infty. \tag{4.3}$$

For some given random experiment with an event A such that $\mathbf{P}(A) = p$ we can think of X_i as the indicator random variable that the event A occurs in the i-th repetition of the random experiment. In this case $\frac{1}{n}\sum_{i=1}^{n} X_i$ is the relative frequency of the event A after n repetitions and (4.3) means that the relative frequencies converge in probability to $\mathbf{P}(A)$ as $n \to \infty$. This way the weak law of large numbers establishes rigorously the convergence of relative frequencies to probabilities, which was discussed next to Example 2.3. For example, we have for the random experiment of rolling a die that the relative frequencies of throwing a six converge in probability to $\frac{1}{6}$.

Example 4.25 (Monte-Carlo integration)**.** Let $g : [0,1] \to [0,c]$ with $c > 0$ be an integrable function. An important problem is to compute the integral $\int_0^1 g(x)\,\mathrm{d}x$. A probabilistic way to approximate this integral is the so-called Monte-Carlo integration. Let $(U_i)_{i\in\mathbb{N}}$ be a sequence of independent $\mathrm{Uniform}([0,1])$-distributed random variables. For $n \in \mathbb{N}$ we approximate $\int_0^1 g(x)\,\mathrm{d}x$ by

$$\frac{1}{n}\sum_{i=1}^{n} g(U_i).$$

In order to obtain this approximation, one can use computer generated random numbers for $U_1, \dots, U_n$. Our goal is now to determine how well this random quantity approximates $\int_0^1 g(x)\,\mathrm{d}x$. First note that

$$\mathbf{E}[g(U_1)] = \int_{-\infty}^{\infty} g(x) f_{U_1}(x)\,\mathrm{d}x = \int_0^1 g(x) \cdot 1\,\mathrm{d}u = \int_0^1 g(x)\,\mathrm{d}x,$$

where f_{U_1} denotes the density of U_1. Since $(g(U_i))_{i\in\mathbb{N}}$ are independent and identically distributed and $\mathbf{E}[g(U_1)^2] \le c^2$, it follows from the weak law of large numbers (see Theorem 4.22) that

$$\frac{1}{n}\sum_{i=1}^{n} g(U_i) \xrightarrow{\mathbf{P}} \int_0^1 g(x)\,\mathrm{d}x \quad \text{as} \quad n \to \infty. \tag{4.4}$$

From Remark 4.23 we even obtain that, for all $n \in \mathbb{N}$ and $\varepsilon > 0$,

$$\mathbf{P}\left(\left|\frac{1}{n}\sum_{i=1}^{n} g(U_i) - \int_0^1 g(x)\,\mathrm{d}x\right| \ge \varepsilon\right) \le \frac{\mathbf{Var}[g(U_1)]}{\varepsilon^2 n} \le \frac{c^2}{\varepsilon^2 n}, \tag{4.5}$$

where we used $\mathbf{Var}[g(U_1)] \le \mathbf{E}[g(U_1)^2] \le c^2$ in the last step. The inequality in (4.5) allows us to control the approximation error for a fixed $n \in \mathbb{N}$, while (4.4) is only an asymptotic statement.

Remark 4.26 (Strong law of large numbers)**.** For a sequence of random variables $(X_n)_{n\in\mathbb{N}}$ and a random variable X we say that $(X_n)_{n\in\mathbb{N}}$ converges $\mathbf{P}$-almost surely to X as $n \to \infty$ if

$$\mathbf{P}(\{\omega \in \Omega : \lim_{n\to\infty} X_n(\omega) = X(\omega)\}) = 1$$

and write $X_n \xrightarrow{\mathbf{P}\text{-a.s.}} X$ as $n \to \infty$. Note that almost sure convergence implies convergence in probability but not vice versa. The so-called strong law of large numbers says that Theorem 4.22 is still true if one replaces convergence in probability by almost sure convergence in (4.1).

4.5 Central limit theorem

In this section we consider the so-called central limit theorem. Roughly speaking, it says that appropriately transformed sums of i.i.d. random variables behave asymptotically like a standard normal random variable. In order to state this result rigorously, we introduce convergence in distribution, which allows us to study the asymptotic behaviour of probability distributions. In particular, we consider the so-called central limit theorem for sums of i.i.d. random variables.

Recall that we write F_Y for the distribution function of a random variable Y, i.e., $F_Y : \mathbb{R} \ni y \mapsto \mathbf{P}(Y \le y)$. We denote $y \in \mathbb{R}$ as continuity point of F_Y if F_Y is continuous in y.

Definition 4.27 (Convergence in distribution)**.** Let $(X_n)_{n\in\mathbb{N}}$ be a sequence of random variables and let X be a random variable. We say that $(X_n)_{n\in\mathbb{N}}$ converges in distribution to X if

$$\lim_{n\to\infty} F_{X_n}(t) = F_X(t)$$

for all $t \in \mathbb{R}$ such that t is a continuity point of F_X. For convergence in distribution we write $X_n \xrightarrow{d} X$ as $n \to \infty$.

Remark 4.28. When we consider collections of random variables throughout this class, we usually assume that they are all defined on the same probability space. For Definition 4.27 this is not necessary. Here, the random variables may be defined on different probability spaces.

One can show that convergence in probability implies convergence in distribution but not vice versa. While $X_n \xrightarrow{\mathbf{P}} X$ as $n \to \infty$ means that X_n and X are similar with high probability for n sufficiently large, $X_n \xrightarrow{d} X$ as $n \to \infty$ only requires that the distribution functions of X_n and X become similar for n sufficiently large.

Results concerning convergence in distribution are often called limit theorems. We start with a Poisson limit theorem, where we study convergence in distribution to a Poisson distributed random variable.

Example 4.29 (Poisson limit theorem for the binomial distribution)**.** For some $\lambda > 0$ let $X_n \sim \text{Binom}(n, \frac{\lambda}{n})$ for $n \in \mathbb{N}$ such that $\lambda/n \leq 1$ and let $Y \sim \text{Poisson}(\lambda)$. In Example 3.8, it is shown that

$$\lim_{n\to\infty} \mathbf{P}(X_n = k) = \mathbf{P}(Y = k)$$

for all $k \in \mathbb{N}_0$. This implies that, for $t \in \mathbb{R}$,

$$\begin{aligned}\lim_{n\to\infty} F_{X_n}(t) &= \lim_{n\to\infty} \mathbf{P}(X_n \leq t) = \lim_{n\to\infty} \sum_{k\in\mathbb{N}_0:k\leq t} \mathbf{P}(X_n = k) = \sum_{k\in\mathbb{N}_0:k\leq t} \mathbf{P}(Y = k) \\ &= \mathbf{P}(Y \leq t) = F_Y(t),\end{aligned}$$

whence

$$X_n \xrightarrow{d} Y \quad \text{as} \quad n \to \infty.$$

The most prominent limit theorem is the central limit theorem, where one considers sums of i.i.d. random variables and convergence in distribution to a standard normal random variable. Recall that we denote the distribution function of the standard normal distribution by Φ.

Theorem 4.30 (Central limit theorem)**.** Let $(X_i)_{i\in\mathbb{N}}$ be a sequence of i.i.d. random variables such that $\mathbf{E}[X_1^2] < \infty$ and $\mathbf{Var}[X_1] > 0$, define $S_n = \sum_{i=1}^n X_i$ for $n \in \mathbb{N}$ and let $N \sim \text{Normal}(0, 1)$. Then,

$$\lim_{n\to\infty} \mathbf{P}\left(\frac{S_n - n\mathbf{E}[X_1]}{\sqrt{\mathbf{Var}[X_1]}\sqrt{n}} \leq t\right) = \Phi(t)$$

for $t \in \mathbb{R}$ and

$$\frac{S_n - n\mathbf{E}[X_1]}{\sqrt{\mathbf{Var}[X_1]}\sqrt{n}} \xrightarrow{d} N \quad \text{as} \quad n \to \infty.$$

We will not provide a proof of the central limit theorem in this class.

Remark 4.31. Let S_n be as in Theorem 4.30.

(i) Note that

$$\mathbf{E}[S_n] = \mathbf{E}\left[\sum_{i=1}^n X_i\right] = \sum_{i=1}^n \mathbf{E}[X_i] = n\mathbf{E}[X_1]$$

and that, by Theorem 3.62 and Remark 3.63,

$$\mathbf{Var}[S_n] = \mathbf{Var}\left[\sum_{i=1}^n X_i\right] = \sum_{i=1}^n \mathbf{Var}[X_i] = n\mathbf{Var}[X_1].$$

Thus, we study in Theorem 4.30 the standardised random variables $\frac{S_n - \mathbf{E}[S_n]}{\sqrt{\mathbf{Var}[S_n]}}$ for $n \in \mathbb{N}$, which have expectation 0 and variance 1 like a standard normal random variable.

(ii) In the central limit theorem we consider the random variables

$$\frac{S_n - n\mathbf{E}[X_1]}{\sqrt{\mathbf{Var}[X_1]}\sqrt{n}} = \frac{\sqrt{n}}{\sqrt{\mathbf{Var}[X_1]}}\left(\frac{S_n}{n} - \mathbf{E}[X_1]\right)$$

for $n \in \mathbb{N}$. It follows from the weak law of large numbers (see Theorem 4.22) that the differences in the parentheses on the right-hand side converge in probability to zero as $n \to \infty$. The central limit theorem tells us that if we rescale them by $\sqrt{n}/\sqrt{\mathbf{Var}[X_1]}$, we have convergence in distribution to a standard normal random variable as $n \to \infty$. This way the central limit theorem describes the fluctuations around zero.

The central limit theorem explains us why the normal distribution plays an important role in probability theory and often arises in applications. Whenever one has a properly transformed sum of i.i.d. random variables satisfying mild assumptions, this sum can be approximated by a normal random variable.

The central limit theorem can be applied to binomially distributed random variables in the following way.

Theorem 4.32 (De Moivre-Laplace theorem)**.** Let $p \in (0,1)$ be fixed and let $Y_n \sim \mathrm{Binom}(n,p)$ for $n \in \mathbb{N}$. Then,

$$\lim_{n\to\infty} \mathbf{P}\left(\frac{Y_n - np}{\sqrt{np(1-p)}} \le t\right) = \Phi(t)$$

for $t \in \mathbb{R}$.

Proof. Let $(X_i)_{i\in\mathbb{N}}$ be a sequence of independent $\mathrm{Ber}(p)$-distributed random variables. Since Y_n has the same distribution as $\sum_{i=1}^n X_i$ for $n \in \mathbb{N}$ as well as $\mathbf{E}[X_1] = p$ and $\mathbf{Var}[X_1] = p(1-p)$, we obtain

$$\lim_{n\to\infty} \mathbf{P}\left(\frac{Y_n - np}{\sqrt{np(1-p)}} \le t\right) = \lim_{n\to\infty} \mathbf{P}\left(\frac{\sum_{i=1}^n X_i - n\mathbf{E}[X_1]}{\sqrt{n}\sqrt{\mathbf{Var}[X_1]}} \le t\right) = \Phi(t)$$

for $t \in \mathbb{R}$, where we used Theorem 4.30 in the last step. □

Let Y_n be as in Theorem 4.32. Motivated by the previous theorem, one uses the approximation

$$\mathbf{P}(Y_n \le u) = \mathbf{P}\left(\frac{Y_n - np}{\sqrt{np(1-p)}} \le \frac{u - np}{\sqrt{np(1-p)}}\right) \approx \Phi\left(\frac{u - np}{\sqrt{np(1-p)}}\right) \tag{4.6}$$

for $u \in \mathbb{R}$ if n is sufficiently large depending on p. For small values of np it is better to approximate Y_n by a Poisson distributed random variable with parameter np (see Examples 3.8 and 4.29). We consider the following example for the application of (4.6).

Example 4.33 (Normal approximation for coin tosses)**.** A fair coin is tossed 100 times. What is the probability that it shows head more than 55 times?

Let H be the number of heads. Note that $H \sim \mathrm{Binom}(100, 0.5)$. It follows from (4.6) that

$$\begin{aligned}\mathbf{P}(H > 55) = 1 - \mathbf{P}(H \le 55) &\approx 1 - \Phi\left(\frac{55 - 100\cdot 0.5}{\sqrt{100}\sqrt{0.5\cdot 0.5}}\right) = 1 - \Phi\left(\frac{5}{5}\right) = 1 - \Phi(1)\\ &\approx 1 - 0.8413 \approx 0.1587.\end{aligned}$$

A calculation using the probabilities of the binomial distribution yields that $\mathbf{P}(H > 55) \approx 0.1356$. This means that (4.6) is here a reasonable approximation.

Chapter 5

Stochastic processes

In many applications one is interested in random quantities that evolve over time, for example, an experiment, where one takes some measurements over time. Such situations can be modelled by stochastic processes.

Definition 5.1 (Stochastic process)**.** Let $T \subseteq \mathbb{R}$ and $E \subseteq \mathbb{R}$. A stochastic process is a collection of random variables $(X_t)_{t \in T}$ with values in E. We denote T as index set and E as state space.

We can think of the t from the index set T as time. A stochastic process $(X_t)_{t \in T}$ provides a value X_t for any time $t \in T$ and describes the evolution of some quantity over time. For each elementary event ω of our underlying probability space we have a map $T \to E, t \mapsto X_t(\omega)$, which we call a path. While a random variable gives us only a single real number for each elementary event, a stochastic process provides with the path a whole function for each elementary event. Thus, we can interpret a stochastic process as a random function.

If T is a discrete subset of $\mathbb{R}$, usually $T = \mathbb{N}_0$ or $T = \mathbb{Z}$, we say that $(X_t)_{t \in T}$ is a discrete-time stochastic process. In case that T is some interval of $\mathbb{R}$, for example, $T = \mathbb{R}$, $T = [0, \infty)$ or $T = [0, 1]$, we denote $(X_t)_{t \in T}$ as continuous-time stochastic process.

A different classification of stochastic processes is based on their state space E. We call stochastic processes with discrete state spaces discrete-valued and, in particular, with $E = \mathbb{Z}$ integer-valued. A stochastic process with $E = \mathbb{R}$ is denoted as real-valued.

Example 5.2 (Random walk)**.** Let $(X_i)_{i \in \mathbb{N}}$ be a collection of i.i.d. random variables. We define

$$S_0 = 0 \quad \text{and} \quad S_t = \sum_{i=1}^{t} X_i \quad \text{for} \quad t \in \mathbb{N}.$$

Then, $(S_t)_{t \in \mathbb{N}_0}$ is a discrete-time stochastic process and is called a random walk. The idea of this stochastic process is that we have a particle on the real line that starts at time zero in the origin and moves by X_i at time i. If $\mathbf{P}(X_1 = 1) = \mathbf{P}(X_1 = -1) = \frac{1}{2}$, we call $(S_t)_{t \in \mathbb{N}_0}$ a simple random walk. A simple random walk is an integer-valued stochastic process, i.e.,

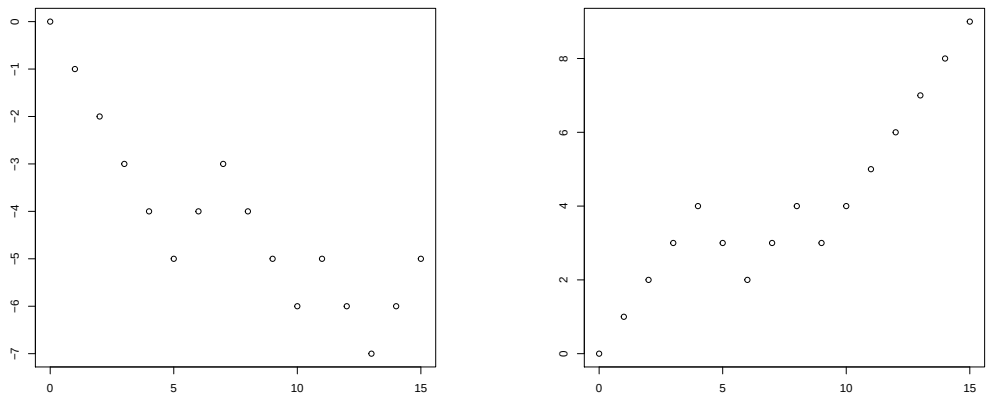


Figure 5.1: Two paths of a simple random walk

our particle moves around the integers. At each time t it flips a fair coin whether it goes one step to the left or one step to the right.

We say that a stochastic process $(X_t)_{t\in T}$ has independent increments if, for all $t_0, \ldots, t_n \in T$ with $t_0 < t_1 < \ldots < t_n$ and $n \in \mathbb{N}$,

$$X_{t_1} - X_{t_0}, X_{t_2} - X_{t_1}, \ldots, X_{t_n} - X_{t_{n-1}}$$

are independent. We denote the differences $X_{t_1} - X_{t_0}, X_{t_2} - X_{t_1}, \ldots, X_{t_n} - X_{t_{n-1}}$ as increments. One of the most important stochastic processes is the so-called Brownian motion.

Definition 5.3 (Brownian motion)**.** A stochastic process $(B_t)_{t\geq 0}$ is called a Brownian motion if

(i) $\mathbf{P}(B_0 = 0) = 1$

(ii) $\mathbf{P}([0,\infty) \ni t \mapsto B_t \text{ is continuous}) = 1$

(iii) $(B_t)_{t\geq 0}$ has independent increments.

(iv) For $s, t \geq 0$ with $s \leq t$, one has $B_t - B_s \sim \text{Normal}(0, t-s)$.

The Brownian motion is also called Wiener process after the mathematician Norbert Wiener. The name Brownian motion is due to the botanist Robert Brown, who described the motion of a particle in a liquid, which can be modelled by a Brownian motion. Note that Definition 5.3 does not imply that such a stochastic process exists. Indeed it is a highly non-trivial result beyond the scope of this class that one can construct a probability space with a stochastic process satisfying Definition 5.3.

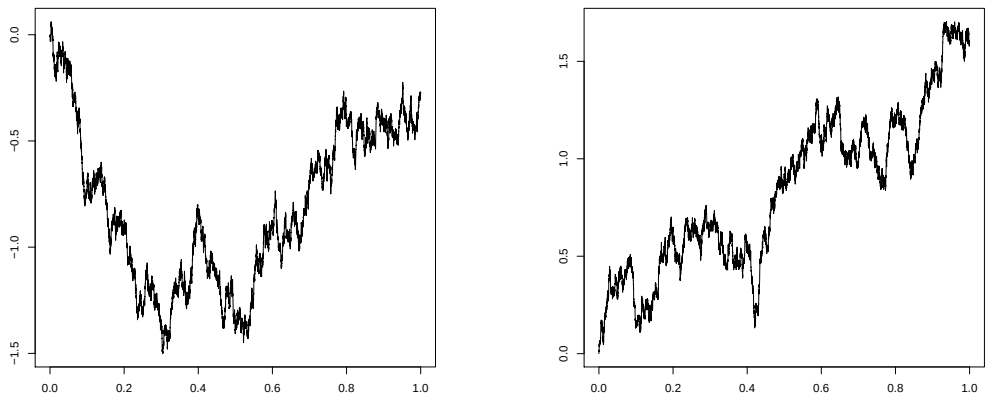


Figure 5.2: Two paths of a Brownian motion

Remark 5.4. Let $(B_t)_{t\geq 0}$ be a Brownian motion.

(i) By the probabilities in (i) and (ii) of Definition 5.3 we mean that

$$\mathbf{P}(\{\omega \in \Omega : B_0(\omega) = 0\}) = 1$$

and

$$\mathbf{P}(\{\omega \in \Omega : [0,\infty) \ni t \mapsto B_t(\omega) \text{ is continuous}\}) = 1,$$

respectively. Thus, we require that the set of all $\omega \in \Omega$ with $B_0(\omega) = 0$ has probability one and that the set of all $\omega \in \Omega$ such that the path $[0,\infty) \ni t \mapsto B_t(\omega)$ is continuous has probability one. We say that an event occurs $\mathbf{P}$-almost surely if it has probability one. This way we can reformulate the assumptions (i) and (ii) of Definition 5.3 as that $B_0 = 0$ $\mathbf{P}$-almost surely and that $[0,\infty) \ni t \mapsto B_t$ is continuous $\mathbf{P}$-almost surely.

(ii) For $t \geq 0$, we have $B_t \sim \text{Normal}(0,t)$. Indeed, it follows from (i) in Definition 5.3 that B_t has the same distribution as $B_t - B_0$, which is $\text{Normal}(0,t)$-distributed by (iv) in Definition 5.3. Since the distribution of the increment $B_t - B_s$ for $s,t \geq 0$ with $s \leq t$ depends only on the difference $t - s$, we say that the Brownian motion has stationary increments.

Example 5.5. Let $(B_t)_{t\geq 0}$ be a Brownian motion and let $\mu \in \mathbb{R}$ and $\sigma \in (0,\infty)$. Then, the stochastic process $(S_t)_{t\geq 0}$ with

$$S_t = e^{\mu t + \sigma B_t} \quad \text{for} \quad t \geq 0$$

is called a geometric Brownian motion with drift μ and volatility σ. The geometric Brownian motion is a common model for stock prices.

In order to describe a stochastic process, one often studies its mean and covariance functions. Note that one denotes the expectation of a random variable also as its mean.

Definition 5.6 (Mean function and covariance function)**.** Let $(X_t)_{t\in T}$ be a stochastic process. If $\mathbf{E}[|X_t|] < \infty$ for all $t \in T$, the mean function of $(X_t)_{t\in T}$ is given by

$$\mu(t) = \mathbf{E}[X_t]$$

for $t \in T$. If $\mathbf{E}[X_t^2] < \infty$ for all $t \in T$, the covariance function of $(X_t)_{t\in T}$ is defined as

$$\varrho(s,t) = \mathbf{Cov}(X_s, X_t)$$

for $s, t \in T$.

As example we consider the mean function and the covariance function of the Brownian motion.

Example 5.7. Let $(B_t)_{t\geq 0}$ be a Brownian motion. Since $B_t \sim \text{Normal}(0,t)$ for $t \geq 0$ (see Remark 5.4 (ii)) we obtain that

$$\mu(t) = \mathbf{E}[B_t] = 0 \quad \text{and} \quad \mathbf{Var}[B_t] = t.$$

For $u, v \geq 0$ with $u \leq v$ we have by (i) in Definition 5.3 that the joint distribution of B_u and B_v equals the joint distribution of $B_u - B_0$ and $B_v - B_0$. This implies that

$$\begin{aligned}\mathbf{Cov}(B_u, B_v) &= \mathbf{Cov}(B_u - B_0, B_v - B_0)\\ &= \mathbf{Cov}(B_u - B_0, B_v - B_u + B_u - B_0)\\ &= \mathbf{Cov}(B_u - B_0, B_v - B_u) + \mathbf{Cov}(B_u - B_0, B_u - B_0).\end{aligned}$$

Since $(B_t)_{t\geq 0}$ has independent increments (see (iii) in Definition 5.3), $B_u - B_0$ and $B_v - B_u$ are independent and $\mathbf{Cov}(B_u - B_0, B_v - B_u) = 0$. Together with $B_u - B_0 \sim \text{Normal}(0,u)$ (see (iv) in Definition 5.3) we see that

$$\mathbf{Cov}(B_u - B_0, B_u - B_0) = \mathbf{Var}[B_u - B_0] = u.$$

Thus, we have shown that the Brownian motion $(B_t)_{t\geq 0}$ has the covariance functions

$$\varrho(s,t) = \mathbf{Cov}(B_s, B_t) = \min\{s,t\}$$

for $s, t \geq 0$.

A further important stochastic process is the so-called Poisson process.

Definition 5.8 (Poisson process)**.** Let $\lambda > 0$ and let $(W_i)_{i\in\mathbb{N}}$ be i.i.d. $\text{Exp}(\lambda)$-distributed random variables. Define

$$X_n = \sum_{i=1}^{n} W_i \quad \text{for} \quad n \in \mathbb{N} \quad \text{and} \quad N_t = \sum_{n=1}^{\infty} \mathbf{1}\{X_n \leq t\} \quad \text{for} \quad t \geq 0.$$

The stochastic process $(N_t)_{t\geq 0}$ is called Poisson process with intensity λ.

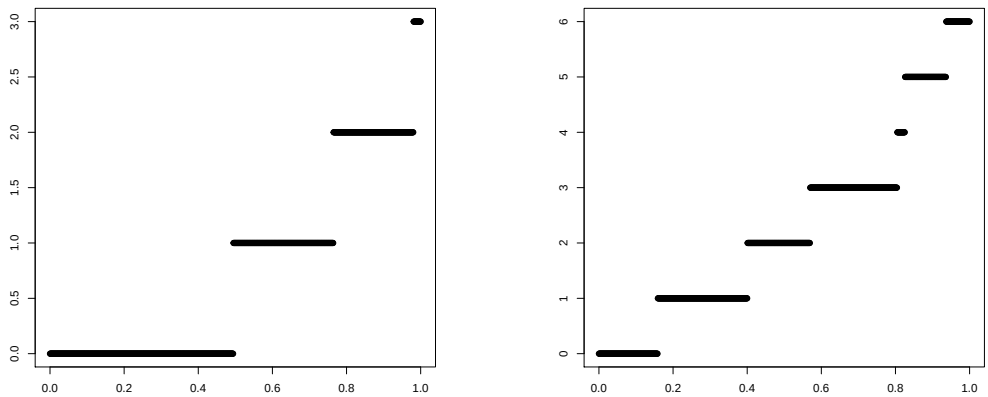


Figure 5.3: Two paths of a Poisson process with intensity 5

Poisson processes are examples for continuous-time integer-valued stochastic processes. In Definition 5.8, we first construct a sequence of random variables $(X_n)_{n\in\mathbb{N}}$. One can think of them as points on $[0,\infty)$, whose distances are independent and $\mathrm{Exp}(\lambda)$-distributed. The Poisson process $(N_t)_{t\geq 0}$ counts the number of points of $(X_n)_{n\in\mathbb{N}}$ in $[0,t]$ for $t \geq 0$. Note that $(X_n)_{n\in\mathbb{N}}$ and $(N_t)_{t\geq 0}$ completely determine each other. Sometimes the collection of points $(X_n)_{n\in\mathbb{N}}$ is also denoted as Poisson process.

Example 5.9 (Geiger counter)**.** A Geiger counter measures radiation by counting particles that hit some tube. The counting process of these particles can be described by a Poisson process.

The following lemma provides a relation between the Poisson process and the Poisson distribution and, thus, explains the name Poisson process.

Lemma 5.10. Let $(N_t)_{t\geq 0}$ be a Poisson process with intensity $\lambda > 0$. Then, for $t \geq 0$, $N_t \sim \mathrm{Poisson}(\lambda t)$.

Proof. For $k \in \mathbb{N}_0$ we have that

$$\mathbf{P}(N_t \leq k) = \mathbf{P}(X_{k+1} > t).$$

Since $X_{k+1} = \sum_{i=1}^{k+1} W_i$, we know from Problem 2 on Exercise Sheet 9 that $X_{k+1} \sim \mathrm{Erl}(k+1,\lambda)$. This Erlang distribution has the density

$$f_{X_{k+1}}(x) = \begin{cases} \frac{\lambda^{k+1}x^k}{k!}e^{-\lambda x}, & x \geq 0, \\ 0, & x < 0. \end{cases}$$

The corresponding distribution function is

$$F_{X_{k+1}}(x) = \begin{cases} 1 - \sum_{j=0}^{k} \frac{(\lambda x)^j}{j!} e^{-\lambda x}, & x \geq 0, \\ 0, & x < 0, \end{cases}$$

because this function is a distribution function and for $x > 0$ we obtain

$$\begin{aligned} F'_{X_{k+1}}(x) &= \sum_{j=0}^{k} \frac{\lambda(\lambda x)^j}{j!} e^{-\lambda x} - \sum_{j=1}^{k} \frac{\lambda(\lambda x)^{j-1}}{(j-1)!} e^{-\lambda x} \\ &= \sum_{j=0}^{k} \frac{\lambda(\lambda x)^j}{j!} e^{-\lambda x} - \sum_{j=0}^{k-1} \frac{\lambda(\lambda x)^j}{j!} e^{-\lambda x} = \frac{\lambda(\lambda x)^k}{k!} e^{-\lambda x}. \end{aligned}$$

This implies that

$$\mathbf{P}(N_t \leq k) = \mathbf{P}(X_{k+1} > t) = 1 - F_{X_{k+1}}(t) = \sum_{j=0}^{k} \frac{(\lambda t)^j}{j!} e^{-\lambda t}$$

for $k \in \mathbb{N}_0$, which completes the proof. □

Remark 5.11. Moreover, one can show that a Poisson process $(N_t)_{t \geq 0}$ of intensity λ has independent increments and that $N_t - N_s \sim \text{Poisson}(\lambda(t-s))$ for $0 \leq s \leq t$.

An often studied class of stochastic processes are so-called stationary processes.

Definition 5.12 (Stationarity of a stochastic process)**.** A stochastic process $(X_t)_{t \in T}$ is called stationary if $T = \mathbb{Z}$ or $T = \mathbb{R}$ and for all $s \in T$, $(X_{t+s})_{t \in T}$ has the same distribution as $(X_t)_{t \in T}$.

This means that a stochastic process $(X_t)_{t \in T}$ is stationary if for any $s \in T$ the shifted stochastic process $(X_{t+s})_{t \in T}$, where t is mapped to X_{t+s} instead of X_t, looks like the original stochastic process $(X_t)_{t \in T}$.

Example 5.13. A collection of i.i.d. random variables $(X_t)_{t \in \mathbb{Z}}$ forms a stationary stochastic process.

A stochastic process is memoryless if it satisfies the so-called Markov property.

Definition 5.14 (Markov property)**.** Let $(X_t)_{t \in T}$ be a stochastic process with an at most countable state space E. We say that $(X_t)_{t \in T}$ has the Markov property if

$$\mathbf{P}(X_t = i | X_{t_n} = i_n, \ldots, X_{t_1} = i_1) = \mathbf{P}(X_t = i | X_{t_n} = i_n)$$

for all $i_1, \ldots, i_n, i \in E$, $t_1, \ldots, t_n, t \in T$ with $t_1 < \ldots < t_n < t$ and $n \in \mathbb{N}$ such that

$$\mathbf{P}(X_{t_n} = i_n, \ldots, X_{t_1} = i_1) > 0.$$

We call $(X_t)_{t \in T}$ a Markov chain if it satisfies the Markov property.

The Markov property means that a future value X_t only depends on the last known value X_{t_n}. All knowledge about further previous values is irrelevant. The simple random walk in Example 5.2 and Poisson processes are examples for stochastic processes with the Markov property. For the simple random walk this follows from the fact that the value at time t is the value at time t_n plus the steps between the times t_n and t. In case of a Poisson process the Markov property follows from the memorylessness of the exponentially distributed random variables in its construction. The random walk is a discrete-time Markov chain, while the Poisson process is a continuous-time Markov chain.

In order to describe discrete-time Markov chains, we introduce their transition probabilities.

Definition 5.15 (Transition probabilities of a discrete-time Markov chain)**.** Let $(X_t)_{t\in\mathbb{N}_0}$ be a discrete-time Markov chain with an at most countable state space E. We call real numbers $(p_{ij})_{i,j\in E}$ transition probabilities of $(X_t)_{t\in\mathbb{N}_0}$ if

$$\mathbf{P}(X_{t+1} = j | X_t = i) = p_{ij} \tag{5.1}$$

for all $t \in \mathbb{N}_0$ and $i, j \in E$ such that $\mathbf{P}(X_t = i) > 0$.

This means that the transition probability p_{ij} is the probability that the Markov chain is in state j in the next step given that it is in state i at the moment.

Remark 5.16. Note that the right-hand side of (5.1) does not depend on t. This situation is called time-homogeneous. In general, one can also have a dependence of the transition probabilities on t, but we will only deal with the time-homogeneous case.

Example 5.17 (Transition probabilities of a simple random walk)**.** A simple random walk $(S_t)_{t\in\mathbb{N}_0}$ is a discrete-time Markov chain with the transition probabilities

$$p_{ij} = \begin{cases} \frac{1}{2}, & j \in \{i-1, i+1\}, \\ 0, & j \notin \{i-1, i+1\}, \end{cases}$$

for $i, j \in \mathbb{Z}$.

The following lemma allows us to compute the distribution of X_{t+1} recursively from that of X_t for $t \in \mathbb{N}_0$.

Lemma 5.18. Let $(X_t)_{t\in\mathbb{N}_0}$ be a discrete-time Markov chain with an at most countable state space E and transition probabilities $(p_{ij})_{i,j\in E}$. Then, for $t \in \mathbb{N}_0$ and $j \in E$,

$$\mathbf{P}(X_{t+1} = j) = \sum_{i\in E} p_{ij}\mathbf{P}(X_t = i).$$

Proof. By the law of total probability and the definition of the transition probabilities (see Definition 5.15) we have

$$\mathbf{P}(X_{t+1} = j) = \sum_{i\in E, \mathbf{P}(X_t=i)>0} \mathbf{P}(X_t = i)\mathbf{P}(X_{t+1} = j | X_t = i) = \sum_{i\in E, \mathbf{P}(X_t=i)>0} p_{ij}\mathbf{P}(X_t = i).$$

Obviously we can drop the assumption $\mathbf{P}(X_t = i) > 0$ for the summation since we have the contribution zero otherwise. This leads to

$$\mathbf{P}(X_{t+1} = j) = \sum_{i \in E} p_{ij} \mathbf{P}(X_t = i)$$

and completes the proof. □

Corollary 5.19. Let $(X_t)_{t \in \mathbb{N}_0}$ be a discrete-time Markov chain with state space $E = \{1, \ldots, m\}$ for some $m \in \mathbb{N}$ and transition probabilities $(p_{ij})_{i,j \in \{1,\ldots,m\}}$. Define the transition matrix

$$P = \begin{pmatrix} p_{11} & \cdots & p_{1m} \\ \vdots & \ddots & \vdots \\ p_{m1} & \cdots & p_{mm} \end{pmatrix}$$

and

$$p(t) = \begin{pmatrix} \mathbf{P}(X_t = 1) & \ldots & \mathbf{P}(X_t = m) \end{pmatrix}$$

for $t \in \mathbb{N}_0$. Then, for $t \in \mathbb{N}_0$,

$$p(t+1) = p(t)P \tag{5.2}$$

and

$$p(t) = p(0)P^t. \tag{5.3}$$

Proof. We obtain (5.2) by rewriting Lemma 5.18. Iterating (5.2) leads to (5.3). □

A time-discrete Markov chain $(X_t)_{t \in \mathbb{N}_0}$ is determined by the distribution of X_0 and the transition probabilities. The latter ones are often described by a transition graph. Here, the vertices are the states from the state space and directed edges represent possible transitions, while the weights of the edges provide the transition probabilities.

We have always assumed that the state space of a Markov chain is a subset of $\mathbb{R}$, although we have never used that the values of the Markov chain are real numbers. In fact, we can allow any arbitrary at most countable set as state space. By indexing the states by the integers or a subset of them, we obtain our framework.

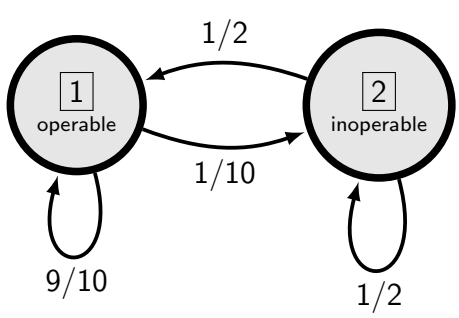


Figure 5.4: Transition graph of the Markov chain from Example 5.20

Example 5.20. A machine is either operable or inoperable and its daily status can be described by a Markov chain. If it is operable, it gets inoperable with probability 1/10. The probability that the machine moves from inoperable to operable is 1/2. At time zero the machine is operable. In the following we abbreviate operable by 1 and inoperable by 2. The Markov chain $(X_t)_{t\in\mathbb{N}_0}$ that describes the status of the machine has the transition matrix

$$P = \begin{pmatrix} \frac{9}{10} & \frac{1}{10} \\ \frac{1}{2} & \frac{1}{2} \end{pmatrix}$$

and the probabilies $p(0) = \begin{pmatrix} 1 & 0 \end{pmatrix}$ for X_0. For the probabilities

$$p(t) = \begin{pmatrix} \mathbf{P}(X_t = 1) & \mathbf{P}(X_t = 2) \end{pmatrix}$$

of X_t with $t \in \mathbb{N}_0$, we obtain from Corollary 5.19

$$p(t) = p(0)P^t = \begin{pmatrix} 1 & 0 \end{pmatrix} \cdot \begin{pmatrix} \frac{9}{10} & \frac{1}{10} \\ \frac{1}{2} & \frac{1}{2} \end{pmatrix}^t .$$

In particular, we have

$$p(1) = \begin{pmatrix} 1 & 0 \end{pmatrix} \cdot \begin{pmatrix} \frac{9}{10} & \frac{1}{10} \\ \frac{1}{2} & \frac{1}{2} \end{pmatrix} = \begin{pmatrix} \frac{9}{10} & \frac{1}{10} \end{pmatrix}$$

and

$$p(2) = \begin{pmatrix} 1 & 0 \end{pmatrix} \cdot \begin{pmatrix} \frac{9}{10} & \frac{1}{10} \\ \frac{1}{2} & \frac{1}{2} \end{pmatrix}^2 = \begin{pmatrix} \frac{9}{10} & \frac{1}{10} \end{pmatrix} \cdot \begin{pmatrix} \frac{9}{10} & \frac{1}{10} \\ \frac{1}{2} & \frac{1}{2} \end{pmatrix} = \begin{pmatrix} \frac{43}{50} & \frac{7}{50} \end{pmatrix} .$$

Chapter 6

Statistics

6.1 Basic notions of statistics

In statistics, usually some observations $x_1, \dots, x_n \in \mathbb{R}$ are given. We call $x = (x_1, \dots, x_n)$ a sample and denote n as sample size. In descriptive statistics, one summarises and visualises the underlying data. Important characteristics to describe our sample $x = (x_1, \dots, x_n)$ are the following.

Definition 6.1. For a sample $x = (x_1, \dots, x_n) \in \mathbb{R}^n$, the sample mean $\overline{x}$ is defined as

$$\overline{x} = \frac{1}{n} \sum_{i=1}^{n} x_i$$

and the sample variance s_x^2 is given by

$$s_x^2 = \frac{1}{n-1} \sum_{i=1}^{n} (x_i - \overline{x})^2$$

if $n \geq 2$. For $i \in \{1, \dots, n\}$ let $x_{(i)}$ be the i-th smallest value of the sample x. The sample median m_x of x is defined as

$$m_x = \begin{cases} x_{((n+1)/2)}, & n \text{ is odd}, \\ x_{(n/2)}/2 + x_{(n/2+1)}/2, & n \text{ is even}. \end{cases}$$

The sample mean is the average of the sample, while the median is the middle number of the sorted sample. The sample variance measures the deviations from the sample mean.

Remark 6.2. The sample mean and the sample variance can be seen as counterparts to the expectation and the variance of a random variable. Why one divides by $n-1$ and not by n in the definition of the sample variance will become clear in Theorem 6.9.

For a random variable X we call $m \in \mathbb{R}$ a median if one has that $\mathbf{P}(X \geq m) \geq \frac{1}{2}$ and $\mathbf{P}(X \leq m) \geq \frac{1}{2}$. Note that a random variable can have more than one median. The

sample median can be seen as a counterpart to the median of a random variable X. The sample median is more robust compared with the sample mean. So it can happen that the sample mean significantly changes if one value of the sample is replaced by a very large value, while the sample median is usually not much affected by this.

In inductive statistics one assumes that the sample is drawn from some underlying population and one wants to infer from the sample to the population, which is described by a probability distribution. In order to formalise this, we introduce statistical models.

Definition 6.3 (Statistical model)**.** A statistical model $(\mathcal{X}, \mathscr{F}_{\mathcal{X}}, (\mathbf{P}_\vartheta)_{\vartheta\in\Theta})$ consists of a sample space $\mathcal{X}$, a σ-algebra $\mathscr{F}_{\mathcal{X}}$ on $\mathcal{X}$ and a family of probability measures $(\mathbf{P}_\vartheta)_{\vartheta\in\Theta}$ on $\mathcal{X}$ with an index set Θ.

The sample space $\mathcal{X}$ is the set of all possible values of our sample $x = (x_1, \dots, x_n)$, i.e., it is a subset of $\mathbb{R}^n$ if we have the sample size n. For each $\vartheta \in \Theta$, $(\mathcal{X}, \mathscr{F}_{\mathcal{X}}, \mathbf{P}_\vartheta)$ is a probability space. Thus, a statistical model does not provide a single probability space but a whole family of probability spaces. Often $(\mathbf{P}_\vartheta)_{\vartheta\in\Theta}$ is a parametric family of distributions where the index set Θ is the parameter space. Our statistical model assumes that $x_1, \dots, x_n$ are realisations of random variables $X_1, \dots, X_n$ with a joint distribution $\mathbf{P}_\vartheta$ for some 'true' $\vartheta \in \Theta$, which is unknown. Our goal is to deduce from the observed sample $x = (x_1, \dots, x_n)$ information about the underlying distribution $\mathbf{P}_\vartheta$.

Since we work with families of probability distributions $(\mathbf{P}_\vartheta)_{\vartheta\in\Theta}$ throughout this chapter, we denote by $\mathbf{E}_\vartheta$ the expectation with respect to $\mathbf{P}_\vartheta$ for $\vartheta \in \Theta$.

Example 6.4. Assume that we observe n repetitions of a random experiment with the possible outcomes success and failure (abbreviated by 1 and 0) and unknown success probability p. This situation is described by a statistical model $(\{0,1\}^n, \mathcal{P}(\{0,1\}^n), (\mathbf{P}_p)_{p\in[0,1]})$, where $\mathbf{P}_p$ is the joint distribution of n independent $\mathrm{Ber}(p)$-distributed random variables $X_1, \dots, X_n$ for $p \in [0,1]$.

Remark 6.5. As in Example 6.4 we often assume that our samples are realisations of i.i.d. random variables, whence $(\mathbf{P}_\vartheta)_{\vartheta\in\Theta}$ are usually families of product measures.

In the following, we study three different types of statistical procedures. Each time one assumes that the sample x is from an underlying probability distribution $\mathbf{P}_\vartheta$ for some unknown parameter $\vartheta \in \Theta$ as described by the underlying statistical model.

- *Point estimation.* One guesses from the sample the parameter ϑ or a quantity $\tau(\vartheta)$ of ϑ.
- *Confidence region.* Instead of a single value for ϑ or $\tau(\vartheta)$, one provides a whole set that is supposed to contain ϑ or $\tau(\vartheta)$ with some certainty. Since in this class the sets will always be intervals, we denote this approach as confidence intervals.
- *Statistical test.* Let $\Theta_0, \Theta_1 \subset \Theta$ be such that $\Theta_0 \cap \Theta_1 = \emptyset$ and $\Theta_0 \cup \Theta_1 = \Theta$. Based on the sample one has to decide with some certainty whether ϑ belongs to Θ_0 or to Θ_1.

6.2 Point estimation

Throughout this section we denote by $(\mathcal{X}, \mathcal{F}_{\mathcal{X}}, (\mathbf{P}_\vartheta)_{\vartheta\in\Theta})$ the underlying statistical model and let $\tau : \Theta \to \mathbb{R}$ if not stated otherwise.

In the following we study so-called point estimators for the quantity $\tau(\vartheta)$ of ϑ. Sometimes one has $\Theta \subseteq \mathbb{R}$ and $\tau(\vartheta) = \vartheta$ so that one considers ϑ itself. For a given sample $x \in \mathcal{X}$ of an underlying distribution $\mathbf{P}_\vartheta$ for some unknown $\vartheta \in \Theta$, a point estimator provides just a single value, which is a guess for $\tau(\vartheta)$. We start with the following definition.

Definition 6.6 (Estimator)**.** A measurable map $T : \mathcal{X} \to \mathbb{R}$ is called a (point) estimator.

Remark 6.7. Throughout this class we only consider estimators with values in $\mathbb{R}$, but it is possible to replace $\mathbb{R}$ by more general spaces. In the sequel we always say only estimator and not point estimator. The term point estimation is used since finding confidence regions is sometimes also called estimation.

For a given sample $x \in \mathcal{X}$ we obtain a real number $T(x)$. In the following we think of an estimator T as a random variable on the probability space $(\mathcal{X}, \mathscr{F}_{\mathcal{X}}, \mathbf{P}_\vartheta)$ for each $\vartheta \in \Theta$.

Examples for estimators are the sample mean, the sample variance or the sample median, but any constant function is also an estimator. Note that Definition 6.6 is a formal definition which does not say when T is a good estimator for $\tau(\vartheta)$. Since T is a random variable, it will usually not equal $\tau(\vartheta)$. In the following we provide some criteria when T is a good estimator for $\tau(\vartheta)$.

Definition 6.8 (Unbiased estimator and bias of an estimator)**.** Let T be an estimator such that $\mathbf{E}_\vartheta[|T|] < \infty$ for all $\vartheta \in \Theta$. We call T an unbiased estimator for $\tau(\vartheta)$ if

$$\mathbf{E}_\vartheta[T] = \tau(\vartheta) \quad \text{for all} \quad \vartheta \in \Theta.$$

Otherwise, we denote $B_T(\vartheta) = \mathbf{E}_\vartheta[T] - \tau(\vartheta)$ as bias of T at ϑ.

Note that an unbiased estimator avoids systematic errors. The following result tells us that the sample mean and the sample variance are unbiased estimators for the expectation and the variance under some mild assumptions. By $\mathbf{Var}_\vartheta[\cdot]$ we denote the variance with respect to $\mathbf{P}_\vartheta$ for $\vartheta \in \Theta$.

Theorem 6.9. Let $(\mathcal{X}, \mathcal{F}_{\mathcal{X}}, (\mathbf{P}_\vartheta)_{\vartheta\in\Theta})$ be a statistical model such that, for each $\vartheta \in \Theta$, $\mathbf{P}_\vartheta$ is the distribution of $X = (X_1, \ldots, X_n)$, where $X_1, \ldots, X_n$ are i.i.d. random variables.

i) Assume that $\mathbf{E}_\vartheta[|X_1|] < \infty$ for all $\vartheta \in \Theta$. Then, the sample mean $\overline{X}$ is an unbiased estimator for $\mathbf{E}_\vartheta[X_1]$, i.e.,

$$\mathbf{E}_\vartheta[\overline{X}] = \mathbf{E}_\vartheta[X_1] \quad \text{for all} \quad \vartheta \in \Theta.$$

ii) Assume that $\mathbf{E}_\vartheta[X_1^2] < \infty$ for all $\vartheta \in \Theta$ and that $n \geq 2$. Then, the sample variance s_X^2 is an unbiased estimator for $\mathbf{Var}_\vartheta[X_1]$, i.e.,

$$\mathbf{E}_\vartheta[s_X^2] = \mathbf{Var}_\vartheta[X_1] \quad \text{for all} \quad \vartheta \in \Theta.$$

Proof. By the linearity of expectation (see Theorem 3.36), we have

$$\mathbf{E}_\vartheta[\overline{X}] = \mathbf{E}_\vartheta\left[\frac{1}{n}\sum_{i=1}^n X_i\right] = \frac{1}{n}\sum_{i=1}^n \mathbf{E}_\vartheta[X_i] = \frac{n\mathbf{E}_\vartheta[X_1]}{n} = \mathbf{E}_\vartheta[X_1],$$

which proves i).

Using again linearity of expectation and the fact that

$$\mathbf{E}_\vartheta[X_i - \overline{X}] = \mathbf{E}_\vartheta[X_i] - \mathbf{E}_\vartheta[\overline{X}] = \mathbf{E}_\vartheta[X_1] - \mathbf{E}_\vartheta[X_1] = 0$$

for $i \in \{1, \dots, n\}$, we derive

$$\mathbf{E}_\vartheta[s_X^2] = \mathbf{E}_\vartheta\left[\frac{1}{n-1}\sum_{i=1}^n (X_i - \overline{X})^2\right] = \frac{1}{n-1}\sum_{i=1}^n \mathbf{E}_\vartheta[(X_i - \overline{X})^2] = \frac{1}{n-1}\sum_{i=1}^n \mathbf{Var}_\vartheta[X_i - \overline{X}].$$

For $i \in \{1, \dots, n\}$, it follows from the independence of $X_1, \dots, X_n$ together with Theorem 3.62 and Remark 3.63 that

$$\begin{aligned}
\mathbf{Var}_\vartheta[X_i - \overline{X}] &= \mathbf{Var}_\vartheta\left[X_i - \frac{1}{n}\sum_{j=1}^n X_j\right] = \mathbf{Var}_\vartheta\left[\left(1 - \frac{1}{n}\right)X_i - \frac{1}{n}\sum_{j=1, j\neq i}^n X_j\right] \\
&= \left(1 - \frac{1}{n}\right)^2 \mathbf{Var}_\vartheta[X_i] + \frac{1}{n^2}\sum_{j=1, i\neq j}^n \mathbf{Var}_\vartheta[X_j] \\
&= \left(\frac{(n-1)^2}{n^2} + \frac{n-1}{n^2}\right)\mathbf{Var}_\vartheta[X_1] = \frac{n-1}{n}\mathbf{Var}_\vartheta[X_1].
\end{aligned}$$

This implies that

$$\mathbf{E}_\vartheta[s_X^2] = \frac{1}{n-1}\sum_{i=1}^n \frac{n-1}{n}\mathbf{Var}_\vartheta[X_1] = \mathbf{Var}_\vartheta[X_1],$$

which completes the proof of ii). □

Note that the sample variance is an unbiased estimator for the variance since we divide only by $n-1$ in its definition (see Definition 6.1), whereas dividing by n leads to an biased estimator. An intuitive explanation for the factor $1/(n-1)$ is that we subtract the sample mean $\overline{X}$ in the sample variance, which is also a random quantity.

Example 6.10. Let the statistical model $(\mathbb{R}^n, \mathcal{B}(\mathbb{R}^n), (\mathbf{P}_\lambda)_{\lambda>0})$ be given, where $\mathbf{P}_\lambda$ is the joint distribution of n independent $\mathrm{Exp}(\lambda)$-distributed random variables $X_1, \dots, X_n$ for $\lambda > 0$. Then, $T : \mathbb{R}^n \ni x \mapsto \overline{x}$ is an unbiased estimator for $\tau(\lambda) = 1/\lambda$ since, by Theorem 6.9 and the fact that an exponential distribution with parameter λ has the expectation $1/\lambda$ (see Example 3.43), for $X = (X_1, \dots, X_n)$ we have

$$\mathbf{E}_\lambda[\overline{X}] = \mathbf{E}_\lambda[X_1] = \frac{1}{\lambda} = \tau(\lambda).$$

An important asymptotic property of estimators is consistency. In the following we denote by $\xrightarrow{\mathbf{P}_\vartheta}$ convergence in probability with respect to $\mathbf{P}_\vartheta$ for $\vartheta \in \Theta$.

Definition 6.11 (Consistency)**.** Let $(\mathcal{X}, \mathscr{F}_\mathcal{X}, (\mathbf{P}_\vartheta)_{\vartheta\in\Theta})$ be a statistical model such that $\mathbf{P}_\vartheta$ is the joint distribution of random variables $(X_i)_{i\in\mathbb{N}}$ for each $\vartheta \in \Theta$. Let $(T_n)_{n\in\mathbb{N}}$ be a sequence of estimators such that T_n only takes the realisations of $X_1, \dots, X_n$ into account. We say that the estimators $(T_n)_{n\in\mathbb{N}}$ are consistent for $\tau(\vartheta)$ if for all $\vartheta \in \Theta$ and $\varepsilon > 0$,

$$\lim_{n\to\infty} \mathbf{P}_\vartheta(|T_n - \tau(\vartheta)| \geq \varepsilon) = 0,$$

i.e., if $T_n \xrightarrow{\mathbf{P}_\vartheta} \tau(\vartheta)$ as $n \to \infty$ for all $\vartheta \in \Theta$.

Since the estimator T_n only depends on a sample of the sample size n, consistency means that we have convergence in probability of the estimators $(T_n)_{n\in\mathbb{N}}$ to $\tau(\vartheta)$ for increasing sample size.

Example 6.12. Let $(\mathcal{X}, \mathscr{F}_\mathcal{X}, (\mathbf{P}_\vartheta)_{\vartheta\in\Theta})$ be a statistical model such that $\mathbf{P}_\vartheta$ is the joint distribution of i.i.d. random variables $(X_i)_{i\in\mathbb{N}}$ with $\mathbf{E}_\vartheta[X_1^2] < \infty$ for each $\vartheta \in \Theta$. For $n \in \mathbb{N}$ let T_n be the sample mean of $X_1, \dots, X_n$. By the weak law of large numbers (see Theorem 4.22) we have that

$$T_n = \frac{1}{n}\sum_{i=1}^n X_i \xrightarrow{\mathbf{P}_\vartheta} \mathbf{E}_\vartheta[X_1] \quad \text{as} \quad n \to \infty$$

for all $\vartheta \in \Theta$, whence the estimators $(T_n)_{n\in\mathbb{N}}$ are consistent for $\mathbf{E}_\vartheta[X_1]$.

Remark 6.13. A further possibility to measure the imprecision of an estimator T for $\tau(\vartheta)$ is its mean squared error

$$\mathbf{E}_\vartheta[(T - \tau(\vartheta))^2]$$

for $\vartheta \in \Theta$. If $\mathbf{E}_\vartheta[T^2] < \infty$ for all $\vartheta \in \Theta$, we have that

$$\mathbf{E}_\vartheta[(T - \tau(\vartheta))^2] = \mathbf{E}_\vartheta[(T - \mathbf{E}_\vartheta[T] + \mathbf{E}_\vartheta[T] - \tau(\vartheta))^2] = \mathbf{Var}_\vartheta[T] + B_T(\vartheta)^2$$

for $\vartheta \in \Theta$, where $B_T(\vartheta)$ is the bias of T at ϑ.

An important technique to construct estimators is the so-called maximum likelihood method, which we introduce here only for discrete statistical models.

Definition 6.14. Let $\mathcal{X}$ be an at most countable set and let $(\mathcal{X}, \mathcal{P}(\mathcal{X}), (\mathbf{P}_\vartheta)_{\vartheta\in\Theta})$ be a statistical model with $\Theta \subseteq \mathbb{R}$. We call $T : \mathcal{X} \to \Theta$ a maximum likelihood estimator for ϑ if

$$\mathbf{P}_{T(x)}(\{x\}) = \max_{\vartheta\in\Theta} \mathbf{P}_\vartheta(\{x\})$$

for all $x \in \mathcal{X}$, i.e., $T(x)$ is a maximiser of the function $\Theta \ni \vartheta \mapsto \mathbf{P}_\vartheta(\{x\})$.

Remark 6.15. For a fixed $x \in \mathcal{X}$ we call the function $L_x : \Theta \ni \vartheta \mapsto \mathbf{P}_\vartheta(\{x\})$ likelihood function. Often it is easier to maximise the logarithm of the likelihood function, the so-called log-likelihood function, than the likelihood function itself.

The idea of maximum likelihood estimation is to choose that ϑ from Θ with the highest probability for the observed sample. This explains the name maximum likelihood.

Example 6.16. We consider the statistical model $(\mathbb{N}, \mathcal{P}(\mathbb{N}), (\mathbf{P}_m)_{m\in\mathbb{N}})$, where $\mathbf{P}_m$ is such that $\mathbf{P}_m(\{i\}) = \frac{1}{m}$ for $i \in \{1, \dots, m\}$ for $m \in \mathbb{N}$. For $x \in \mathbb{N}$ and $m \in \mathbb{N}$ we have

$$\mathbf{P}_m(\{x\}) = \begin{cases} \frac{1}{m}, & x \le m, \\ 0, & x > m. \end{cases}$$

This means that $\mathbb{N} \ni m \mapsto \mathbf{P}_m(\{x\})$ is maximised for $m = x$. Thus, $T(x) = x$ is the maximum likelihood estimator for m. Since T is Laplace distributed on $\{1, \dots, m\}$ under $\mathbf{P}_m$, we obtain that $\mathbf{E}_m[T] = \frac{m+1}{2}$. Thus, T has the bias $B_T(m) = \frac{m+1}{2} - m = \frac{1-m}{2}$ at m.

Remark 6.17. So far we often used T as notation for an estimator. In statistics it is quite common to denote an estimator for ϑ by $\hat{\vartheta}$. For example, one can write $\hat{m}$ for the maximum likelihood estimator for m in Example 6.16.

Example 6.18. For $n \in \mathbb{N}$ consider the statistical model $(\{0,1\}^n, \mathcal{P}(\{0,1\}^n), (\mathbf{P}_p)_{p\in[0,1]})$, where $\mathbf{P}_p$ is the joint distribution of n independent $\mathrm{Ber}(p)$-distributed random variables $X_1, \dots, X_n$ for $p \in [0,1]$. For a given sample $x = (x_1, \dots, x_n) \in \{0,1\}^n$ we have that

$$\begin{aligned} \mathbf{P}_p(\{(x_1, \dots, x_n)\}) &= \mathbf{P}_p(X_1 = x_1, \dots, X_n = x_n) = \prod_{i=1}^n \mathbf{P}_p(X_i = x_i) = \prod_{i=1}^n p^{x_i}(1-p)^{1-x_i} \\ &= p^{\sum_{i=1}^n x_i}(1-p)^{n-\sum_{i=1}^n x_i}. \end{aligned} \tag{6.1}$$

Next we maximise the function $[0,1] \ni p \mapsto p^m(1-p)^{n-m}$ for a fixed $m \in \{0, \dots, n\}$ by putting its derivative to zero, i.e.,

$$mp^{m-1}(1-p)^{n-m} - p^m(n-m)(1-p)^{n-m-1} = 0,$$

which is equivalent to

$$m(1-p) - (n-m)p = 0 \quad \Longleftrightarrow \quad m - np = 0 \quad \Longleftrightarrow \quad p = \frac{m}{n}.$$

This implies that $[0,1] \ni p \mapsto \mathbf{P}_p(\{(x_1, \dots, x_n)\})$ has a maximum at $p = \frac{1}{n}\sum_{i=1}^n x_i$. Thus, we see that

$$\hat{p} = \frac{1}{n}\sum_{i=1}^n x_i$$

is the maximum likelihood estimator for p. Note that this is exactly the sample mean $\overline{x}$ of $x = (x_1, \dots, x_n)$. By Theorem 6.9 the sample mean is an unbiased estimator for $\mathbf{E}_p[X_1]$, whence we have $\mathbf{E}_p[\hat{p}] = p$.

In this case, we were able to maximise the function in (6.1) directly. As mentioned in Remark 6.15, the maximisation of the log-likelihood function is often easier. We provide the respective calculations in the current setting. As before, we consider a fixed value $m = \sum_{i=1}^n x_i$ to simplify notation. From (6.1) we deduce that the log-likelihood function is given by

$$\log\big(\mathbf{P}_p(\{(x_1,\dots,x_n)\})\big) = \log\big(p^m(1-p)^{n-m}\big) = m\log(p) + (n-m)\log(1-p),$$

where we used the logarithm rules $\log(ab) = \log(a) + \log(b)$ and $\log(a^c) = c\log(a)$ for $a, b > 0$ and $c \in \mathbb{R}$. Now we maximise this function with respect to p. As before, we consider its derivative and put it to zero, i.e.

$$\frac{m}{p} - \frac{n-m}{1-p} = 0$$

and obtain as above

$$m(1-p) - (n-m)p = 0 \Longleftrightarrow m - np = 0 \Longleftrightarrow p = \frac{m}{n},$$

resulting in the maximum likelihood estimator

$$\hat{p} = \frac{1}{n}\sum_{i=1}^n x_i$$

as before.

The main advantage of considering the log-likelihood function instead of the likelihood function is that products turn into sums, which are easier to differentiate. In this case, it was not much easier because the likelihood function could be simplified to only two factors in (6.1). Quite often this is not possible and one has an expression of the form

$$\prod_{i=1}^n \mathbf{P}_p(X_i = x_i)$$

which cannot be simplified to fewer than n factors. Then, the log-likelihood

$$\sum_{i=1}^n \log\big(\mathbf{P}_p(X_i = x_i)\big)$$

is indeed much easier to differentiate.

6.3 Confidence intervals

In the previous section we studied point estimators $T : \mathcal{X} \to \mathbb{R}$ for a quantity $\tau(\vartheta)$, which for a given sample x provide a guess $T(x)$ for $\tau(\vartheta)$. The disadvantage of this procedure is that in general one has no information about the error $T(x) - \tau(\vartheta)$. In this section,

we derive from the sample a whole set of values that is supposed to contain $\tau(\vartheta)$ with some certainty. Such sets are called confidence regions. In this class, we consider only confidence regions that are intervals, so-called confidence intervals.

First we give a precise definition of a confidence interval. We denote by $\mathcal{I}$ the set of all intervals on $\mathbb{R}$.

Definition 6.19 (Confidence interval)**.** Let $(\mathcal{X}, \mathscr{F}_{\mathcal{X}}, (\mathbf{P}_\vartheta)_{\vartheta\in\Theta})$ be a statistical model and let $\tau : \Theta \to \mathbb{R}$. We call a map $C : \mathcal{X} \to \mathcal{I}$ a confidence interval for $\tau(\vartheta)$ with error level $\alpha \in (0, 1)$ (or confidence level $1 - \alpha$) if

$$\mathbf{P}_\vartheta(\{x \in \mathcal{X} : \tau(\vartheta) \in C(x)\}) \geq 1 - \alpha$$

for all $\vartheta \in \Theta$.

For each $\vartheta \in \Theta$ we can think of C as a random interval on the probability space $(\mathcal{X}, \mathscr{F}_{\mathcal{X}}, \mathbf{P}_\vartheta)$, while $\tau(\vartheta)$ is deterministic. If C is a confidence interval with error level α, for all possible $\vartheta \in \Theta$ the probability that $\tau(\vartheta)$ is contained in C is at least $1 - \alpha$. Here, α can be interpreted as upper bound for the probability of the error that $\tau(\vartheta)$ does not belong to C.

Example 6.20. For $\sigma_0 > 0$ and $n \in \mathbb{N}$ let $(\mathbb{R}^n, \mathcal{B}(\mathbb{R}^n), (\mathbf{P}_\mu)_{\mu\in\mathbb{R}})$ be the statistical model where $\mathbf{P}_\mu$ is the joint distribution of n independent Normal(μ, σ_0^2)-distributed random variables $X_1, \dots, X_n$ and let $X = (X_1, \dots, X_n)$. In the following we construct a confidence interval for μ with error level $\alpha \in (0, 1)$. Let $z_{1-\alpha/2} \in \mathbb{R}$ be such that $\Phi(z_{1-\alpha/2}) = 1-\alpha/2$, where Φ is the distribution function of the standard normal distribution. Define

$$C : \mathbb{R}^n \ni x \mapsto (\overline{x} - z_{1-\alpha/2}\sigma_0/\sqrt{n}, \overline{x} + z_{1-\alpha/2}\sigma_0/\sqrt{n}),$$

where $\overline{x}$ is the sample mean of $x = (x_1, \dots, x_n)$. For any $\mu \in \mathbb{R}$, we have

$$\begin{aligned}
\mathbf{P}_\mu(\{x \in \mathbb{R}^n : \mu \in C(x)\}) &= \mathbf{P}_\mu(\{x \in \mathbb{R}^n : \mu \in (\overline{x} - z_{1-\alpha/2}\sigma_0/\sqrt{n}, \overline{x} + z_{1-\alpha/2}\sigma_0/\sqrt{n})\}) \\
&= \mathbf{P}_\mu(\{x \in \mathbb{R}^n : \overline{x} \in (\mu - z_{1-\alpha/2}\sigma_0/\sqrt{n}, \mu + z_{1-\alpha/2}\sigma_0/\sqrt{n})\}) \\
&= \mathbf{P}_\mu\big(\overline{X} \in (\mu - z_{1-\alpha/2}\sigma_0/\sqrt{n}, \mu + z_{1-\alpha/2}\sigma_0/\sqrt{n})\big).
\end{aligned}$$

Since $X_1, \dots, X_n$ are independent and Normal(μ, σ_0^2)-distributed, we have that $\sum_{i=1}^n X_i \sim$ Normal$(n\mu, n\sigma_0^2)$ (see Example 4.16) and, thus,

$$\overline{X} = \frac{1}{n}\sum_{i=1}^n X_i \sim \text{Normal}(\mu, \sigma_0^2/n).$$

This means that $\frac{\overline{X}-\mu}{\sigma_0/\sqrt{n}}$ has a standard normal distribution under $\mathbf{P}_\mu$, which leads to

$$\begin{aligned}
\mathbf{P}_\mu(\{x \in \mathbb{R}^n : \mu \in C(x)\}) &= \mathbf{P}_\mu\big(\overline{X} - \mu \in (-z_{1-\alpha/2}\sigma_0/\sqrt{n}, z_{1-\alpha/2}\sigma_0/\sqrt{n})\big) \\
&= \mathbf{P}_\mu\left(\frac{\overline{X} - \mu}{\sigma_0/\sqrt{n}} \in (-z_{1-\alpha/2}, z_{1-\alpha/2})\right) \\
&= \mathbf{P}_\mu\left(\frac{\overline{X} - \mu}{\sigma_0/\sqrt{n}} < z_{1-\alpha/2}\right) - \mathbf{P}_\mu\left(\frac{\overline{X} - \mu}{\sigma_0/\sqrt{n}} \leq -z_{1-\alpha/2}\right) \\
&= \Phi(z_{1-\alpha/2}) - \Phi(-z_{1-\alpha/2}).
\end{aligned} \tag{6.2}$$

Since, by symmetry, $\Phi(-u) = 1 - \Phi(u)$ for $u \in \mathbb{R}$, we obtain

$$\begin{aligned}\mathbf{P}_\mu(\{x \in \mathbb{R}^n : \mu \in C(x)\}) &= \Phi(z_{1-\alpha/2}) - (1 - \Phi(z_{1-\alpha/2})) = 2\Phi(z_{1-\alpha/2}) - 1 \\ &= 2\left(1 - \frac{\alpha}{2}\right) - 1 = 1 - \alpha,\end{aligned} \tag{6.3}$$

where we used the definition of $z_{1-\alpha/2}$ in the second last step. Thus, we can conclude that C is a confidence interval for μ with error level α.

The quantity $z_{1-\alpha/2}$ that was used in the previous example is the so-called $(1-\alpha/2)$-quantile of the standard normal distribution. Quantiles can be defined for arbitrary random variables.

Definition 6.21 (Quantile)**.** Let X be a random variable on a probability space $(\Omega, \mathscr{F}, \mathbf{P})$ and let $p \in (0,1)$. A real number $q_p \in \mathbb{R}$ is called a p-quantile of X if

$$\mathbf{P}(X \leq q_p) \geq p \quad \text{and} \quad \mathbf{P}(X \geq q_p) \geq 1 - p. \tag{6.4}$$

Note that the quantiles only depend on the distribution of X, whence we will often refer to the quantiles of a distribution.

Remark 6.22. (i) A median of a random variable (see Remark 6.2) is a 0.5-quantile.

(ii) The p-quantiles do not need to be unique. For small $u \in (0, 0.5)$ one often studies the quantiles q_u and q_{1-u} of X, for which one has $\mathbf{P}(X < q_u) \leq u$ and $\mathbf{P}(X > q_{1-u}) \leq u$. Thus, these quantiles are lower and upper bounds for X that are only exceeded with small probability.

(iii) For a random variable X with an absolutely continuous distribution and $p \in (0,1)$, a p-quantile q_p satisfies by definition

$$1 = p + 1 - p \leq \mathbf{P}(X \leq q_p) + \mathbf{P}(X \geq q_p) = 1 - \mathbf{P}(X = q_p) = 1 - 0 = 1,$$

whence both inequalities in (6.4) must be equalities and q_p is a p-quantile of X if and only if

$$\mathbf{P}(X \leq q_p) = p \quad \text{or} \quad \mathbf{P}(X \geq q_p) = 1 - p.$$

Example 6.23. For each $p \in (0,1)$, a p-quantile q_p of the standard normal distribution is a solution of

$$\Phi(q_p) = p.$$

Since Φ is strictly increasing, this equation has a unique solution. We denote the unique p-quantile of the standard normal distribution by z_p. The following table provides some important values for z_p.

p	0.8	0.9	0.95	0.975	0.99	0.995
z_p	0.84	1.28	1.64	1.96	2.33	2.58

Example 6.24. A quantity that can be measured in an experiment follows a normal distribution with unknown expectation μ and known standard deviation $\sigma = 10$. From 25 independent measurements we obtain the sample mean $\overline{x} = 16$. Our goal is to derive a confidence interval for μ with error level 0.05. The underlying statistical model is exactly as in Example 6.20. Since $z_{1-0.05/2} = z_{0.975} = 1.96$, from Example 6.20 we obtain

$$(16 - 1.96 \cdot 10/\sqrt{25}, 16 + 1.96 \cdot 10/\sqrt{25}) = (12.08, 19.92)$$

as confidence interval for μ with error level 0.05.

6.4 Statistical tests

Let $(\mathcal{X}, \mathscr{F}_{\mathcal{X}}, (\mathbf{P}_\vartheta)_{\vartheta\in\Theta})$ be a statistical model and let $\Theta_0, \Theta_1 \subset \Theta$ be such that $\Theta_0 \cap \Theta_1 = \emptyset$ and $\Theta_0 \cup \Theta_1 = \Theta$ (i.e., Θ_0 and Θ_1 form a partition of Θ). Assume that we have a sample x that is supposed to come from a distribution $\mathbf{P}_\vartheta$ for some unknown $\vartheta \in \Theta$. Our goal is to decide based on the sample x if

$$H_0 : \vartheta \in \Theta_0 \quad \text{or} \quad H_1 : \vartheta \in \Theta_1.$$

We denote H_0 as null hypothesis and H_1 as alternative. So far one could think that the roles of Θ_0 and Θ_1 or H_0 and H_1 are interchangeable, but this will not be the case in the following.

When analysing data one usually wants to verify that there is a certain effect or structure in the sample. In this situation the null hypothesis is that the effect or structure in question is not present, while the alternative describes the situation one wants to verify. This means that the alternative is usually the research hypothesis. One should regard the null hypothesis as the true statement until the data show convincingly the opposite. Using the procedure described in the following, one can disprove the null hypothesis and this way establish the alternative, but one cannot prove the null hypothesis.

Note that the described situation occurs in many research fields ranging from natural sciences and medicine to economics and social sciences.

Example 6.25. In order to test a new drug, a study with n patients was conducted. For each of the patients it is known whether the treatment was successful or not. Since an existing drug has a success rate of 50%, one would like to decide if the new drug is better or not. Here, the statistical model is $(\{0,1\}^n, \mathcal{P}(\{0,1\}^n), (\mathbf{P}_p)_{p\in[0,1]})$, where $\mathbf{P}_p$ is the joint distribution of n independent $\mathrm{Ber}(p)$-distributed random variables $X_1, \ldots, X_n$. Since the aim is to verify that the new drug has a better success rate, one chooses the null hypothesis $H_0 : p \leq 0.5$ and the alternative $H_1 : p > 0.5$.

In order to decide between the null hypothesis and the alternative, we will use a so-called test.

Definition 6.26 (Test)**.** A measurable map $\varphi : \mathcal{X} \to \{0, 1\}$ is called a test.

We can think of a test as a decision rule between the null hypothesis and the alternative. For a sample $x \in \mathcal{X}$ we reject H_0 in favour of H_1 if $\varphi(x) = 1$, while we cannot reject H_0 if $\varphi(x) = 0$.

When applying a statistical test φ for $H_0 : \vartheta \in \Theta_0$ against $H_1 : \vartheta \in \Theta_1$, two kinds of errors can occur:

(i) *Type I error*: The null hypothesis is true, but it is rejected, i.e., $\vartheta \in \Theta_0$ but $\varphi(x) = 1$.

(ii) *Type II error*: The null hypothesis is wrong, but it is not rejected, i.e., $\vartheta \in \Theta_1$ but $\varphi(x) = 0$.

	$\vartheta \in \Theta_0$	$\vartheta \in \Theta_1$
$\varphi(x) = 0$ (i.e., H_0 is not rejected)	correct decision	type II error
$\varphi(x) = 1$ (i.e., H_0 is rejected)	type I error	correct decision

For $\vartheta \in \Theta_0$ the probability of a type I error is

$$\mathbf{P}_\vartheta(\{x \in \mathcal{X} : \varphi(x) = 1\}),$$

while for $\vartheta \in \Theta_1$ a type I error cannot happen. On the other hand, the probability of a type II error is

$$1 - \mathbf{P}_\vartheta(\{x \in \mathcal{X} : \varphi(x) = 1\})$$

for $\vartheta \in \Theta_1$ (for $\vartheta \in \Theta_0$ there is no type II error). The type I error and the type II error are also denoted as α-error and β-error, respectively.

We see that $\mathbf{P}_\vartheta(\{x \in \mathcal{X} : \varphi(x) = 1\})$, which is the probability to reject the null hypothesis, should be close to zero for $\vartheta \in \Theta_0$ and close to one for $\vartheta \in \Theta_1$. In general it is not possible to achieve this simultaneously. Thus, we focus on controlling the probability of a type I error. This means that a type I error is considered more serious than a type II error. The philosophy here is that it is worse to verify something wrong than to fail to discover something true. So in the context of Example 6.25 it might be worse to treat many people with a drug that is weaker than an existing one than not to find a new better drug.

Definition 6.27 (Significance level and power of a test)**.** Let $\varphi : \mathcal{X} \to \{0, 1\}$ be a test for $H_0 : \vartheta \in \Theta_0$ against $H_1 : \vartheta \in \Theta_1$.

(i) We say that φ is a test of (significance) level $\alpha \in (0, 1)$ if

$$\mathbf{P}_\vartheta(\{x \in \mathcal{X} : \varphi(x) = 1\}) \leq \alpha$$

for all $\vartheta \in \Theta_0$.

(ii) For $\vartheta \in \Theta_1$ we call

$$\mathbf{P}_\vartheta(\{x \in \mathcal{X} : \varphi(x) = 1\})$$

the power of φ at ϑ.

In the following we will always provide the level α of a test. For a test φ with a fixed level α, which we also denote as a level-α test, we know that the probability of a type I error is at most α. Moreover, it is desirable that the null hypothesis is rejected for $\vartheta \in \Theta_1$. The probability of this event is given by the power of φ at ϑ.

The level α of a test is typically chosen between 0.01 and 0.1. The quantity $1 - \alpha$ is sometimes called the confidence level of the test. When computing the necessary sample size, one provides an upper bound such as 0.2 for the type II error at a particular $\vartheta \in \Theta_1$ one is interested in.

The trivial test $\varphi = 0$, i.e., we never reject the null hypothesis, has level α for all $\alpha \in (0, 1)$, but it has the power 0 for all $\vartheta \in \Theta_1$.

Let φ be a test of level $\alpha \in (0, 1)$ for $H_0 : \vartheta \in \Theta_0$ against $H_1 : \vartheta \in \Theta_1$ and let $x \in \mathcal{X}$ be a sample. If $\varphi(x) = 1$, we reject the null hypothesis at level α. In this case, we can conclude that the null hypothesis is wrong and that the alternative must be true. If $\varphi(x) = 0$, we cannot reject the null hypothesis at level α. This does not imply that the null hypothesis is true, it only means that there is not sufficient evidence that it is wrong. For example, it can happen that the null hypothesis is rejected at level 0.05, but not at level 0.01. This also shows that it is important to choose the level α in advance.

Usually a test $\varphi : \mathcal{X} \to \{0, 1\}$ is of the form that one has a so-called test statistic $S : \mathcal{X} \to \mathbb{R}$ and a critical region (or rejection region) $\mathcal{K} \subset \mathbb{R}$ and

$$\varphi : \mathcal{X} \ni x \mapsto \begin{cases} 0, & S(x) \notin \mathcal{K}, \\ 1, & S(x) \in \mathcal{K}. \end{cases}$$

For the situation that $\Theta \subseteq \mathbb{R}$ and $\vartheta_0 \in \mathbb{R}$ we study one-sided tests for

$$H_0 : \vartheta \leq \vartheta_0 \quad \text{and} \quad H_1 : \vartheta > \vartheta_0$$

and two-sided tests for

$$H_0 : \vartheta = \vartheta_0 \quad \text{and} \quad H_1 : \vartheta \neq \vartheta_0.$$

The one-sided case includes also the situation that the inequality signs in H_0 and in H_1 are in the opposite directions.

Example 6.28 (One-sided binomial test)**.** For the statistical model from Example 6.25 and some $p_0 \in (0, 1)$ we consider the null hypothesis $H_0 : p \leq p_0$ and the alternative $H_1 : p > p_0$. Note that we have $p_0 = 0.5$ in Example 6.25. Our goal is to construct a test of level $\alpha \in (0, 1)$ for $H_0 : p \leq p_0$ against $H_1 : p > p_0$. Let $m_0 \in \mathbb{N}_0$ be the smallest $m \in \mathbb{N}_0$ such that

$$\sum_{k=m+1}^{n} \binom{n}{k} p_0^k (1 - p_0)^{n-k} \leq \alpha,$$

i.e., m_0 is an $(1-\alpha)$-quantile of a Binom(n, p_0)-distribution. We define

$$\varphi : \{0,1\}^n \ni x = (x_1, \dots, x_n) \mapsto \begin{cases} 0, & \sum_{i=1}^n x_i \le m_0, \\ 1, & \sum_{i=1}^n x_i > m_0. \end{cases}$$

For $p \in [0,1]$ with $p \le p_0$ we obtain that

$$\mathbf{P}_p(\{x \in \{0,1\}^n : \varphi(x) = 1\}) = \mathbf{P}_p\left(\left\{x \in \{0,1\}^n : \sum_{i=1}^n x_i > m_0\right\}\right) = \mathbf{P}_p\left(\sum_{i=1}^n X_i > m_0\right).$$

Here, we have that $\sum_{i=1}^n X_i \sim \text{Binom}(n,p)$. Since the distribution function of a binomial distribution with parameters n and p at a fixed $t \in \mathbb{R}$ is non-increasing in p (this is left to the reader as an exercise), we have that

$$\mathbf{P}_p\left(\sum_{i=1}^n X_i > m_0\right) \le \mathbf{P}_{p_0}\left(\sum_{i=1}^n X_i > m_0\right) = \sum_{k=m_0+1}^n \binom{n}{k} p_0^k (1-p_0)^{n-k} \le \alpha.$$

In the last step we used the definition of m_0. This implies

$$\mathbf{P}_p(\{x \in \{0,1\}^n : \varphi(x) = 1\}) \le \alpha$$

for $p \in [0,1]$ with $p \le p_0$ so that φ is a level-α test for $H_0 : p \le p_0$ against $H_1 : p > p_0$.

Example 6.29 (Two-sided binomial test)**.** For $n \in \mathbb{N}$ we consider the statistical model $(\{0,1\}^n, \mathcal{P}(\{0,1\}^n), (\mathbf{P}_p)_{p\in[0,1]})$, where $\mathbf{P}_p$ is the joint distribution of n independent Ber(p)-distributed random variables $X_1, \dots, X_n$ for $p \in [0,1]$. Our goal is to find a test of level $\alpha \in (0,1)$ for $H_0 : p = p_0$ against $H_1 : p \ne p_0$ for a fixed $p_0 \in (0,1)$. Let $\ell_0 \in \mathbb{N}_0$ be the largest $\ell \in \mathbb{N}_0$ such that

$$\sum_{k=0}^{\ell-1} \binom{n}{k} p_0^k (1-p_0)^{n-k} \le \frac{\alpha}{2}$$

and let $u_0 \in \mathbb{N}_0$ be the smallest $u \in \mathbb{N}_0$ such that

$$\sum_{k=u+1}^{n} \binom{n}{k} p_0^k (1-p_0)^{n-k} \le \frac{\alpha}{2}.$$

This means that ℓ_0 is an $\alpha/2$-quantile and u_0 is an $(1-\alpha/2)$-quantile of a Binomial(n, p_0)-distribution. Then,

$$\varphi : \{0,1\}^n \ni x = (x_1, \dots, x_n) \mapsto \begin{cases} 0, & \sum_{i=1}^n x_i \in [\ell_0, u_0], \\ 1, & \sum_{i=1}^n x_i \notin [\ell_0, u_0], \end{cases}$$

is a level-α test for $H_0 : p = p_0$ against $H_1 : p \neq p_0$ since

$$\begin{aligned}
\mathbf{P}_{p_0}(\{x \in \{0,1\}^n : \varphi(x) = 1\}) &= \mathbf{P}_{p_0}\left(\left\{x = (x_1, \dots, x_n) \in \{0,1\}^n : \sum_{i=1}^n x_i \notin [\ell_0, u_0]\right\}\right) \\
&= \mathbf{P}_{p_0}\left(\sum_{i=1}^n X_i \notin [\ell_0, u_0]\right) \\
&\leq \mathbf{P}_{p_0}\left(\sum_{i=1}^n X_i < \ell_0\right) + \mathbf{P}_{p_0}\left(\sum_{i=1}^n X_i > u_0\right) \\
&\leq \frac{\alpha}{2} + \frac{\alpha}{2} = \alpha,
\end{aligned}$$

where we used $\sum_{i=1}^n X_i \sim \mathrm{Binom}(n, p_0)$ and the definitions of ℓ_0 and u_0 in the second last step.

Example 6.30. Let $(\{0,1\}^n, \mathcal{P}(\{0,1\}^n), (\mathbf{P}_\vartheta)_{\vartheta \in \Theta})$ be a statistical model with a $\vartheta_0 \in \Theta$ such that $\mathbf{P}_{\vartheta_0}$ is the joint distribution of n independent Ber(1/2)-distributed random variables $X_1, \dots, X_n$. Our goal is to test $H_0 : \vartheta = \vartheta_0$ against $H_1 : \vartheta \neq \vartheta_0$ at level $\alpha \in (0,1)$. Under $\mathbf{P}_{\vartheta_0}$ we have that the indicator random variables

$$\mathbf{1}\{X_1 = X_2\}, \mathbf{1}\{X_2 = X_3\}, \dots, \mathbf{1}\{X_{n-1} = X_n\}$$

are $n-1$ independent Ber(1/2)-distributed random variables. Let $\ell_0 \in \mathbb{N}_0$ be the largest $\ell \in \mathbb{N}_0$ such that

$$\sum_{k=0}^{\ell-1} \binom{n-1}{k} \frac{1}{2^{n-1}} \leq \frac{\alpha}{2}$$

and let $u_0 \in \mathbb{N}_0$ be the smallest $u \in \mathbb{N}_0$ such that

$$\sum_{k=u+1}^{n-1} \binom{n-1}{k} \frac{1}{2^{n-1}} \leq \frac{\alpha}{2}.$$

Then, it follows from Example 6.29 that

$$\varphi : \{0,1\}^n \ni x = (x_1, \dots, x_n) \mapsto \begin{cases} 0, & \sum_{i=1}^{n-1} \mathbf{1}\{x_i = x_{i+1}\} \in [\ell_0, u_0], \\ 1, & \sum_{i=1}^{n-1} \mathbf{1}\{x_i = x_{i+1}\} \notin [\ell_0, u_0], \end{cases}$$

is a level-α test for $H_0 : \vartheta = \vartheta_0$ against $H_1 : \vartheta \neq \vartheta_0$.

For $\alpha = 0.9$ and $n = 40$ a calculation with R leads to $\ell_0 = 14$ and $u_0 = 25$.

Example 6.31 (One-sided Gauss test)**.** For $n \in \mathbb{N}$ and $\sigma_0 > 0$ let $(\mathbb{R}^n, \mathcal{B}(\mathbb{R}^n), (\mathbf{P}_\mu)_{\mu \in \mathbb{R}})$ be a statistical model where $\mathbf{P}_\mu$ is the joint distribution of n independent Normal(μ, σ_0^2)-distributed random variables $X_1, \dots, X_n$ for $\mu \in \mathbb{R}$. This means that the sample comes from independent normally distributed random variables with known standard deviation

σ_0 and unknown expectation μ. Our goal is to test the null hypothesis $H_0 : \mu \leq \mu_0$ against the alternative $H_1 : \mu > \mu_0$ for some given $\mu_0 \in \mathbb{R}$.

For $\alpha \in (0,1)$ we consider the test

$$\varphi : \mathbb{R}^n \ni x \mapsto \begin{cases} 0, & \overline{x} < \mu_0 + z_{1-\alpha}\sigma_0/\sqrt{n}, \\ 1, & \overline{x} \geq \mu_0 + z_{1-\alpha}\sigma_0/\sqrt{n}, \end{cases}$$

where $\overline{x}$ is the sample mean of $x = (x_1, \ldots, x_n)$ and $z_{1-\alpha}$ is the $(1-\alpha)$-quantile of the standard normal distribution. For $\mu \in \mathbb{R}$ we obtain that

$$\begin{aligned} \mathbf{P}_\mu(\{x \in \mathbb{R}^n : \varphi(x) = 1\}) &= \mathbf{P}_\mu(\{x \in \mathbb{R}^n : \overline{x} \geq \mu_0 + z_{1-\alpha}\sigma_0/\sqrt{n}\}) \\ &= \mathbf{P}_\mu\big(\overline{X} \geq \mu_0 + z_{1-\alpha}\sigma_0/\sqrt{n}\big), \end{aligned}$$

where the sample mean $\overline{X} = \frac{1}{n}\sum_{i=1}^n X_i$ follows a Normal$(\mu, \sigma_0^2/n)$-distribution. This implies that

$$\begin{aligned} \mathbf{P}_\mu(\{x \in \mathbb{R}^n : \varphi(x) = 1\}) &= \mathbf{P}_\mu\big(\overline{X} - \mu \geq \mu_0 - \mu + z_{1-\alpha}\sigma_0/\sqrt{n}\big) \\ &= \mathbf{P}_\mu\left(\frac{\overline{X} - \mu}{\sigma_0/\sqrt{n}} \geq \frac{\mu_0 - \mu}{\sigma_0/\sqrt{n}} + z_{1-\alpha}\right) \\ &= 1 - \mathbf{P}_\mu\left(\frac{\overline{X} - \mu}{\sigma_0/\sqrt{n}} < \frac{\mu_0 - \mu}{\sigma_0/\sqrt{n}} + z_{1-\alpha}\right) \\ &= 1 - \Phi\left(\frac{\mu_0 - \mu}{\sigma_0/\sqrt{n}} + z_{1-\alpha}\right). \end{aligned}$$

For $\mu \leq \mu_0$ the fraction in the argument of Φ is non-negative so that, by the definition of $z_{1-\alpha}$,

$$\mathbf{P}_\mu(\{x \in \mathbb{R}^n : \varphi(x) = 1\}) \leq 1 - \Phi(z_{1-\alpha}) = 1 - (1-\alpha) = \alpha.$$

Thus, we have shown that φ is a test of level α for $H_0 : \mu \leq \mu_0$ against $H_1 : \mu > \mu_0$.

The above computations imply that the test φ has the power

$$\mathbf{P}_\mu(\{x \in \mathbb{R}^n : \varphi(x) = 1\}) = 1 - \Phi\left(\frac{\mu_0 - \mu}{\sigma_0/\sqrt{n}} + z_{1-\alpha}\right)$$

for $\mu > \mu_0$, whence the power converges to α as μ goes to μ_0 from above. Thus, the rejection probability of the null hypothesis is rather small if the true parameter μ is only slightly larger than μ_0.

Example 6.32 (Two-sided Gauss test)**.** For the same statistical model as in Example 6.31 we consider the null hypothesis $H_0 : \mu = \mu_0$ and the alternative $H_1 : \mu \neq \mu_0$ for some fixed $\mu_0 \in \mathbb{R}$. For $\alpha \in (0,1)$ define the test

$$\varphi : \mathbb{R}^n \ni x \mapsto \begin{cases} 0, & \overline{x} \in (\mu_0 - z_{1-\alpha/2}\sigma_0/\sqrt{n}, \mu_0 + z_{1-\alpha/2}\sigma_0/\sqrt{n}), \\ 1, & \overline{x} \notin (\mu_0 - z_{1-\alpha/2}\sigma_0/\sqrt{n}, \mu_0 + z_{1-\alpha/2}\sigma_0/\sqrt{n}). \end{cases}$$

We have that

$$\begin{aligned}\mathbf{P}_{\mu_0}(\{x \in \mathbb{R}^n : \varphi(x) = 1\}) &= \mathbf{P}_{\mu_0}\big(\overline{X} \notin (\mu_0 - z_{1-\alpha/2}\sigma_0/\sqrt{n}, \mu_0 + z_{1-\alpha/2}\sigma_0/\sqrt{n})\big) \\ &= 1 - \mathbf{P}_{\mu_0}\big(\overline{X} \in (\mu_0 - z_{1-\alpha/2}\sigma_0/\sqrt{n}, \mu_0 + z_{1-\alpha/2}\sigma_0/\sqrt{n})\big) \\ &= 1 - (1-\alpha) = \alpha,\end{aligned}$$

where we used (6.2) and (6.3) from Example 6.20 in the second last step. This proves that φ is a level-α test for $H_0 : \mu = \mu_0$ against $H_1 : \mu \neq \mu_0$.

This example shows that constructing confidence intervals and constructing tests are closely related problems.

Remark 6.33 (p-value)**.** Throughout this section we always choose a level α for our tests. In statistical software such as R often so-called p-values are used instead of specifying the level α. For a given sample the p-value is the probability to obtain under the null hypothesis a result that is at least as extreme as the sample. This way the p-value measures how strongly the sample contradicts the null hypothesis. For a given sample with a p-value p the null hypothesis can be rejected for all levels α such that $\alpha \geq p$ and cannot be rejected for levels α with $\alpha < p$.

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